



# Catchment Overview: Garren Brook

*A report prepared on behalf of Wye Salmon Association (WSA)*

|                |                        |
|----------------|------------------------|
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## Executive Summary

The objective of this report is to review the current ecological status of the Garren Brook Catchment. This is the second such report, commissioned by Wye Salmon Association (WSA), focussing on specific sub catchments as exemplars within the wider River Wye context. WSA believe that following such a 'bottom-up' approach will support the development of well targeted local Nutrient Management Plans (NMPs) led by The Wye Nutrient Management Board. A key intention is to leverage the growing body of citizen science data and provide appropriate analysis in the hope that the responsible organisations will respond with meaningful improvement actions. It will clearly not be possible to maintain the current enthusiasm for citizen science unless volunteers see that the work leads to measurable improvements to the health of the rivers.

The Garren Brook joins the River Wye just downstream of Goodrich and drains approximately 90 km<sup>2</sup> of rolling Herefordshire countryside. Most of the catchment sits above old red sandstone bedrock, which leads to high surface water alkalinity and a propensity for fine sedimentation. Economic activity within the catchment is almost exclusively agricultural.

Farming activities are roughly evenly split between grassland and arable cropping. There is no unmanaged, rough land within the catchment and little in the way of woodland; agriculture is intensive. Based on DEFRA statistics for the whole of Herefordshire, we estimate the current livestock population within the catchment to be around 4,000 cattle and 23,000 sheep, spread over about 120 farms. Importantly, the long-term trend on ruminants has been downwards, with sheep numbers falling 21% and cattle numbers falling 32% since 2000. In the meantime, poultry numbers have increased dramatically, rising from around 500k birds in 2000 to approximately 2,300k birds today.

There are no large settlements within the Garren Catchment. The human population, estimated to be about 3,500, is thus relatively dispersed. Consequently, there is little access to mains drainage, and we find that 96% of the population are served by private septic tanks. Academic research in other English regions has shown that a high proportion of rural septic tanks are poorly maintained and often do not comply with current regulations, resulting in increased phosphorus emissions.

The Gamber joins the Garren Brook part way down the catchment and this confluence naturally splits the catchment into three waterbodies for administrative purposes, and for our analysis. The Garren Brook is not part of the River Wye SAC, therefore default Water Framework Directive (WFD) targets apply for water quality metric. In their 2019 WFD assessment, EA determined that the catchment failed to achieve *good* ecological status for three elements: *fish*, *macrophytes* and *orthophosphate*. Reasons cited for these failures centre around poor riparian management, poor stock management and poor nutrient management. The Garren also fails on chemical status, with high levels of mercury, nonylphenol, PFOS & PBDE - none of which are ideal in a food growing area.

A further important pressure on the Garren system is abstraction. We show how abstraction is heavily concentrated within the arable dominated Gamber valley and how total licensed volumes represent a significant part of summer flow volumes on that tributary. Moreover, the abstraction is very remote from the sole EA flow measuring station at Marstow Mill on the lower Garren.

Our analysis of historical water quality data from EA shows some important long-term trends. Levels of orthophosphate have increased significantly since 2000, especially in the upper catchment. Moreover, the annual average orthophosphate fraction (of total phosphorus) has fallen significantly. Combining these two factors, we estimate that the total phosphorus load has increased since 2000 by 68% on the Upper Garren, by 49% on the Gamber and by 25% on the Lower Garren. Against the backdrop of falling cattle and sheep numbers, the only plausible explanation for these dramatic increases would be poultry manure spreading. Although not a WFD water quality element, we also note that annual mean

concentrations of nitrate are relatively high, being approximately five times the value observed on the pasture dominated Llynfi. Furthermore, the spatial variation correlates with arable cropping.

Whilst citizen science data may lack the formal traceability of EA data, it compensates with improved spatial and temporal frequency. This opens up the possibility of more sophisticated analysis techniques, in particular load apportionment modelling (LAM). WSA have been sampling in the Garren Catchment since June 2021 at predominantly weekly intervals across eight locations. We report here on measurements of reactive phosphorus (RP), acidity (pH) and specific conductivity (SC), plus visual observations of turbidity in the rivers. To better interpret water quality RP data, we also make use of the results of our recent soil phosphorus survey across the Garren Catchment. Key findings from the WSA CS data are as follow:

- SC is consistent with data reported by EA for matching locations, on an annual mean basis. Analysis of flowrate dependence yields a well-defined, linear relationship characteristic of the drainage pathways in each location.
- pH is also consistent with EA data on an annual mean basis, but LAM analysis highlights some low pH outliers, which are suggestive of contamination by slurry spreading for winter crops.
- RP is also consistent with EA orthophosphate data on an annual mean basis. LAM analysis shows that all areas of the catchment are dominated by diffuse sources of RP. However, we find strong evidence of fixed flow sources in certain locations, notably in the Garren above Orcop. These signals are indicative of a significant number directly discharging (non-compliant) septic tanks in the area. Measurements on a small un-named stream in this area exhibit by far the highest diffuse signal observed within the catchment and correlate with the presence of two IPUs and an anaerobic digestion plant.
- The observed rise in soil Olsen-P fits well with what would be expected from the spreading of 33% of the poultry manure generated within the catchment on the grasslands above the Gamber confluence.

Next, we build a simple source apportionment model using our population estimates to compute sewage and manure volumes within the catchment. Using export coefficients similar to those employed in our Llynfi report, we then model the phosphorus export to surface water. Key conclusions are as follow:

- Manure loads within the catchment total 887 kgP/day, of which 78% is from poultry, 17% is from cattle and 6% from sheep. We show that this volume of manure is only marginally capable of being spread within the Garren Catchment within the limits of NVZ regulations.
- Exports to surface water are dominated by agriculture, with human sewage contribution around 20% of the total.
- The rise in river RP levels observed since 2000 is consistent with spreading 39% of the total poultry manure load – a result in remarkable agreement with our estimate from the rise in soil Olsen-P.

In conclusion, we summarise the main pressures on the Garren as abstraction, turbidity and phosphorus. Phosphorus is the most intractable of these, with soil P being driven inexorably upwards, mainly as a result of poultry manure spreading, which will drive increasing export of P into the rivers.

This report makes several recommendations for action by key actors. Of particular importance is the need for the poultry sector to properly deal with its manure waste stream, rather than dumping it on the land. More generally, there is a pressing need for farmers to proactively lower soil Olsen-P in line with best practice guidance and fence livestock away from surface water to reduce sedimentation. We also suggest that EA should organise an annual soil survey to monitor progress on the phosphorus problem.

## Introduction

This report is the second in a series commissioned by Wye Salmon Association (WSA) with the purpose of leveraging newly available citizen science (CS) data to better understand specific sub catchments flowing into the River Wye SAC. As specific sub-catchments are all uniquely different, the development of individual sub-catchment Nutrient Management Plans (NMPs) is recognised to be a seminal step for addressing remediation of nutrient imbalance in SAC rivers in both England and Wales. Consequently, sub-catchment NMPs form an integral part of the “River Wye SAC Nutrient Management Plan, Phosphate Action Plan” published by Natural Resources Wales, Environment Agency & Natural England in September 2021 with accountability for delivery devolved to the Wye Nutrient Management Board (NMB). The list of generic actions for the development of a NMP can be found in Appendix 4. Our reports do not purport to be peer reviewed science as none of us are professional ecologists. They are simply the reporting of citizen science work and an attempt to fit the results into the wider context of sub-catchments, presenting conclusions and recommendations to be addressed by those accountable for producing NMPs and for delivering remediation, including for example the NMB and Herefordshire County Council, the EA, NRW & DCWW.

A key intention of this work is to exploit the citizen science data now becoming available. It is obvious that, if the engagement and enthusiasm of citizen scientists is to be maintained into the future, then the data collected must be used to trigger meaningful actions which result in measurable improvements to the health of the Wye and its tributaries. Our volunteers are motivated to collect this data to help improve the fortunes of our beloved rivers, not as an interesting hobby and certainly not to cover over the cracks in our under-resourced Environment Agency (EA). Thus, this work attempts to provide a thorough analysis of the citizen science data and combine resulting information with what can be gleaned from complementary EA data, in the hope that the responsible organisations will respond with appropriate action.

The first of our reports (Green, 2022a) examined the Llynfi Catchment in rural Wales, falling within The Brecon Beacons National Park, which was found to be dominated by livestock farming. In this report, we examine a sub catchment<sup>1</sup> in the lower reaches of Herefordshire; an area including both arable and livestock farming, plus a significant number of intensive poultry units (IPUs).

The Garren Brook is one of a number of significant Wye tributaries which, for reasons not well understood, are not included within the River Wye SAC. As such, the Garren is subject to default Water Framework Directive (WFD) water quality targets, rather than the more stringent targets which apply within the SAC. However, attempting to improve water quality metrics within the SAC, without considering all tributaries is clearly a nonsense. Therefore, WSA consider a study of this sub catchment to be highly relevant to the issues facing the failing SAC, as well as to the Garren Catchment in and of itself.

This work examines the continuous, background pollution which impacts the Garren Catchment. In the rural setting of this catchment, surface water pollution originates from two principal sources: agriculture and human sewage. We shall focus primarily on phosphorus pollution as a key aspect, although it is understood that there may be other pollutants of concern. We shall touch on some of the other anthropogenic pressures where these emerge from the data.

A key philosophical aspect of this work is to examine the catchment at a high level of abstraction. It is certainly true that CS data, with higher spatial and temporal frequency, may uncover specific point pollution sources; indeed, we shall discuss one such case later. But primarily we shall be working at the

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<sup>1</sup> Terminology: although the Garren is a *sub catchment* of the Wye, within this report it is convenient to refer to the *Garren Catchment* and reserve the term *sub catchment* to refer to parts thereof.

sub catchment level. Phosphorus export from sewage treatment works (STWs) is well quantified insofar as fluxes come from pipes which are monitored (in the case of STWs with P-stripping), or otherwise well characterised (in terms of input load). On the other hand, we shall show that the Garren Catchment is dominated by diffuse sources, and for these, fluxes are impossible to measure. Moreover, the science of how phosphorus added to the land is transported to the river, and how it behaves within the river, is highly complex. These topics have been the subject of decades of academic research and will doubtless be the subject of decades more. However, detailed spatial and temporal knowledge of nutrient flows is neither possible, nor necessary to addressing the problems of the Wye Catchment, or the sub catchments within it. Crucially, we cannot afford to wait for ‘perfect science’ before beginning to address these chronic diffuse pollution issues.

The recent RePhOKUs report (Withers et al., 2022) teaches us that the fundamental problem facing the River Wye is a systemic one. The soils within the Wye Catchment are accumulating phosphorus at the rate of around 3,000 tonnes P per year. Of this, around 2,600 tonnes (85%) come from poultry manure; the unmanaged waste stream of an industry which has grown massively over the last quarter century. This accumulation is driving an ever-increasing rate of leaching and run-off into the rivers. The farm-scale mitigation measures which have historically been deployed to address phosphorus pollution are entirely inadequate to the scale of this problem (Davey et al., 2020) and it is important to note that none of the generic NMP actions listed in Appendix 4 adequately address this issue. What is needed is a catchment scale project to eliminate the phosphorus flow imbalance and progressively drive down the soil phosphorus concentration. To this end, it appears axiomatic that we must begin to systematically measure soil phosphorus concentrations.

Within this report, we shall firstly describe the river and its catchment, including the relevant activities taking place within it. We shall then review what the Environment Agency (EA) has reported about the river as part of their WFD compliance assessment. Sources include both the EA’s high-level compliance reports and also the water quality measurements which underpin these. We shall then examine the WSA CS data. Because of the higher temporal frequency of the CS water quality sampling compared to EA sampling, we are able to utilise the technique of load apportionment modelling (LAM) to extract useful information from this data, beyond simple annual means and bar charts. We shall also make use of soil phosphorus concentrations across the catchment, derived from recent CS soil sampling. Finally, we shall present a source apportionment model (SAM) of the catchment and see how well it accounts for the observed data.

## Catchment overview

### The river

The Garren Brook is a relatively small river, draining approximately 90 km<sup>2</sup> of Herefordshire farmland. It flows around 30 km in a south-easterly direction to join the River Wye just downstream of Goodrich. The topography of the catchment is shown in Figure 1.

The catchment, outlined in red, is defined with respect to the EA gauging station, (ref. 55036) at Marstow Mill, approximately 1km upstream of the Wye confluence. It is flanked by the Wye Valley to the east and south, by the Monnow Valley to the south-west, and by the Worm Valley to the North. A distinct ridge of hills defines the western edge of the catchment, peaking at 366m elevation on Garway Hill, but the eastern edge is an altogether gentler affair. The source of the Garren Brook lies on the eastern slopes of Garway Hill. The river drops fairly steeply down to Orcop and from there maintains a gentler pace as it meanders down to the Gamber confluence north of Llangarron. The Gamber flows over the lower ground to east and has a lower average gradient than the Garren. Below the confluence, the gradient remains moderate until the last few kilometres, where the river drops more steeply into

the Wye Valley, flowing beneath the A40 near Whitchurch. The elevation profiles for the two rivers are shown in Appendix 1.

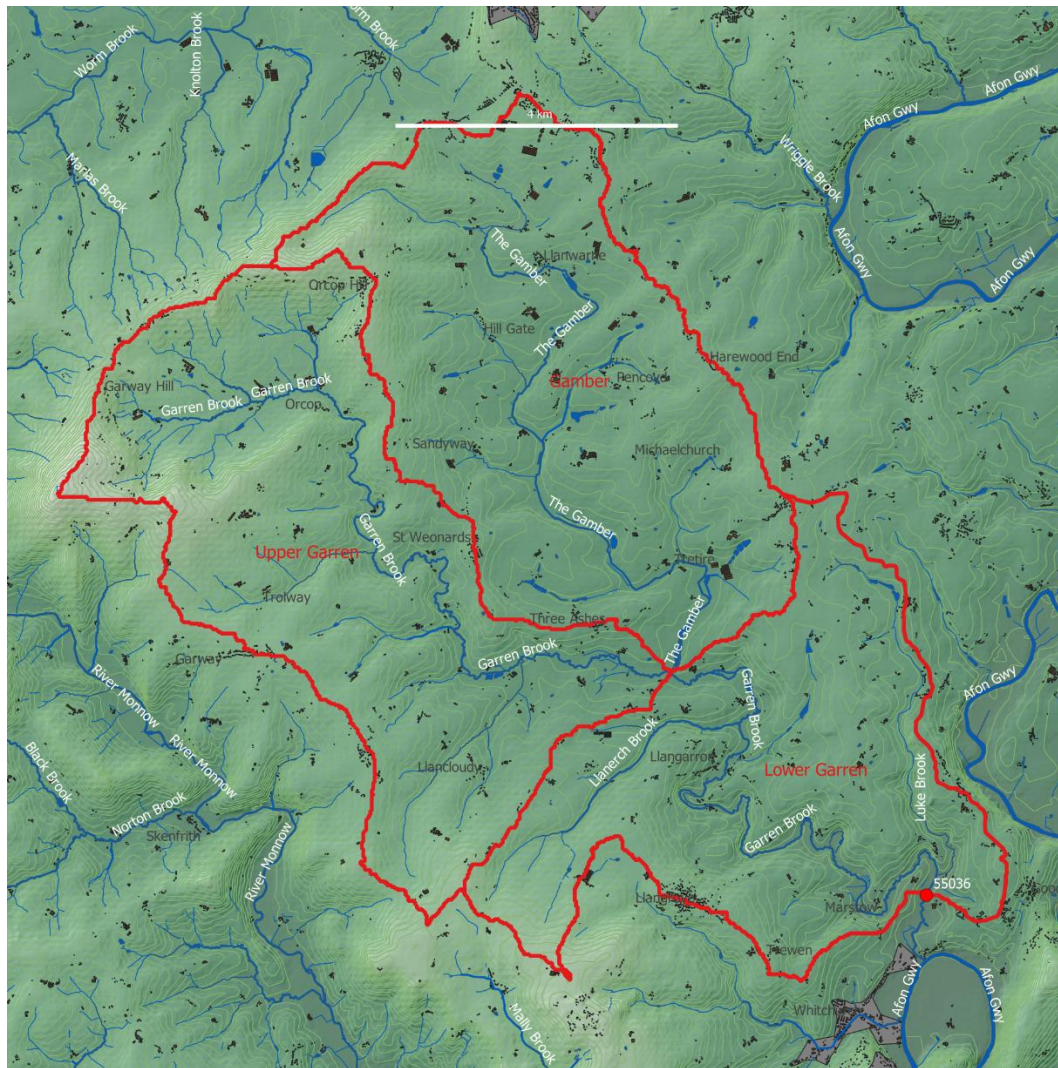


Figure 1: topography of the Garren Catchment; the red outline is the catchment above EA gauging station 55036 at Marstow Mill (red circle, bottom right); also shown are the three sub catchments.

For the analysis to follow, it is convenient to break the catchment into three sub catchments, as shown in Figure 1. The *Upper Garren* sub catchment is defined as the Garren above the Gamber confluence. The *Gamber* sub catchment, by definition, only exists above this confluence. The *Lower Garren* sub catchment is everything below the Gamber confluence and includes the Llanerch Brook and Luke Brook. Note that this sub catchment delineation matches the EA's water body definitions.

## Geology

The entire Garren Catchment lies over *lower Old Red Sandstone* within the *Anglo-Welsh Basin*; this is a very low permeability bedrock. The sandstone mineralogy is responsible for high values of alkalinity in the groundwater-fed streams of the catchment, in common with several other lower Wye tributaries (*inter alia* the Monnow, Trothy, lower Lugg & Frome). Further details are provided in Appendix 3.

Over most of the Garren Catchment, soil depth is classified as *shallow* (soil and subsoil can be dug to a depth of approximately 0.5m) and the soil texture is classified as *loam to sandy loam* over a sand rich subsoil. However, immediately adjacent to the streams, we find fluvial deposits of deep *clay to sandy loam*.

## Hydrology

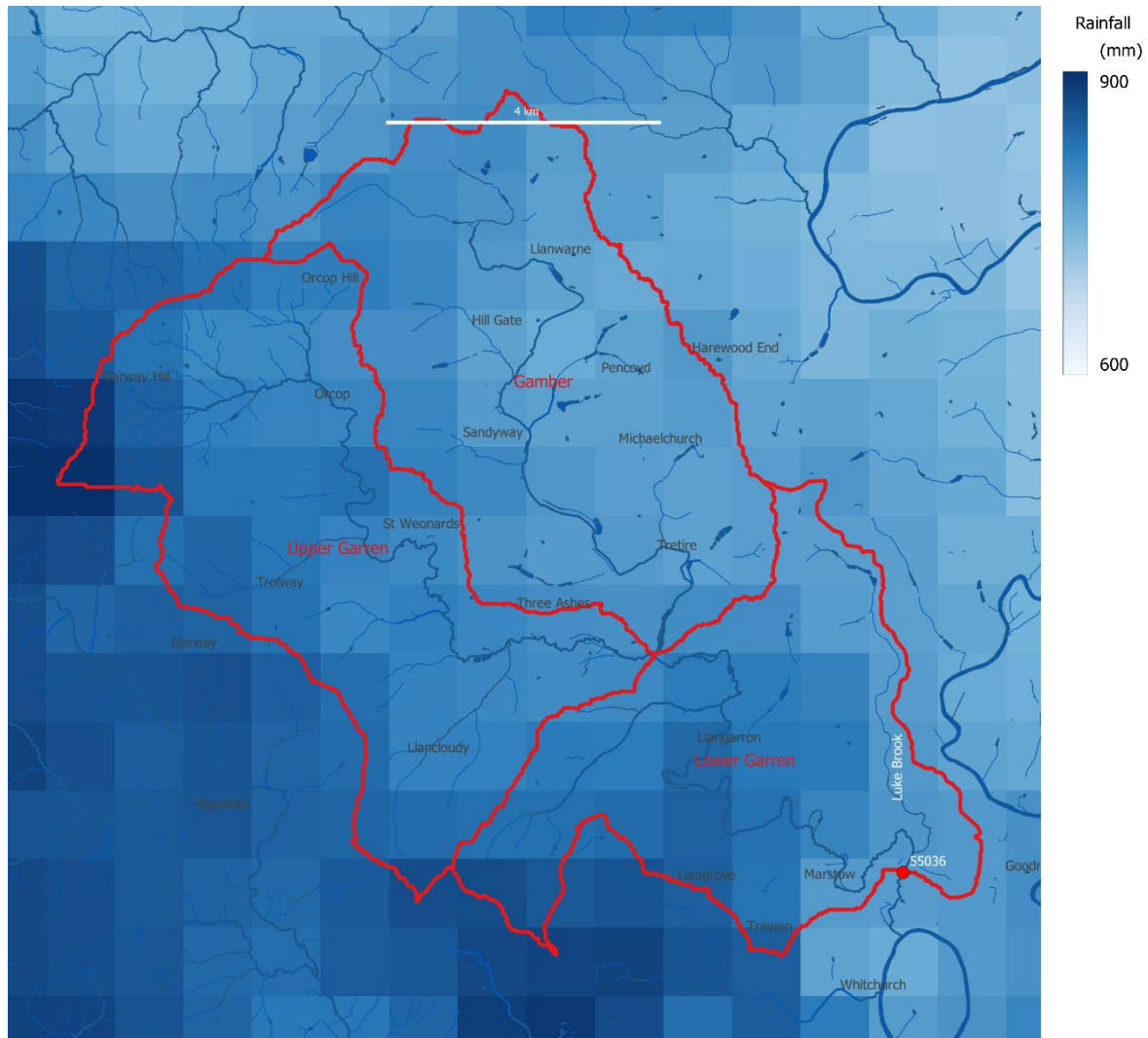


Figure 2: annual average rainfall (2015-19) over the Garren Catchment; data from UK Centre for Ecology & Hydrology (CEH)

Rainfall across the Garren Catchment is shown in Figure 2 above. It can be seen that the heaviest rainfall is to the west and south of the catchment, with the Gamber sub catchment being noticeably drier. Annual average rainfall ranges from a maximum of 908mm on Garway Hill down to a minimum of 758mm near Llanwarne. Statistics for each of the sub catchments are listed in Table 1 below.

|                               | Area<br>(ha) | Average rainfall<br>(mm. yr <sup>-1</sup> ) | Volume<br>(10 <sup>6</sup> .m <sup>3</sup> .yr <sup>-1</sup> ) | Fraction    |
|-------------------------------|--------------|---------------------------------------------|----------------------------------------------------------------|-------------|
| Upper Garren                  | 3,244        | 824                                         | 26.7                                                           | 37%         |
| Gamber                        | 3,217        | 786                                         | 25.3                                                           | 35%         |
| Lower Garren                  | 2,499        | 819                                         | 20.5                                                           | 28%         |
| <b>Total Garren Catchment</b> | <b>8,959</b> | <b>809</b>                                  | <b>72.5</b>                                                    | <b>100%</b> |

Table 1: annual rainfall statistics for the Garren Catchment and the sub catchments

The discharge, or volumetric flowrate, of the Garren Brook is measured by EA at Marston Mill (see Figure 1) using an ultrasonic velocity sensor. Key flow statistics from this gauging station, reported via the [National River Flow Archive](#) website (operated by CEH), are as follow:

|                                        |                                      |
|----------------------------------------|--------------------------------------|
| Base flow index <sup>2</sup>           | 0.73                                 |
| 95% exceedance flow (Q95) <sup>3</sup> | 0.09 m <sup>3</sup> .s <sup>-1</sup> |
| 50% exceedance flow (Q50) <sup>4</sup> | 0.61 m <sup>3</sup> .s <sup>-1</sup> |
| Mean flow                              | 1.02 m <sup>3</sup> .s <sup>-1</sup> |

Table 2: key flow parameters for the Garren Brook at Marstow Mill

This base flow index for the Garren is high by the standards of the Wye (cf. the Irfon at 0.38 and Elan at 0.34). This is indicative of a lowland, groundwater-fed river.

Based on the mean flow value, the Garren carries 44% of the rain landing on its catchment; this is often referred to as the hydrologically effective rainfall (HER), the fraction of which is calculated in Box 1 below. The remaining 56% is lost to evaporation and transpiration.

#### Box 1: effective rainfall calculation

Volume of rain = 72.5 x 10<sup>6</sup> m<sup>3</sup>.yr<sup>-1</sup>  
= 72.5 x 10<sup>6</sup> / (3600 x 24 x 365) m<sup>3</sup>.s<sup>-1</sup>  
= 2.30 m<sup>3</sup>.s<sup>-1</sup>

HER fraction = Garren mean flow / volume of rain  
= 1.02 / 2.30 = 44%

It will be useful later to know how much water flows in the tributaries. In the absence of flow measuring stations, we must estimate these. By approximating that the HER fraction is constant across the catchment, we can then estimate the flow parameters for the sub catchments as follows:

|              | Fraction<br>(from Table 1) | Mean flow<br>(m <sup>3</sup> .s <sup>-1</sup> ) | Q95 flow<br>(m <sup>3</sup> .s <sup>-1</sup> ) |
|--------------|----------------------------|-------------------------------------------------|------------------------------------------------|
| Upper Garren | 37%                        | 0.38                                            | 0.035                                          |
| Gamber       | 35%                        | 0.36                                            | 0.033                                          |
| Lower Garren | 28%                        | 0.29                                            | 0.027                                          |

Table 3: estimated flow parameters for the sub catchments, assuming constant HER ratio; each sub catchment flow equals the full catchment figure multiplied by 'Fraction'.

## Land use

Figure 3 below shows the distribution of land use within the Garren Catchment. Apart from some acid grassland on the summit of Garway Hill, the whole of the catchment is essentially managed farmland. There is relatively little broadleaved woodland, almost no coniferous woodland and negligible amount of suburban development. There is a noticeable differentiation of land use between the three sub catchments. Grassland predominates in the Upper Garren, whereas arable and horticulture predominates in the Gamber, particularly in the flatter land in the south. The Lower Garren is a more even split between the two. The breakdown is illustrated in Figure 4.

Agricultural land use is not a static thing, however. It evolves over time to reflect changing economic and industrial factors. By comparing the above dataset for the full Garren Catchment to the corresponding 2000 dataset, we can see some changes. Since 2000, arable and horticultural land use has increased by approximately 9%, of which 2% used to be grassland. The balance is primarily due to the elimination of one third of the broadleaved woodland, almost all the coniferous woodland and all the remaining bracken and bare ground areas. The 2% reduction in total grassland however is not all it seems. In 2000, approximately one quarter of grassland, representing 10% of the total Garren Catchment, was unimproved, neutral grassland. By 2021, we find almost no unimproved grassland at all. So, the overall picture is one of increasing agricultural intensification, where arable farmers have

<sup>2</sup> Base flow index is a measure of 'flashiness' of a river. A chalk-stream would have a value close to 1, a mountain stream below 0.5.

<sup>3</sup> Q95 is the flow which is exceeded for 95% of the time. It represents the river 'on its bones' and is important for worst-case dilution considerations.

<sup>4</sup> Q50 flow is exceeded 50% of the time. It is arguably closer to what we would intuitively call 'normal flow' than is the mean flow, which is influenced by brief but high spate events.

maximised acreage, largely at the expense of natural habitats, and the livestock farmers have ‘improved’ all grassland to maximise yields.

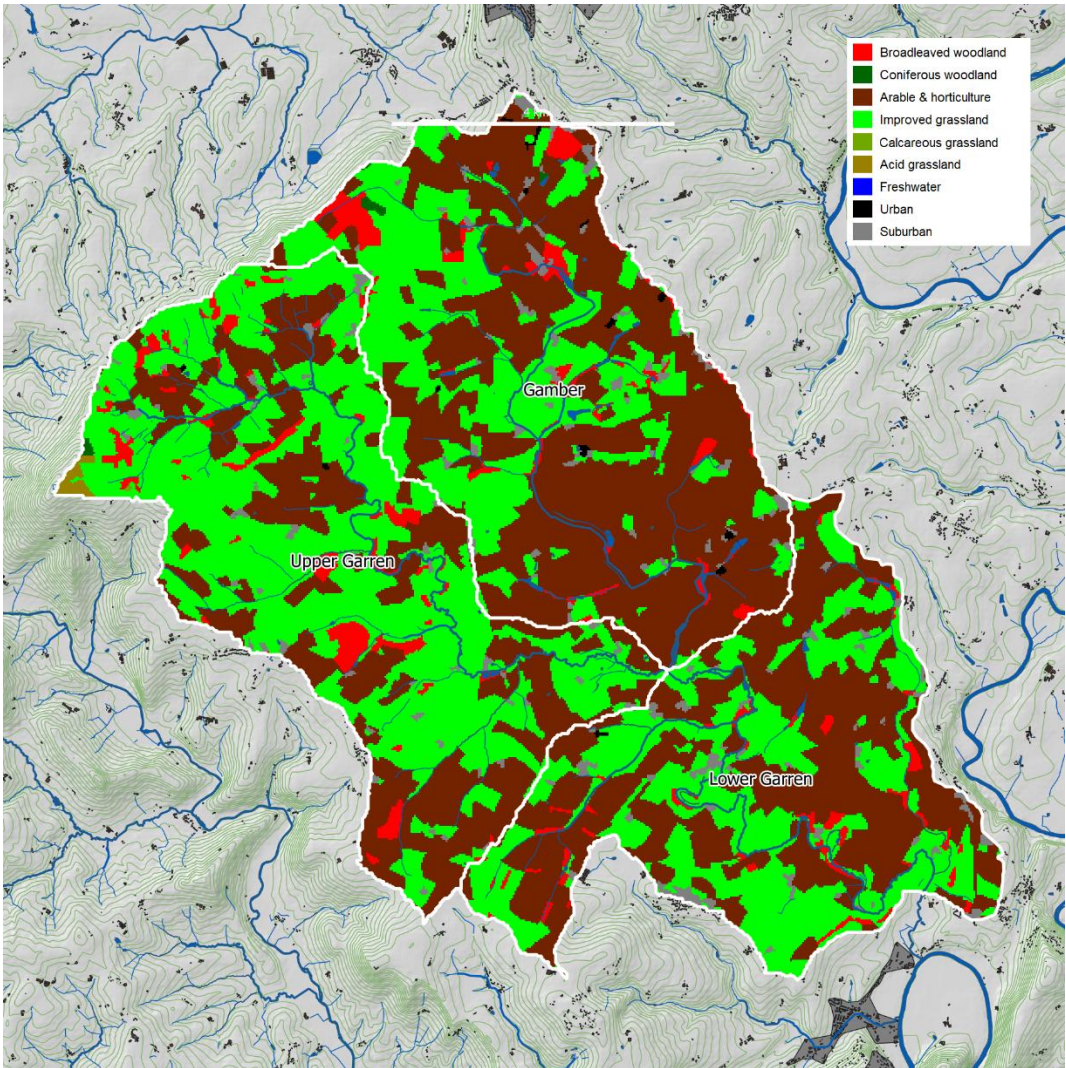


Figure 3: land use in the Garren Catchment, extracted from 2020 CEH data

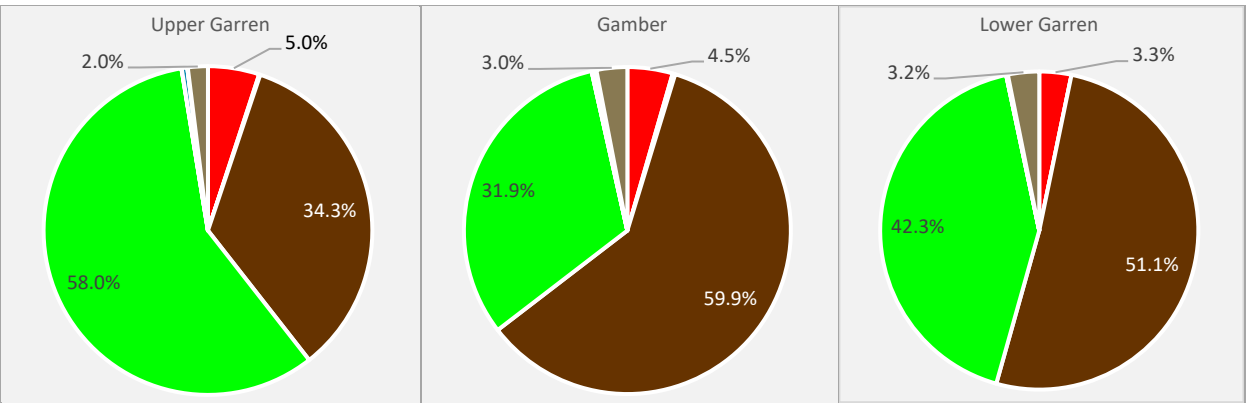


Figure 4: pie charts of the land use within each Garren sub catchment, from 2020 CEH data; colours as Figure 3

The principal crop types found on the arable land within the Garren Catchment are shown in Table 4 below.

| Crop type                     | Fraction |
|-------------------------------|----------|
| Wheat (winter + spring)       | 26%      |
| Maize                         | 15%      |
| Field beans (winter + spring) | 13%      |
| Barley (winter + spring)      | 12%      |
| Lucerne                       | 11%      |
| Potato                        | 8%       |
| Oilseed rape                  | 8%       |
| Other                         | 8%       |

Table 4: major arable crop types in the Garren Catchment (from 2020 CEH data)

## Abstraction

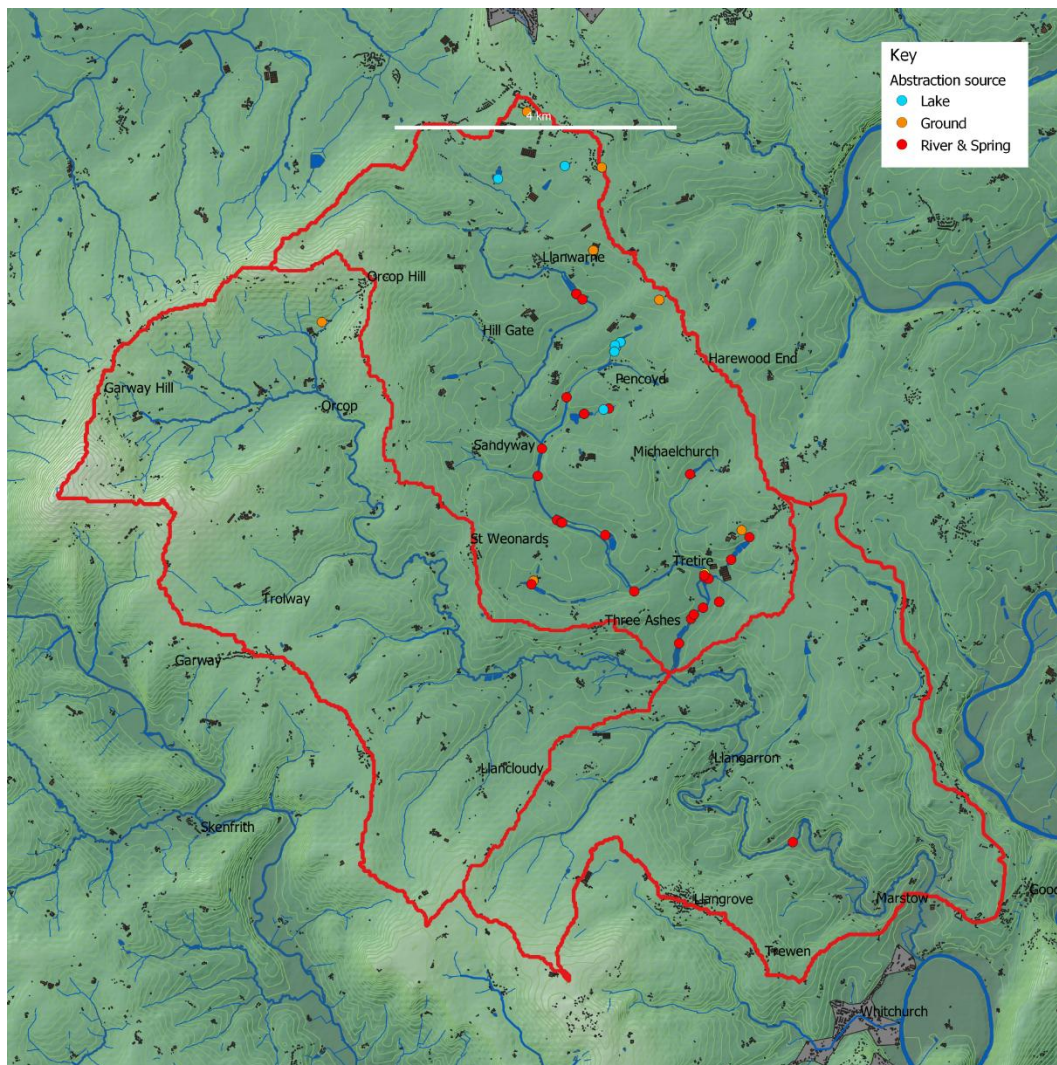


Figure 5: agricultural abstraction locations within the Garren Catchment; data provided by EA (Fol request)

A considerable amount of abstraction takes place within the Garren Catchment, for a variety of purposes and from a range of sources. In total, 52 extraction points are permitted by EA (data from Fol request). It can be seen in Figure 5 above that all but five of these are located in the Gamber sub catchment, so let us focus on this area. These comprise: 31 points of extraction from surface water (the Gamber or a tributary), 9 from lakes and 9 from boreholes, totalling 49. However, there are no aquifers in this area, so boreholes are merely intercepting groundwater en route to the river.

Of the 31 direct surface water extraction points within the Gamber sub catchment, 15 are for direct irrigation. A smaller but significant number (12) are abstracting from the Gamber during the winter months to fill storage lakes. One provides flow-through for a fish and cress farm; we assume this water finds its way back to the river. The remaining 3 are for domestic and general farming use.

The use of winter flow to top up storage lakes for summer abstraction is clearly preferable to direct summer abstraction. Anecdotally, it is understood that the practice is becoming more widespread. Certainly, it is to be encouraged.

Licensing for abstraction is complex. A single farm may have multiple licenses, each with a maximum daily volume and maximum annual volume. Each license may then cover multiple extraction points (as shown in Figure 5). Licenses also have conditions in relation to a reference flowrate, below which abstraction is prohibited. From the information provided by EA, this threshold is typically quite low; by definition, irrigation is needed in dry weather.

Excluding the fish farm, we can sum the daily maximum volumes for summer abstraction from the Gamber and its tributaries; this corresponds to a total flowrate of  $0.071 \text{ m}^3.\text{s}^{-1}$ . If we now compare this to the Q95 flowrate for the Gamber ( $0.033 \text{ m}^3.\text{s}^{-1}$ ), we can see how significant is the abstraction pressure on the Gamber. Now admittedly, summing the license limits as we have just done is not really fair. There are undoubtedly diversity factors to apply: not all farmers will pump at the maximum permissible level and will not pump every day. But, nevertheless, it is fair to conclude that a significant fraction of the Gamber summer flow is being abstracted and that this represents a significant anthropogenic pressure on the river.

### Human population

Unfortunately, census data does not correspond directly to river catchments, therefore we must estimate the catchment population based on the available data. The smallest unit for which population statistics are reported in England is the civil parish (CP); those which overlap the Garren Catchment are shown in Figure 6 below. For each of these, we can determine the area of overlap with each of the three Garren sub catchments and multiply by the corresponding population density, taken from the [City Population](#) website (here, we use official 2020 estimates based on the 2011 census). Summing the result over each sub catchment then yields an estimate for the total population, as shown in Table 5 below.

|                               | Population   |
|-------------------------------|--------------|
| Upper Garren                  | 1,140        |
| Gamber                        | 1,209        |
| Lower Garren                  | 1,341        |
| <b>Total Garren Catchment</b> | <b>3,690</b> |

Table 5: estimated human populations within the Garren Catchment, using official 2020 population data for CPs which overlap the catchment

Clearly, this estimate assumes uniform population density across each CP, whereas the reality is that populations are clustered in villages and hamlets. Nevertheless, this represents a reasonable estimate which is close enough for the purposes of calculating sewage loads.

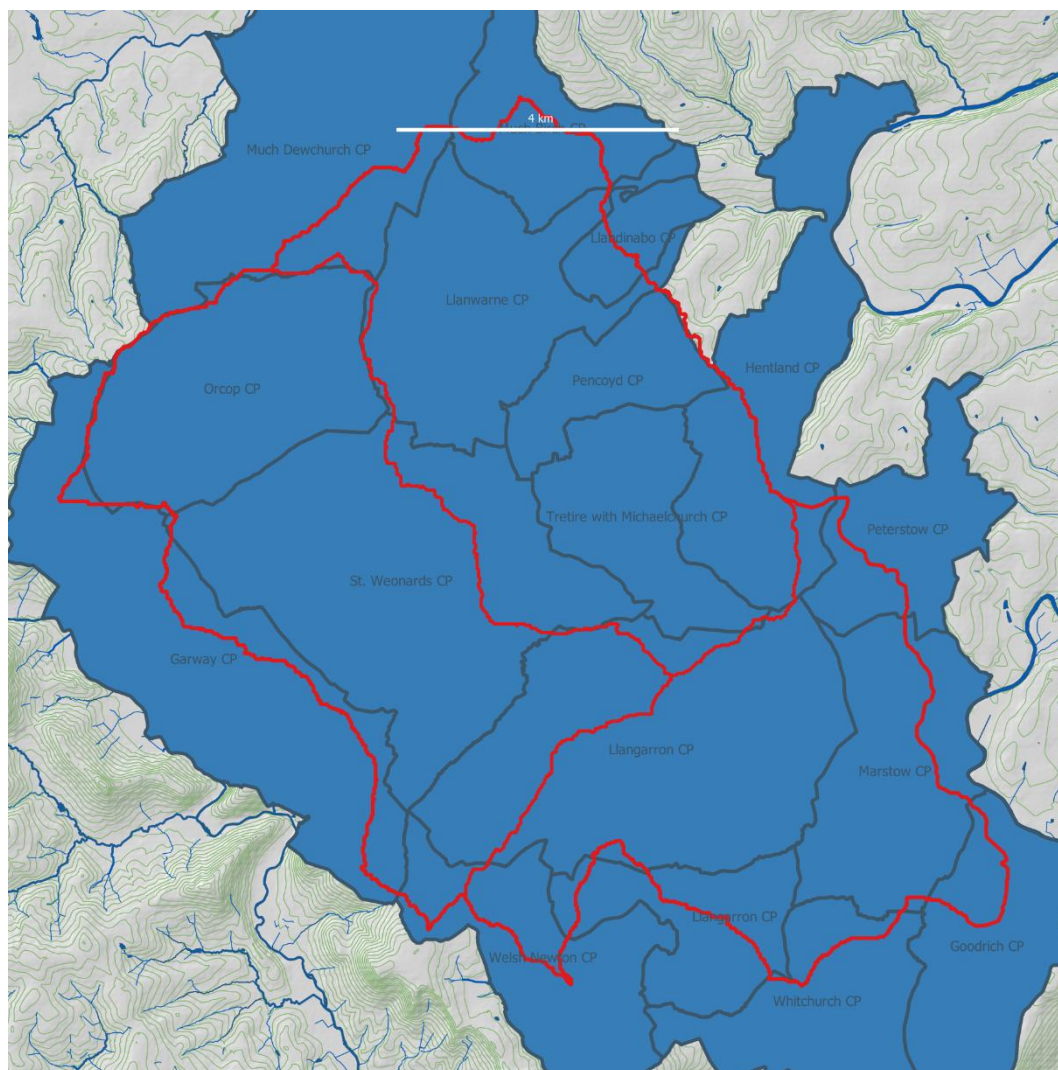


Figure 6: civil parishes which overlap the Garren Catchment

### Grazing livestock population

We are able to estimate livestock numbers in similar fashion. DEFRA publish detailed statistics for [livestock numbers](#), down to county level. So, on the basis that Herefordshire is a tolerably uniform agricultural area, we shall assume that the improved grassland of the Garren Catchment operates with the same average stock densities reported by DEFRA for the improved grassland of Herefordshire as a whole. Results are shown in Table 6 below.

|                               | Dairy      | Beef         | Sheep         |
|-------------------------------|------------|--------------|---------------|
| Upper Garren                  | 351        | 1,679        | 10,818        |
| Gamber                        | 192        | 917          | 5,904         |
| Lower Garren                  | 197        | 944          | 6,081         |
| <b>Total Garren Catchment</b> | <b>740</b> | <b>3,540</b> | <b>22,803</b> |

Table 6: estimated head of grazing livestock within the Garren Catchment, using DEFRA statistics for Herefordshire from the 2021 survey

We present here only the headline numbers. Behind these, the DEFRA statistics include a detailed breakdown by age and sex, which we shall use later to calculate manure loads. Comparing to the statistics for Herefordshire in 2000, we find that cattle numbers have fallen by 21% since then, a considerable fraction in the dairy sector. Sheep numbers have also fallen by 32%.

### Poultry population

Although the DEFRA statistics include poultry numbers, we are fortunately able to be more precise for poultry. WSA maintain a database of intensive poultry units (IPUs) within the Wye Catchment, which is in turn derived from the data compiled by members of the Campaign for Protection of Rural Wales

(CPRW). The data is primarily extracted from planning applications for IPU and, in addition to bird types and numbers, includes geographical coordinates. Thus, we are able to place IPU precisely within the Garren Catchment, as shown in Figure 7 below.

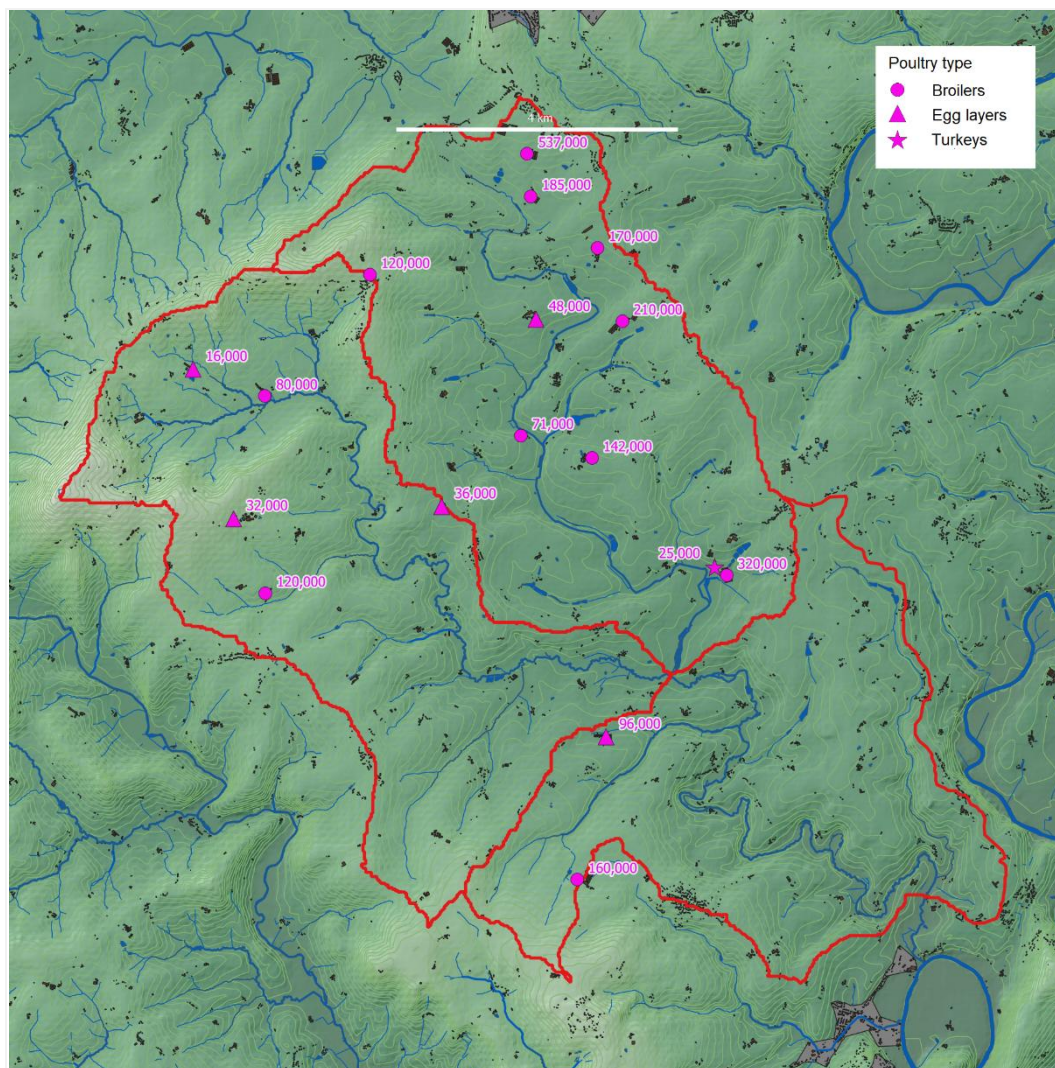


Figure 7: IPU locations within the Garren Catchment; bird numbers are shown adjacent to each location

There appear to be 15 IPU within the Garren Catchment, with a total population of approximately 2.3 million birds. As in the rest of Herefordshire, broilers dominate, with relatively few egg layers. Sites are concentrated in the Gamber sub catchment, with a smaller number in the Upper Garren sub catchment. The two more southerly IPU lie within the catchment of the Llanerch Brook. A breakdown of the total bird numbers is shown in Table 7 below.

Note that on a per hectare basis, this stock density is roughly three times the average for Herefordshire as a whole. This is a local hotspot within a national hotspot for poultry.

|                               | Egg layers     | Broilers         | Turkeys       |
|-------------------------------|----------------|------------------|---------------|
| Upper Garren                  | 64,000         | 200,000          | -             |
| Gamber                        | 84,000         | 1,653,000        | 25,000        |
| Lower Garren                  | 96,000         | 160,000          | -             |
| <b>Total Garren Catchment</b> | <b>244,000</b> | <b>2,013,000</b> | <b>25,000</b> |

Table 7: estimated head of poultry within the Garren Catchment, based primarily on a search of Herefordshire CC planning applications

From the planning application dataset, we can also estimate the approximate year of commencement for each IPU and thus examine the time evolution of the population within the catchment. This is shown in Figure 8 below.

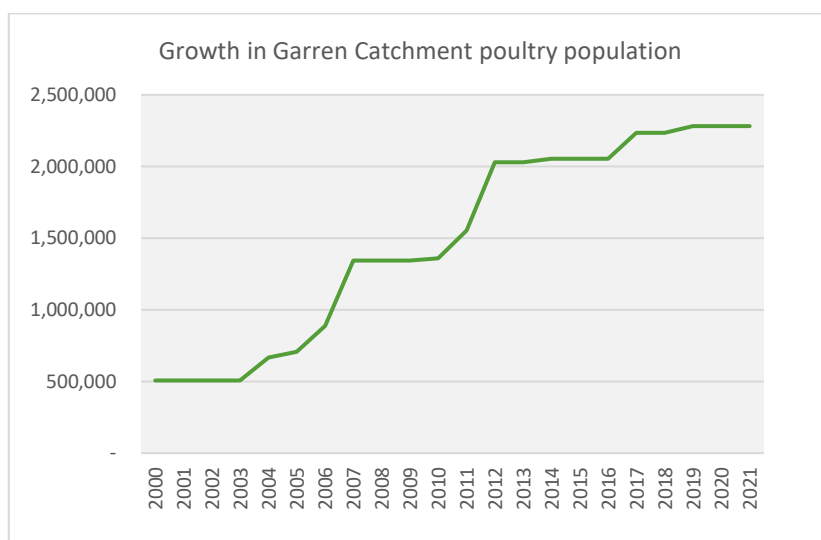


Figure 8 estimated Garren Catchment poultry population by year

## EA information

### WFD compliance

EA have a duty to monitor and classify the state of English rivers under the WFD regulations. This was originally an annual process, however since 2016, the frequency has dropped to three-yearly. The classification results for the Garren over the period 2009 – 2019 are available on the [EA Catchment Data Explorer](#). A high-level summary of selected classifications is shown in Table 8 below.

|              | 2009       |          | 2014       |          | 2019       |          |
|--------------|------------|----------|------------|----------|------------|----------|
|              | Ecological | Chemical | Ecological | Chemical | Ecological | Chemical |
| Upper Garren | Good       |          | Moderate   | Good     | Moderate   | Fail     |
| Gamber       | Moderate   |          | Moderate   | Good     | Moderate   | Fail     |
| Lower Garren | Poor       |          | Moderate   | Good     | Poor       | Fail     |

Table 8: WFD classifications for the Garren waterbodies (sub catchments); for 'Ecological' elements, the classification goes High > Good > Moderate > Poor; for 'Chemical' elements, the classification is simply Good or Fail. The target is Good for both ecological and chemical elements. Note that there was no requirement to measure chemical elements in 2009.

For the *ecological quality elements*, the Upper Garren has degraded from good to moderate over the last ten years, whereas the Gamber and Lower Garren have remained poor or moderate throughout. Specific elements driving the classification are *Fish* in the Lower Garren and *Macrophytes & Phytobenthos* throughout the catchment. Stated simply, there are almost no plants in the rivers and very few fish in the Lower Garren (*Fish* have not been assessed on the two upper sub catchments). Invertebrates generally score high or good. Within the *Physico-chemical elements*, ammonia, dissolved oxygen, temperature and pH are all classified as high status. Phosphate is classified as good throughout the catchment, other than on the Upper Garren where it dropped to moderate in 2019.

Reasons for not achieving good status on ecological quality elements are as follow. Low fish numbers are attributed to riparian activities; specifically poor livestock management with bankside erosion caused by livestock. Barriers to fish movement are also cited. Lack of in-river plants is likewise attributed to rural land management with poor agricultural nutrient management and poor livestock management cited.

The *chemical quality elements* have only been included since 2013 (WFD Cycle 2); moreover, the bar has been progressively raised since then. The good classification in 2014 probably reflects the fact that not many *priority hazardous substances* were being measured at this time. Substances failing in 2019

include mercury, nonylphenol<sup>5</sup>, PFOS<sup>6</sup> and PBDE<sup>7</sup>. The *reasons for not achieving good status* do not offer specific origins or mitigations for these substances.

Overall, though, the most immediate problems facing the river are the ecological elements. The lack of fish and macrophytes is immediately obvious to anyone who looks into the river. Also obvious, is the persistent turbidity, as witnessed by our citizen scientists and shown in Figure 9 below.



Figure 9: the Garren Brook viewed from Langstone Bridge during dry weather in June-22; note the cloudy, grey/brown appearance; visibility is less than 0.5m

This is not the typical turbidity following heavy rainfall, but is present all the time, even in dry weather when the river is low – as it was in June when the above photograph was taken. It is therefore reasonable to think that the turbidity is a key factor in the lack of macrophytes: the light simply doesn't penetrate to the bed of the river.

WSA recently queried the turbidity issue with EA, who stated that “[EA] suspect the turbidity is primarily due to sedimentation and agricultural run-off within the catchment” and that “sedimentation and nutrients are the main pressures within the catchment”. By way of explanation, they consider that “the

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<sup>5</sup> Nonylphenols have historically been used to manufacture a variety of products, but their use has now been phased out in Europe. They are endocrine disruptors and oestrogen mimics.

<sup>6</sup> PFOS is a fluorosurfactant and was the active ingredient in Scotchguard, now restricted; a *forever chemical*.

<sup>7</sup> PBDE was widely used as a flame retardant, also now restricted; another *forever chemical* suspected of causing reduced fertility in humans.

*underlying geology of red sandstone and diffuse agricultural pollution are thought to be the major issue with large areas of maize and potatoes being grown”.*

Combining this clarification with the more cryptic language in the *reasons for not achieving good* list discussed above, brings us to the following summary. The high-clay soils of the riparian zone (see *Geology* section) are trampled into the stream by livestock. This is supplemented by runoff from arable fields during heavy rainfall. The clay particles<sup>8</sup> are mixed into the water column by livestock wading in the river and possibly also by natural water turbulence. This results in persistent turbidity which suppresses macrophytes by blocking the light.

In short, the Garren is not a healthy river – and it is largely down to agriculture.

### Water quality data

Routine measurements made by EA (including, but not restricted to, the physico-chemical determinands listed above) are also available to download from the [EA Water Quality Archive](#). These records span the period from 2000 up to today, which provides the opportunity to look for longer-term trends.

Determinands recorded include alkalinity, specific conductance, biological oxygen demand, unionised ammonia, nitrate, nitrite, orthophosphate, total phosphorus, pH, temperature, suspended solids, dissolved oxygen. However, not all determinands are recorded consistently throughout the period.

Having downloaded the dataset and filtered down to the Garren and Gamber locations, some immediate observations can be made.

- Samples have been taken from three primary locations:
  - Upper Garren at Llangunnoch Bridge
  - Gamber at Kilbreece Bridge
  - Lower Garren at the Wye confluence
- In 2021, an additional location was used on the Lower Garren at Trebandy (see Figure 15 for locations)
- Routine measurement of total phosphorus ceased in 2002

### EA sampling regime

Before 2010, most years have a relatively full suite of determinands recorded on a monthly basis. From 2010 onwards (the start of the UK Government’s austerity programme), we find a reduced set of determinands, at much reduced frequency, and from a reduced set of locations. In 2018, there were no measurements whatsoever within the Garren Catchment. In Table 9 below, we present the number of samples taken by EA each year for the last five years.

|                          | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
|--------------------------|------|------|------|------|------|------|
| Upper Garren             | 1    | -    | -    | -    | -    | -    |
| Gamber                   | -    | -    | 6    | 5    | 12   | -    |
| Lower Garren (Wye conf.) | 2    | -    | 9    | 4    | 12   | 8    |
| Lower Garren (Trebandy)  | -    | -    | -    | -    | 12   | 11   |

Table 9: EA sampling frequency (samples per year) over the last 6 years; 2022 is as of November

The last full set of samples on the Upper Garren at Llangunnoch Bridge was in 2007. Since then, coverage has been sporadic on this waterbody, and discontinued entirely after 2017. It is unclear how WFD compliance was assessed in 2019 and perplexing insofar as this is a failing waterbody.

The Gamber and Lower Garren have likewise received sporadic coverage. Six and nine samples respectively would seem to be entirely inadequate for calculating the annual statistics required for WFD compliance in 2019. The continued lack of sampling on the Upper Garren waterbody is concerning. At

<sup>8</sup> Clay particles are very small and, once in the water column, will remain in suspension for great distances.

the time of writing, we note that regular monthly sampling during 2022 is limited to Trebandy, with only partial sampling at the Wye confluence; the complete lack of upper catchment sampling continues and will ultimately remove any ability to judge long-term trends on the upper catchment.

We shall now review some of the more salient determinands and, where appropriate, compare them to the WFD water quality standards (The Water Framework Directive (Standards and Classification) Directions (England and Wales) 2015, 2015), the requirements of which are tabulated in Appendix 2.

### Alkalinity

Alkalinity refers to  $\text{CaCO}_3$ , which is released into groundwater from the underlying sandstone bedrock. As the term suggests, the effect of  $\text{CaCO}_3$  is to raise the pH value of the river above 7. In common with some of the other lowland tributaries of the Wye which share the same geology, the Garren has a high level of alkalinity. Moreover, the high concentration of  $\text{CaCO}_3$  can represent a significant fraction of the total ionic load, meaning that high alkalinity is also associated with high values of electrical conductivity. A further aspect of alkalinity is that it influences the WFD water quality standards for certain determinands (orthophosphate, unionised ammonia & dissolved oxygen), therefore it is useful to extract an average value for each of the sub catchments, as shown below.

|              | Alkalinity<br>( $\text{mgCaCO}_3\text{.l}^{-1}$ ) |
|--------------|---------------------------------------------------|
| Upper Garren | 228                                               |
| Gamber       | 242                                               |
| Lower Garren | 233                                               |

Table 10: annual mean alkalinity values within the Garren Catchment (average over all data 2000-2021)

### Acidity (pH)

As expected, pH values reflect the high alkalinity. The WFD standard for high status requires the 5-percentile and 95-percentile values to fall within the range 6-9. The results shown in Table 11 below comply with this requirement. Note that the Upper Garren at Llangunnoch Bridge has not been comprehensively monitored in recent years, so we have inserted 2007 data here.

|              | pH      |             |         |
|--------------|---------|-------------|---------|
|              | Minimum | Annual Mean | Maximum |
| Upper Garren | 7.66    | 7.97        | 8.22    |
| Gamber       | 7.12    | 7.75        | 8.08    |
| Lower Garren | 7.07    | 8.09        | 8.85    |

Table 11: pH values within the Garren Catchment (2021 data except Upper Garren which is 2007)

### Temperature

As a salmonid river, water temperature is required to be below 20°C for high status and 23°C for good status, both at the 98-percentile level. The results shown in Table 12 below comply with the requirement for high status.

|              | Water Temperature (°C) |             |         |
|--------------|------------------------|-------------|---------|
|              | Minimum                | Annual Mean | Maximum |
| Upper Garren | 5.2                    | 9.8         | 14.2    |
| Gamber       | 6.2                    | 10.7        | 15.2    |
| Lower Garren | 4.5                    | 11.0        | 15.1    |

Table 12: water temperature values within the Garren Catchment (2021 data except Upper Garren which is 2007)

### Dissolved oxygen saturation

Since the Garren is a salmonid river, this fact takes precedence over alkalinity and elevation in determining WFD classification standards for dissolved oxygen saturation (DO). All salmonid rivers require 90% of measurements to fall above 80% saturation for high status, or above 75% saturation for good status. Results are shown in Table 13 below.

|              | Dissolved Oxygen (%) |             |         |
|--------------|----------------------|-------------|---------|
|              | Minimum              | Annual Mean | Maximum |
| Upper Garren | 83.9%                | 95.7%       | 112.0%  |
| Gamber       | 78.2%                | 91.7%       | 115.8%  |
| Lower Garren | 72.1%                | 98.7%       | 132.3%  |

Table 13: DO values within the Garren Catchment (2021 data except Upper Garren which is 2007)

As for pH, we substitute 2007 data for the Upper Garren. The minima for the Gamber and Lower Garren are seen to fall below the 80% target. However, in both cases, this is due to a single datapoint out of twelve, therefore the full year's data does in fact meet the criterion for over 90% of the samples, and thus qualifies for high status on DO.

### Electrical conductivity

Electrical conductivity is very temperature dependent and so is commonly referenced to a 'standard' temperature of 25°C. This value is known as the *specific conductivity* (SC), or sometimes *specific conductance*. SC is not a WFD water quality element but is included here for comparison to the CS data presented later. As discussed above, the Garren is seen to have comparatively high values of SC, resulting from its high level of alkalinity. Annual average values are shown in Table 14 below.

|              | SC ( $\mu\text{S}\cdot\text{cm}^{-1}$ ) |
|--------------|-----------------------------------------|
| Upper Garren | 551                                     |
| Gamber       | 660                                     |
| Lower Garren | 621                                     |

Table 14: Annual average specific conductivity values within the Garren Catchment (2021 data except Upper Garren which is 2014-17 average)

### Unionised ammonia

Unionised ammonia ( $\text{NH}_3$ ) is toxic to aquatic life, including fish, and is included as a WFD water quality element. Because of its high alkalinity, The Garren is classified as a *type 7* waterbody, resulting in 90 percentile limits of 300  $\mu\text{gN/l}$  for high status and 600  $\mu\text{gN/l}$  for good status.

|              | NH <sub>3</sub> ( $\mu\text{gN}\cdot\text{l}^{-1}$ ) |         |
|--------------|------------------------------------------------------|---------|
|              | Annual Mean                                          | Maximum |
| Upper Garren | 1.0                                                  | 2.1     |
| Gamber       | 0.7                                                  | 3.6     |
| Lower Garren | 0.9                                                  | 4.3     |

Table 15: NH<sub>3</sub> concentrations within the Garren Catchment (2021 data except Upper Garren which is 2007)

The maximum levels shown in Table 15 above are two orders of magnitude below the limit for high status. Moreover, the data for unionised ammonia shows no evidence of any long-term change.

### Nitrate

Nitrate is a nutrient, generally indicative of agricultural pollution however it is not a WFD water quality element. The most recent nitrate measurements are shown in Table 16 below.

|              | Nitrate ( $\text{mgN}\cdot\text{l}^{-1}$ ) |         |
|--------------|--------------------------------------------|---------|
|              | Annual Mean                                | Maximum |
| Upper Garren | 10.7                                       | 12.7    |
| Gamber       | 11.4                                       | 13.0    |
| Lower Garren | 9.6                                        | 11.0    |

Table 16: nitrate concentrations within the Garren Catchment (2021 data except Upper Garren which is 2007)

Perhaps more interesting is the long-term trend for nitrate shown in Figure 10 below. All three sub catchments show a marked dip around 2001-2, presumably caused by the foot and mouth outbreak. However, over the last eight years, the nitrate signal appears to have been stable (which is more than can be said of the EA sampling frequency).

It is worth noting that nitrate concentration appears to correlate with the fraction of arable land cover. Highest in the Gamber sub catchment, lowest in the Upper Garren and an intermediate value on the Lower Garren.

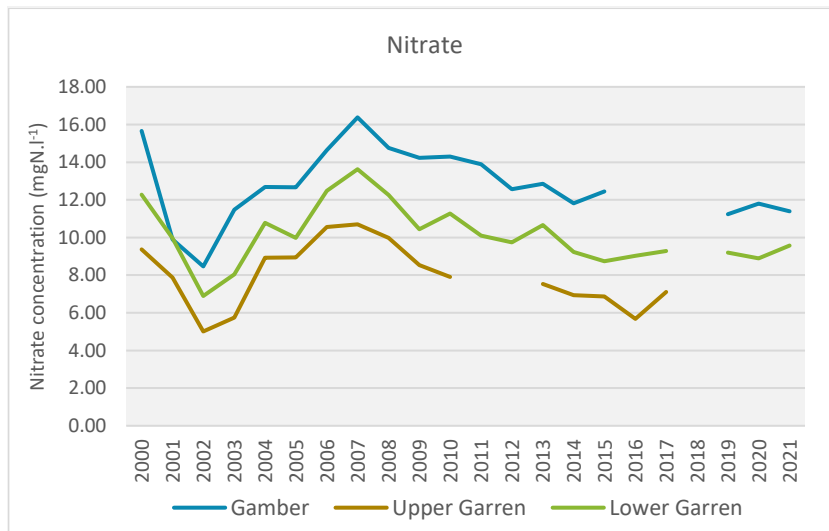


Figure 10: annual average nitrate concentrations in the Garren Catchment

### Nitrite

Nitrite is not directly added to soil but is an intermediate stage in the nitrification of ammonium compounds, which in turn derive either from synthetic fertilizers, or from the mineralisation of organic matter in manures. It might therefore be reasonable to expect a correlation with nitrate, however this is not obviously the case in Figure 11 below. We find that concentrations were generally higher pre-2006 but have been fairly stable since then. The peak in 2001 possibly reflects a switch from slurry to synthetic fertilisers during the foot and mouth outbreak.

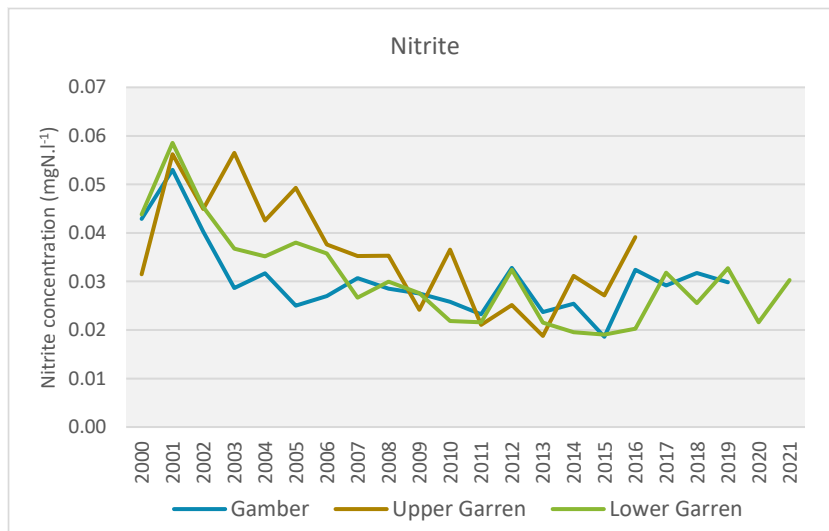


Figure 11: annual average nitrite concentrations in the Garren Catchment

### Phosphorus

Phosphorus is a nutrient and is the primary cause of eutrophication in rivers. It originates from agricultural sources as well as human sewage. It is a WFD water quality element having classification targets which are a function of both elevation and alkalinity. For convenience of measurement, the targets are specified in terms of reactive phosphorus (RP)<sup>9</sup> rather than total phosphorus (TP). The EA

<sup>9</sup> RP is defined as the fraction of phosphorus which is indicated by the photo-molybdenum blue test, which is widely used by laboratories and field instruments.

refer to this as *orthophosphate*. For most of the Garren Catchment, the limits are approximately 0.04 mgP/l for high status and 0.08 mgP/l for good status. In the upper reaches of the Garren, above 100m elevation (not sampled by EA), the limits reduce to 0.03 and 0.06 mgP/l. Note that the target for good status in the lower Garren (0.09 at the Wye confluence) is substantially higher than the Wye SAC target at this point of 0.05 mgP/l. Furthermore, from 2022 onwards, the Wye target has reduced to 0.035mgP/l.

The most recent EA data, which is consistent with the WFD classification for these waterbodies, is summarised in Table 17 below.

|              | RP (mgP.l <sup>-1</sup> ) |
|--------------|---------------------------|
| Upper Garren | 0.15                      |
| Gamber       | 0.06                      |
| Lower Garren | 0.06                      |

Table 17: Annual average RP values within the Garren Catchment (2021 data except Upper Garren which is 2016 average)

As for nitrate, it is instructive to look at the long-term trends. The first graph, figure12, shows the annual averages for the Upper Garren.

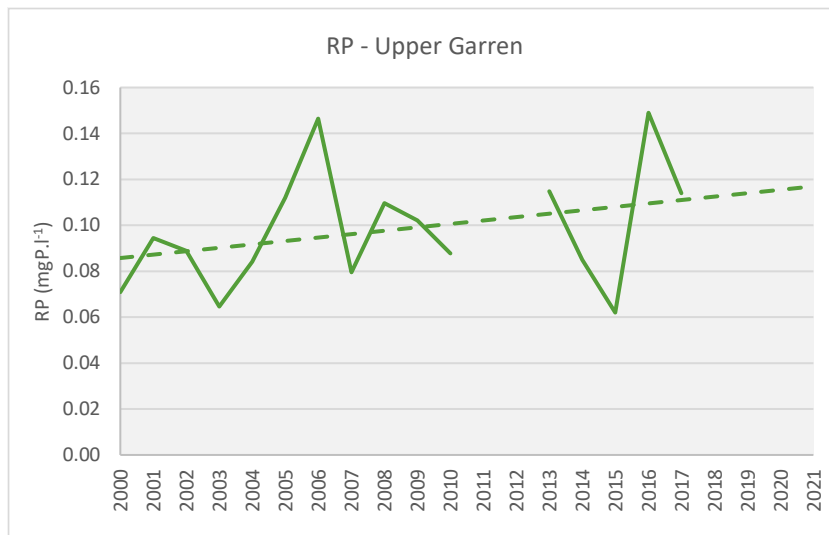


Figure 12: annual average RP concentration on the Upper Garren; the dashed line is the best-fit linear regression

Not only does the Upper Garren have a very high average RP concentration, but it has been rising steadily since 2000 at an average rate of 15% per decade.

Figure 13 below shows the corresponding graph for the Gamber. Here we see a generally lower level, but again a rising trend – although less rapid than the Upper Garren: around 8% per decade.

Figure 14 shows the result for the Lower Garren sampled at the Wye confluence. The overall average is around 0.08 mgP/l – effectively splitting the difference between the two upper sub catchments. The 2021 annual average of 0.061 mgP/l clearly falls below the long-term average. However, the long-term rising trend is not exhibited at this sampling location. Herein lies the difficulty with reactive phosphorus measurement. In-river processes serve to modify the concentration signal as it flows downstream, generally having the effect of reducing it.

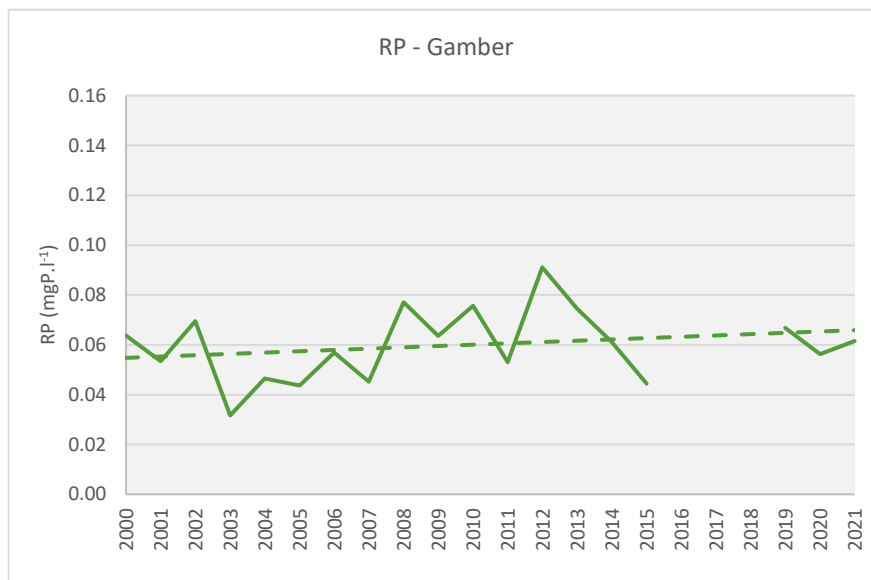


Figure 13: annual average RP concentration on the Gamber; the dashed line is the best-fit linear regression

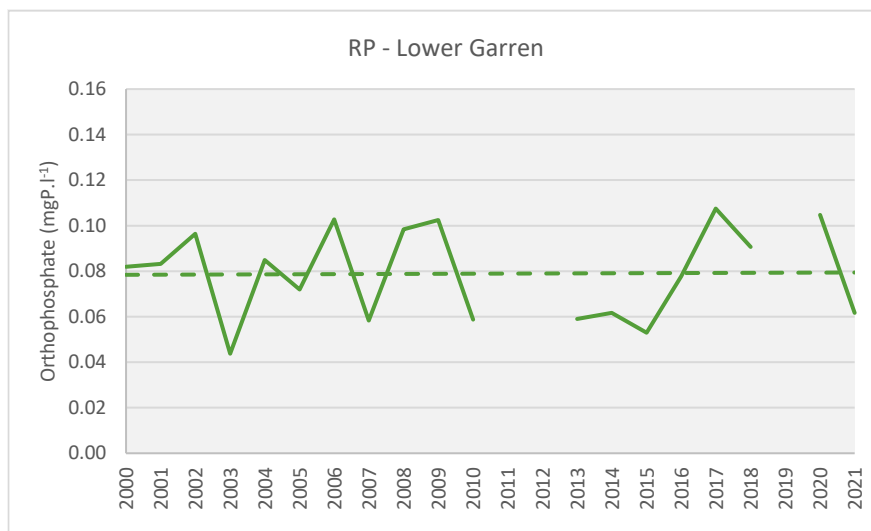


Figure 14: annual average RP concentration on the Lower Garren; the dashed line is the best-fit linear regression

This analysis serves to illustrate the dangers of sampling phosphorus predominantly at the lower reaches of a catchment. It is not indicative of what is happening many miles upstream. The decision by EA to cease sampling the Upper Garren after 2017 is somewhat baffling, given that this waterbody has the highest RP concentrations and the highest rate of increase in the Garren Catchment. The more recent decision to commence sampling at a second location (Trebandy) on the Lower Garren seems odd when the Upper Garren remains un-monitored.

A final point which warrants discussion is the marked difference between the Upper Garren and Gamber sub catchments. The Upper Garren carries roughly twice the concentration of phosphorus as the Gamber. Why? If we take the values of the long-term trend lines (the regression lines in figures 12 & 13) in 2021 (0.118 & 0.065 mgP/l), we can derive a ratio of 1.82:1 between the two values. As observed earlier, the most significant difference between the two sub catchments is the proportion of arable farming versus grassland (see Figure 5). The Upper Garren has more improved grassland cover than the Gamber – actually at a ratio of 1.83:1. Now, clearly, we cannot read too much into this mathematical coincidence, but it does at least suggest that the majority of the river phosphorus loads could be coming from the grassland and/or associated livestock, rather than the arable land.

## RP/TP ratio

For the modelling presented later in this report, we shall need a reasonable estimate of the fraction of total phosphorus indicated by a measurement of reactive phosphorus; however, RP/TP is a highly variable parameter. Within the EA dataset, we find individual samples having a ratio of anywhere between 30% and 90%. Fortunately, the annual average values do come out to something fairly consistent. We have both RP and TP recorded every month of the year in the three sub catchments once in 2000 and at Trebandy in the Lower Garren during 2021. The results are presented in Table 18 below.

|                          | 2000 | 2021 |
|--------------------------|------|------|
| Upper Garren             | 65%  |      |
| Gamber                   | 63%  |      |
| Lower Garren (Wye conf.) | 67%  |      |
| Lower Garren (Trebandy)  |      | 53%  |

Table 18: Annual average RP/TP ratios

This rather suggests that something has changed since 2000. For the subsequent analysis we shall use the most recent figure of 53%.

### Key points

- WFD compliance
  - Over the last decade, the Garren has degraded from good ecological status to moderate
  - Main challenges are sedimentation and nutrient management (phosphorus); the latter most noticeable in the Upper Garren sub catchment
  - Principal failing elements are fish and macrophytes
  - Main causes are agricultural (combined with local geology); fish also impacted by channel obstructions
- EA water quality data (2000-2021)
  - Over the last decade, monitoring has been severely cut back, resulting in fewer locations and reduced sampling frequency; statistical adequacy for compliance is now questionable
  - The Upper Garren sub catchment has not had a complete dataset (12 monthly samples) since 2007; it is unclear on what basis compliance was assessed for this waterbody in 2019
  - Temperature, electrical conductivity, pH and unionised ammonia show no surprises
  - Long-term trends on nitrate & nitrite show foot and mouth transient in 2001-2, but last few years appear stable. Relative nitrate concentration in the three sub catchments correlates with arable land cover
  - Phosphorus concentration is highest in the Upper Garren sub catchment; the long-term trends are upwards, again most significantly on the Upper Garren; phosphorus concentration is thus correlating with improved grassland cover
  - The most recent RP/TP value is 53%; this is noticeably lower than the value of 65% in 2000.

## WSA information

### Soil sampling

If we wish to understand, and ultimately control, diffuse agricultural phosphorus pollution, then we must understand phosphorus concentrations within the catchment soils. This is surely axiomatic, and indeed was one of the key messages from the recent RePhOKUs report (Withers et al., 2022). For this reason, in June 2022, WSA & CPRE undertook a survey of agricultural soils within the Garren Catchment. The details of the survey have been reported elsewhere (Green, 2022b), so here we shall reproduce just the key findings and conclusions.

Samples were taken at 22 randomly selected locations (roughly 1 per 4km<sup>2</sup>). These were then assayed by an agricultural laboratory using the Olsen-P<sup>10</sup> test for phosphorus and compared to results from the

<sup>10</sup> The Olsen-P test is widely used in UK agriculture to assess soil phosphorus. It uses a relatively weak desorption chemistry, and the result is generally considered to approximate to plant available (and therefore leachable) phosphorus. It forms the basis of the best practice guidance on soil phosphorus management (ADHB, 2022)

Countryside Survey of 2007. The average Olsen-P results for the different land uses samples are shown in Table 19 below.

| Land use  | No. samples | Olsen-P (mg.l <sup>-1</sup> ) | Change from 2007 |
|-----------|-------------|-------------------------------|------------------|
| Grassland | 12          | 48                            | +17              |
| Arable    | 9           | 54                            | 0                |
| Orchard   | 1           | 15                            |                  |

Table 19: mean Olsen-P values compared to 2007 values

We see immediately that grassland has risen significantly since 2007. According to RB209 (ADHB, 2022), the target range for pasture is 16-25 mg.l<sup>-1</sup>, so the current value has no agronomic justification whatsoever. Arable land has remained stable but, again, is way above the target range. All the crops growing in the arable fields which were sampled also have a target range of 16-25 mg.l<sup>-1</sup>.

Within the data are some significant outliers. Arable samples included one location at 157 mg.l<sup>-1</sup> and grassland samples included locations at 107, 167 & 82 mg.l<sup>-1</sup>. Our earlier report (Green, 2022b) demonstrates a clear correlation between the outlier locations and the locations of IPUs having a stated intention of spreading manure on their own or neighbour's land up to the nitrate vulnerable zone (NVZ) nitrogen limit. To be clear, this means spreading as a means of disposal, not spreading in line with crop needs. This begs the question, what would we expect to happen to phosphorus, if we dumped poultry manure at the NVZ limit of 170 kgN.ha<sup>-1</sup>.yr<sup>-1</sup>? A quick feasibility calculation is in order.

RB209 tells us that a phosphorus application rate of 100 kgP.ha<sup>-1</sup> will raise Olsen-P by around 10 mg.l<sup>-1</sup>; an estimate which Withers et al. confirmed as being applicable to soils within the Wye Catchment. RB209 also tells us the N:P ratio for poultry manure. The calculation in Box 2 below combines these different pieces of information.

#### Box 2: rate of change of Olsen-P

1 tonne of poultry manure contains: 28 kgN  
7.4 kgP

NVZ max. application rate for N: 170 kgN.ha<sup>-1</sup>.year<sup>-1</sup>

Associated rate for P:  $170 \times 7.4 / 28 = 45 \text{ kgP.ha}^{-1}.\text{year}^{-1}$

Rate of change (RB209): 10 mg.l<sup>-1</sup> Olsen-P per 100 kgP.ha<sup>-1</sup>

Olsen-P increase at NVZ rate of application is then:  
 $10 \times 45 / 100 = 4.5 \text{ mg.l}^{-1}.\text{year}^{-1}$

So, if we started with a field at 30 mg.l<sup>-1</sup> Olsen-P in 2007 and applied poultry manure at the NVZ nitrogen limit, Olsen-P would increase by 4.5 mg.l<sup>-1</sup> each year and would now be around 100 mg.l<sup>-1</sup>. But, in fact, Figure 7 shows that many IPUs have been operating since the mid-nineties – so one can readily see how the observed outliers arise from the practice of spreading as a means of disposal – especially with real-world uncertainties around actual volumes spread.

Looking at the number of these high-phosphorus outliers, we can infer that approximately 18% of agricultural land within the Garren Catchment is being abused in this way, comprising 25% of the grassland and a smaller fraction of arable land.

### Water quality data

WSA have pioneered citizen science within the Wye Catchment and have been actively capturing water quality data since July 2020, progressively adding to the number of sites monitored and also to the parameters measured. Following the establishment of the Wye Catchment Monitoring Network coordinated by Cardiff University, data is now uploaded onto the *Epicollect* portal and pooled with data from an ever-growing list of volunteers, organised by Friend of the Upper Wye (FOUW), Friends of the Lugg (FOL), Campaign to protect Rural England – Herefordshire (CPRE) and others.

Sampling within the Garren Catchment commenced in June 2021 and currently includes eight regular sites. The locations of the sites are shown in Figure 15 below.

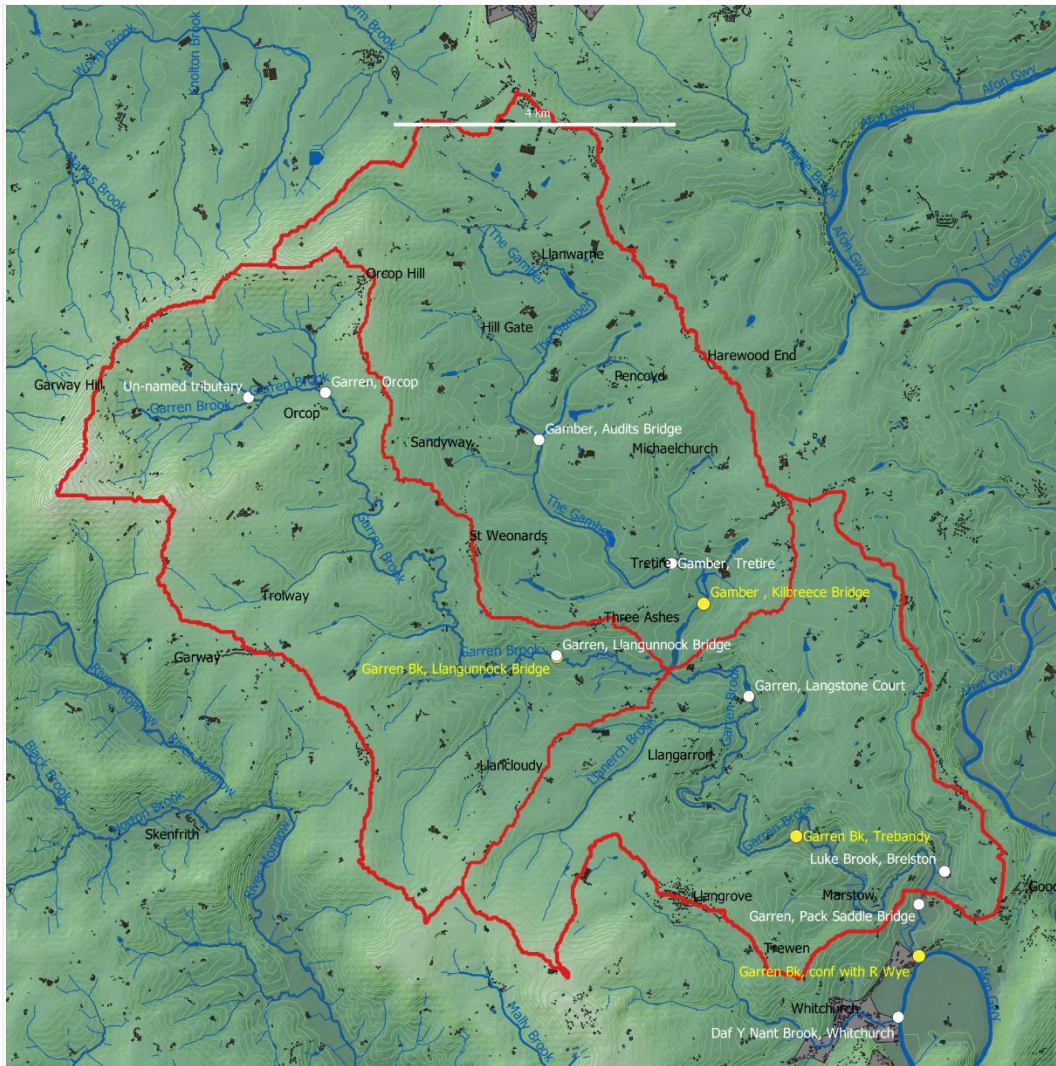


Figure 15: locations of WSA sampling sites (white) and EA sampling sites (yellow)

There are two sites in each of the sub catchments, plus one on the Luke Brook and one on a small, unnamed stream draining land adjacent to two IPU's above Orcop. The two upper sub catchments are monitored immediately above their confluence (Llangunnock Bridge and Tretire) and in the upper reaches (Orcop and Audits Bridge). The lower sub catchment is similarly monitored just below the Gamber confluence at Langstone Court and at the bottom of the catchment at Pack Saddle Bridge. Note the close correspondence to the EA sites: Llangunnock Bridge is shared; Tretire is close to Kilbreece Bridge and Pack Saddle Bridge is close to the EA Wye Confluence site.

Parameters measured include temperature, pH, SC/EC-TDS<sup>11</sup>, PO<sub>4</sub> and gauge board height plus an observation of colour/turbidity and weather conditions. Sampling frequency is typically weekly, and data reported here covers the period June 2021 through to June 2022. This data has been augmented by daily average volumetric flowrates at Marstow Mill, obtained from EA via a freedom of information request.

<sup>11</sup> EC-TDS refers to total dissolved solids derived from a measure of electrical conductivity, as opposed to a gravimetric assay

We shall now present the data for each salient parameter, excluding the special case of the Un-named tributary, which we shall consider separately.

### Acidity (pH)

pH was measured using an *HM Digital pH-80* handheld instrument. Key statistics are shown in Table 20 below.

|                     | No. samples | pH min | pH max |
|---------------------|-------------|--------|--------|
| <b>Upper Garren</b> |             |        |        |
| Orcop               | 28          | 8.0    | 8.6    |
| Llangunnock Bridge  | 30          | 6.5    | 9.0    |
| <b>Gamber</b>       |             |        |        |
| Audits Bridge       | 26          | 7.9    | 9.6    |
| Tretire             | 32          | 6.5    | 9.1    |
| <b>Lower Garren</b> |             |        |        |
| Langstone Court     | 34          | 6.0    | 8.7    |
| Luke Brook (trib.)  | 33          | 6.5    | 8.9    |
| Pack Saddle Bridge  | 40          | 6.0    | 9.4    |

Table 20: pH recorded from June-21 to June-22

Three sites, Audits Bridge, Tretire and Pack Saddle Bridge, have maxima falling outside the WFD range for high status. For the first two sites, only 1 sample exceeded pH 9.0, so over 95% of the data falls within the acceptable range. At Pack Saddle Bridge, there were 2 samples over pH 9.0, representing 5% of the data. Thus, all sites are found to be compliant with WFD high status for acidity over the period monitored, but Pack Saddle Bridge is marginal.

Looking more closely at Pack Saddle Bridge, an interesting pattern emerges. In Figure 16 below, we examine how pH varies with the volumetric flowrate (discharge) of the river.

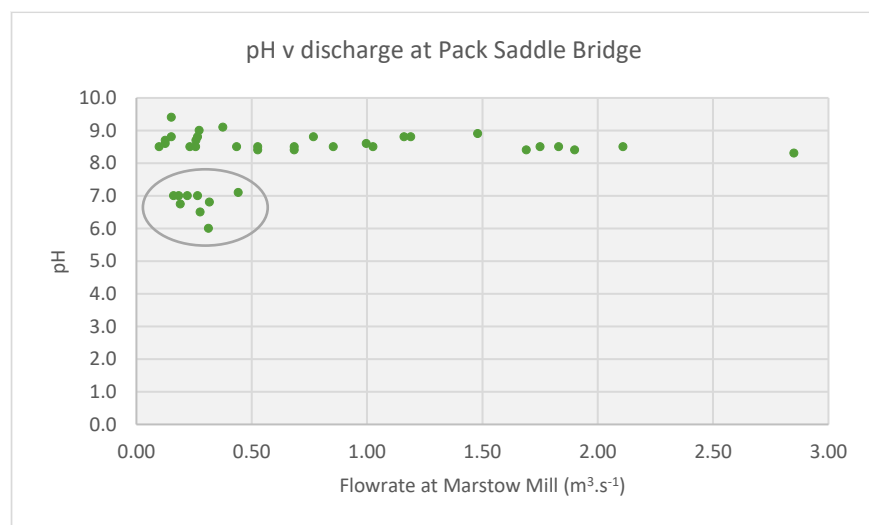


Figure 16: Lower Garren pH dependence on discharge; note the circled cluster of low-pH samples

It is immediately apparent that the distribution of pH is bimodal. For most of the time, the pH falls within the range 8-9, but less frequently drops to 6-7. Moreover, all of the excursions into acidity occur while the river is below Q50 flowrate ( $0.6 \text{ m}^3 \cdot \text{s}^{-1}$ ). The most obvious source of acidity in a rural catchment is slurry, which could be a single point source caused by poor slurry management, or a more widespread diffuse problem due to spreading practices. We should now look upstream to the sites immediately above the Gamber confluence, Llangunnock Bridge on the Upper Garren and Tretire on the Gamber.

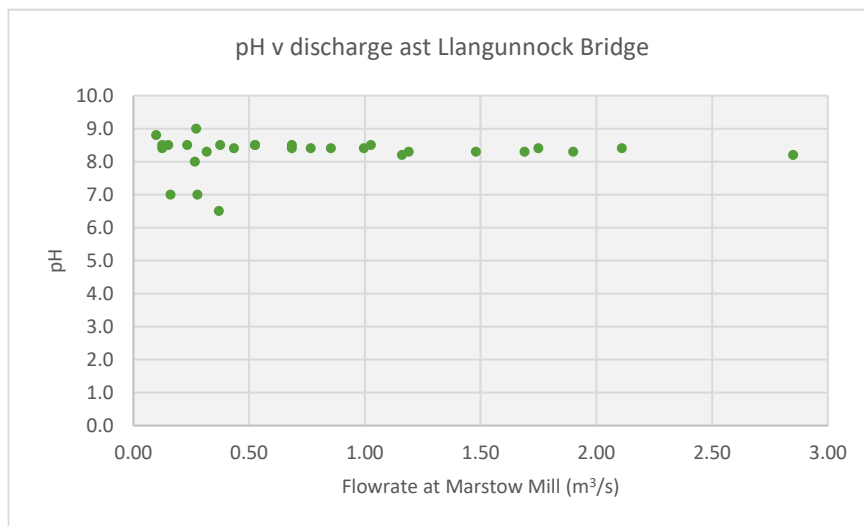


Figure 17: Upper Garren  
pH dependence on discharge

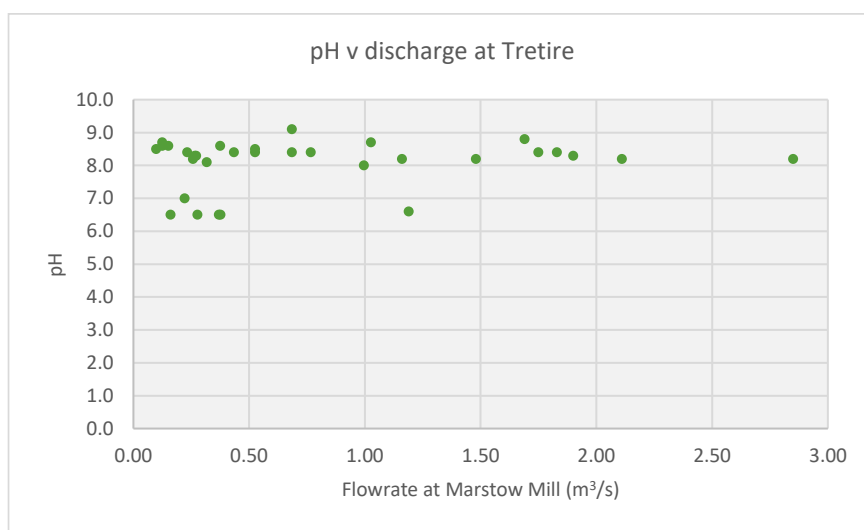


Figure 18: Gamber  
pH dependence on discharge

Both upper sub catchments (Figures 17 & 18) show the same bi-modal distribution, with low pH excursions during periods of low flow and, although not shown here, the data from the Luke Brook and Langstone Court exhibits the same pattern; Orcop and Audits bridge do not. It can be confirmed from the WSA records that 16 out of 17 low pH events occurred in dry weather (the 17<sup>th</sup> was light drizzle). The consistency between these sites is strongly suggestive of a diffuse origin affecting wide areas of the catchment, in all likelihood slurry spreading practices. If so, the good news is that spreading appears to be occurring in dry weather, as required by NVZ regulations, but the bad news is that slurry is somehow getting into the river, which is not so good. A further useful piece of information is that all low pH readings were taken within the period September-November – winter cereal sowing time (wheat, barley, maize, etc.).

To summarise, we find all monitored sites, for most of the year, showing a healthy pH in the range 8-9. Llangunnock Bridge, Tretire and all sites downstream have excursions into the range 6-7, which are correlated with dry weather in the autumn. Slurry spreading for winter cereal crops would seem to be the most likely origin.

#### Electrical conductivity

Electrical conductivity was recorded using an *HM Digital TDS-3* handheld instrument which reports total dissolved solids (EC-TDS) derived from a measure of specific conductance (SC), the two being related by

a linear proportionality constant (0.50 for NaCl). Here, we report results as SC (conductivity at 25°C). Table 21 below shows the annual average values for this parameter.

|                     | No. samples | SC ( $\mu\text{S}\cdot\text{cm}^{-1}$ ) |
|---------------------|-------------|-----------------------------------------|
| <b>Upper Garren</b> |             |                                         |
| Orcop               | 28          | 476                                     |
| Llangunnock Bridge  | 45          | 551                                     |
| <b>Gamber</b>       |             |                                         |
| Audits Bridge       | 30          | 618                                     |
| Tretire             | 48          | 594                                     |
| <b>Lower Garren</b> |             |                                         |
| Langstone Court     | 49          | 592                                     |
| Luke Brook (trib.)  | 46          | 684                                     |
| Pack Saddle Bridge  | 55          | 611                                     |

Table 21: annual average electrical conductivity recorded from June-21 to June-22; SC refers to conductivity at 25°C

The values above for sites at Llangunnock Bridge, Tretire and Pack Saddle Bridge correspond closely with EA measurements on the three sub catchments (Table 14) since these locations are in close proximity to the EA measurements sites.

It can be seen that results around the Garren Catchment are fairly uniform, in the range 592 – 684  $\mu\text{S}\cdot\text{cm}^{-1}$ , with the exception of the upper reaches of the Garren at Orcop. Orcop is significantly lower at 476  $\mu\text{S}\cdot\text{cm}^{-1}$  and this has a knock-on effect downstream at Llangunnock Bridge. The reason for this most probably has its origins in the underlying geology (see Appendix 3). The small area around Orcop has different lithology to the rest of the Garren Catchment, apparently resulting in slightly reduced mineral uptake.

It is interesting to make use of the relatively high sampling frequency of WSA data, by examining the relationship between SC and volumetric flowrate. In Figure 19 below we show a single example at Langstone Court, below the Gamber confluence.

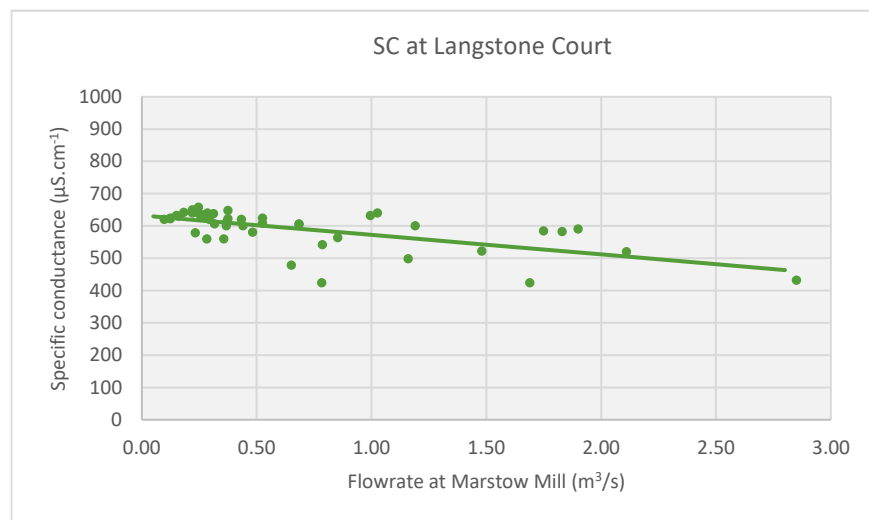


Figure 19: specific conductance versus flowrate at Langstone Court recorded from June-21 to June-22

This is typical of such plots and shows a substantially linear fall in SC as flowrate increases. The solid line represents a regression model which characterises the data in terms of 3 parameters according to the equation:

$$SC = A \cdot Q^{-1} + B(1 - C \cdot Q)$$

In this equation, Q is the volumetric flowrate, as per the horizontal axis in Figure 18. The other parameters are as follow:

- A represents fixed flow pollution sources, such as significant STWs, etc.

- *B* is the intercept with the vertical axis of Figure 18 (a theoretical zero flow value); its value is a characteristic of the catchment geology
- *C* is the rate of fall-off with flow (the gradient); its value is indicative of catchment drainage whereby a groundwater fed river would have a low gradient and a catchment where additional, more rapid drainage pathways turn on during rainfall (e.g., a hilly area) would have a higher gradient

Table 22 below lists the three parameters obtained for each of the sites monitored:

|                     | No. samples | A | B   | C     |
|---------------------|-------------|---|-----|-------|
| <b>Upper Garren</b> |             |   |     |       |
| Orcop               | 28          | 0 | 532 | 0.117 |
| Llangunnoch Bridge  | 45          | 0 | 598 | 0.114 |
| <b>Gamber</b>       |             |   |     |       |
| Audits Bridge       | 30          | 0 | 622 | 0.009 |
| Tretire             | 48          | 0 | 598 | 0.012 |
| <b>Lower Garren</b> |             |   |     |       |
| Langstone Court     | 49          | 0 | 633 | 0.096 |
| Luke Brook (trib.)  | 46          | 0 | 686 | 0.005 |
| Pack Saddle Bridge  | 55          | 0 | 654 | 0.105 |

Table 22: SC regression parameters for each WSA site using data from June-21 to June-22

The *A* parameter is zero throughout, consistent with what we already know about the catchment: there are no large STWs which might inject a fixed flow signal.

The *B* parameter essentially tells a similar story to the mean values presented in Table 21. Orcop is a low outlier, Luke Brook is a high outlier, and the rest are clustered in the range 598 – 654. Because the *B* parameter removes the effect of the flowrate dependence, it is arguably telling us something more fundamental than a simple annual mean.

The *C* parameter shows a significant divergence across the catchment. The Upper Garren and Lower Garren have values which are an order of magnitude larger than the values for the Gamber and adjacent Luke Brook. This most probably reflects the flatter topography found in the eastern side of the catchment which would be expected to result in a reduced flow dependence.

### Phosphorus

Reactive phosphorus (RP) was recorded using a *Hanna Instruments HI-713 Low Range Phosphate Checker*. This instrument provides results as ppm (= mg.l<sup>-1</sup>) phosphate (as PO<sub>4</sub>). All data presented here has been converted to mg.l<sup>-1</sup> as P (abbreviated to mgP.l<sup>-1</sup>), this being the unit used for compliance limits and by the EA.

The annual average values for each site are shown in Table 23 below.

|                     | No. samples | RP <sub>mean</sub> (mgP.l <sup>-1</sup> ) |
|---------------------|-------------|-------------------------------------------|
| <b>Upper Garren</b> |             |                                           |
| Un-named tributary  | 48          | 0.192                                     |
| Orcop               | 28          | 0.101                                     |
| Llangunnoch Bridge  | 46          | 0.071                                     |
| <b>Gamber</b>       |             |                                           |
| Audits Bridge       | 30          | 0.049                                     |
| Tretire             | 49          | 0.058                                     |
| <b>Lower Garren</b> |             |                                           |
| Langstone Court     | 50          | 0.066                                     |
| Luke Brook (trib.)  | 46          | 0.068                                     |
| Pack Saddle Bridge  | 57          | 0.071                                     |

Table 23: annual average reactive phosphorus (RP) recorded from June-21 to June-22

Based on these results, all locations marginally achieve WFD good status (see Appendix 2), with the exception of the Upper Garren at Orcop, which achieves only a moderate. This is broadly in line with the 2019 EA assessment. However, as previously pointed out, we should also consider that this water flows into the River Wye SAC at a point where the 2022 RP target calls for 0.035 mgP/l, a target which is currently only marginally being met.

We now present a load apportionment analysis for each of the sub catchments in the manner developed by Bowes and co-workers (Bowes et al., 2008). In each case, we fit the data to a function defined by the following equation:

$$C_{RP} = A \cdot Q^{B-1} + C \cdot Q^{D-1}$$

where  $C_{RP}$  is river phosphorus concentration,  $Q$  is volumetric flow rate and  $A$ ,  $B$ ,  $C$  &  $D$  are load coefficients to be determined empirically.

Coefficient  $B$  relates to instream (silt) absorption and release processes, which are not usually significant in eutrophic rivers. So, we can set  $B=0$  and the function simplifies to the following:

$$C_{RP} = A \cdot Q^{-1} + C \cdot Q^{D-1}$$

- The first term (parameter  $A$ ) is the nutrient<sup>12</sup> concentration originating from fixed mass flow sources, primarily point sources such as STWs, PTPs and 'misconnected' septic tanks.
- The second term (parameters  $C$  &  $D$ ) is the nutrient concentration originates in response to rainfall and flow-related sources and will largely equate to diffuse sources including agriculture, road run-off and septic tank soakaways.

The transition point, where the two terms in the model are equal is given by:

$$Q_e = \left( \frac{A}{C} \right)^{(1/D)}$$

The model can readily be programmed into Microsoft Excel and the parameters  $A$ ,  $C$  &  $D$  optimised by the Excel Solver function to minimise residual variance in the dataset. This process is referred to as load apportionment modelling (LAM). Since 2008, it has been successfully applied to various catchments, including the Dorset Frome (Bowes et al., 2009), the Thames Basin (Bowes et al., 2014) and more recently the Great Lakes (Rattan et al., 2021).

<sup>12</sup> Although we are concerned here with phosphorus, the analysis method can be applied to other nutrients

We shall present the analysis for each monitoring site, grouped by sub catchment. Note that, for ease of comparison, all sites are plotted against the flowrate at Marstow Mill.

### Upper Garren

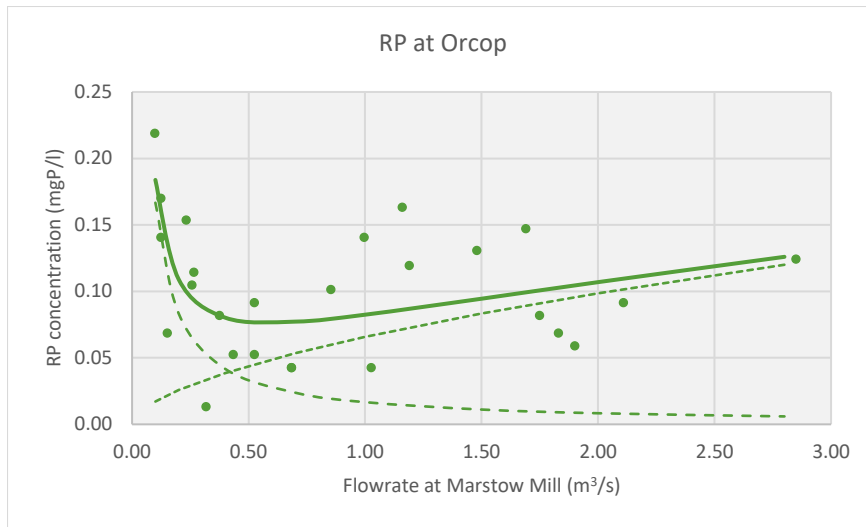


Figure 20: RP v flowrate on the Upper Garren at Orcop; the solid line is the LAM curve; dashed lines represent the 2 terms of the LAM equation:

- fixed flow - longer dashes
- diffuse - shorter dashes.

This site shows a significant fixed flow RP load but is diffuse source dominated for 60% of the year

LAM is significant at 95% confidence level.

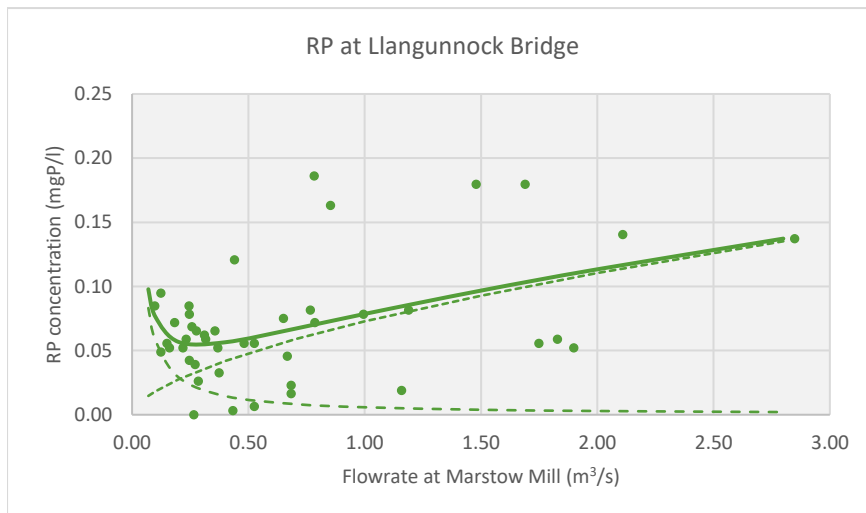


Figure 21: RP v flowrate on the Upper Garren at Llangunnock Bridge; the solid line is the LAM curve; dashed lines represent the 2 terms of the LAM equation:

- fixed flow - longer dashes,
- diffuse - shorter dashes.

This site shows a significant fixed flow RP load but nevertheless is diffuse source dominated for most of the year

LAM is significant at 95% confidence level.

Orcop (Figure 20) is the further upstream of the two monitoring sites on the Upper Garren. At higher flows, its LAM curve shows a gentle upward gradient which is typical of diffuse pollution from lowland agricultural catchments. However, there is also a pronounced upswing at low flows which is indicative of significant fixed flow pollution.

The transition point,  $Q_e = 0.42 \text{ m}^3 \cdot \text{s}^{-1}$  is a little below the Marstow Mill Q50 value (Table 2), a level which is exceeded for about 60% of the year. Thus, the headwaters of the Upper Garren are dominated by diffuse pollution sources for 60% of the year but become fixed flow dominated for the remainder.

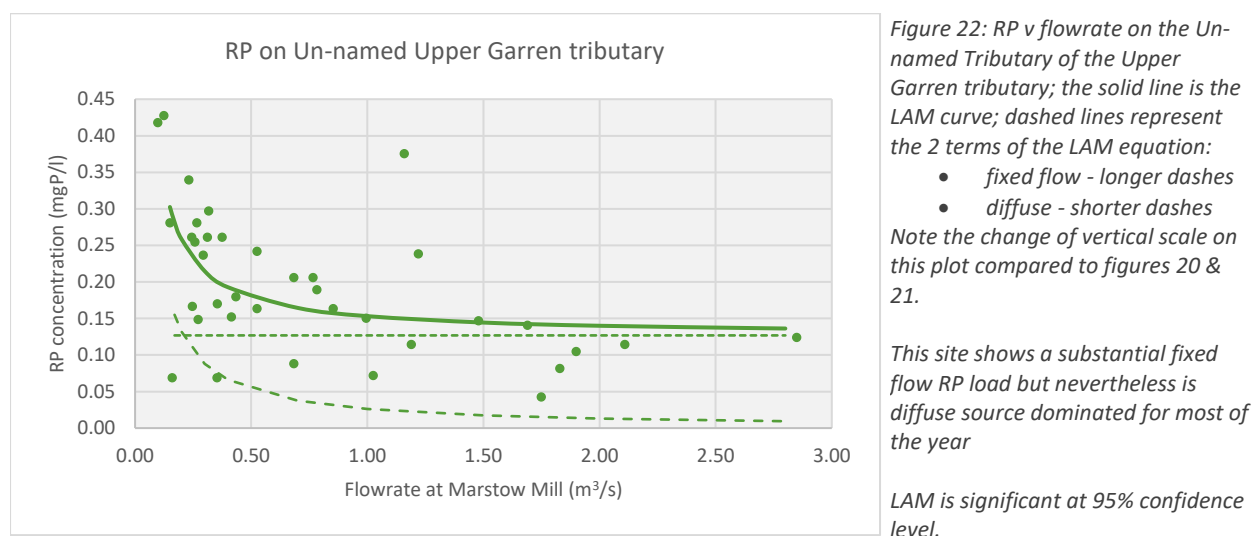
Using QGIS mapping software, we can determine that the Garren above the WSA monitoring site at Orcop drains an area of  $9.8 \text{ km}^2$  and has an average annual rainfall of 832mm. Assuming a fixed HER fraction, this represents 11.2% of the total catchment flow at Marstow Mill. Hence, the fixed flow RP signal represents 11.2% of the A parameter: 0.16kgP/day. This is not readily explained by licensed direct discharge sewage plants in the area.

Downstream at Llangunnock Bridge (Figure 21), just above the Gamber confluence, the fixed flow signal is still present, but smaller.

At this point, we have  $Q_e = 0.20 \text{ m}^3.\text{s}^{-1}$ , which is approximately the 85% exceedance level. Thus, the Upper Garren sub catchment as a whole is dominated by diffuse pollution sources for 85% of the time, with fixed flow sources dominating only at very low flowrates.

We find that the Garren at Llangunnock Bridge is draining an area of  $30.5 \text{ km}^2$  and has an average annual rainfall of 827mm. This equates to 34.8% of the flow at Marstow Mill. This fraction of the A parameter equates to  $0.18 \text{ kgP/day}$  – not very different to the value for Orcop. However, fixed flow signals like this tend to dissipate as they flow downstream, so the similarity suggests that the Upper Garren is receiving further fixed flow inputs below Orcop.

Before leaving the Upper Garren sub catchment, we shall examine the Un-named tributary. This site was selected by our local volunteer because it drains land adjacent to 2 IPUs. After 12 months of monitoring, the site has the dubious honour of possessing the highest average RP level of 72 sites being regularly monitored by WSA.



The stream is small and drains approximately  $1.5 \text{ km}^2$  of land above Orcop. Figure 22 above shows a very strong fixed flow RP signal, representing an RP flow of  $0.04 \text{ kgP/day}$ . Additionally, there is a very high diffuse source background which is substantially flowrate independent. Reading off the diffuse flow curves at  $1 \text{ m}^3.\text{s}^{-1}$  (mean flow), we see that this is roughly twice the value found at the two Upper Garren sites (Figures 20 & 21). The transition point  $Q_e = 0.20 \text{ m}^3.\text{s}^{-1}$ , meaning again that the RP signal is diffuse source dominated for 85% of the year, becoming fixed flow dominated for just 15%.

In Table 25 below, we summarise the RP mass flows at the three monitoring sites within the Upper Garren sub catchment, based on their LAM parameters.

|                     | $Q_e$<br>( $\text{m}^3.\text{s}^{-1}$ ) | Fixed flow RP<br>( $\text{kgP/day}$ ) | Diffuse RP<br>( $\text{kgP/day}$ ) |
|---------------------|-----------------------------------------|---------------------------------------|------------------------------------|
| <b>Upper Garren</b> |                                         |                                       |                                    |
| Un-named tributary  | 0.21                                    | 0.04                                  | 0.15                               |
| Orcop               | 0.42                                    | 0.16                                  | 0.59                               |
| Llangunnock Bridge  | 0.21                                    | 0.18                                  | 2.02                               |

Table 24: annual average reactive phosphorus (RP) flows on the Upper Garren, calculated from the LAM parameters.

### Gamber

Audits Bridge is the further upstream of the two monitoring points on the Gamber, being situated approximately halfway down the catchment. Its LAM curve, shown in figure 23, shows no evidence of

fixed flow sources and is typical of a catchment dominated by diffuse pollution. The curve has noticeably less gradient than the two curves from the Upper Garren.

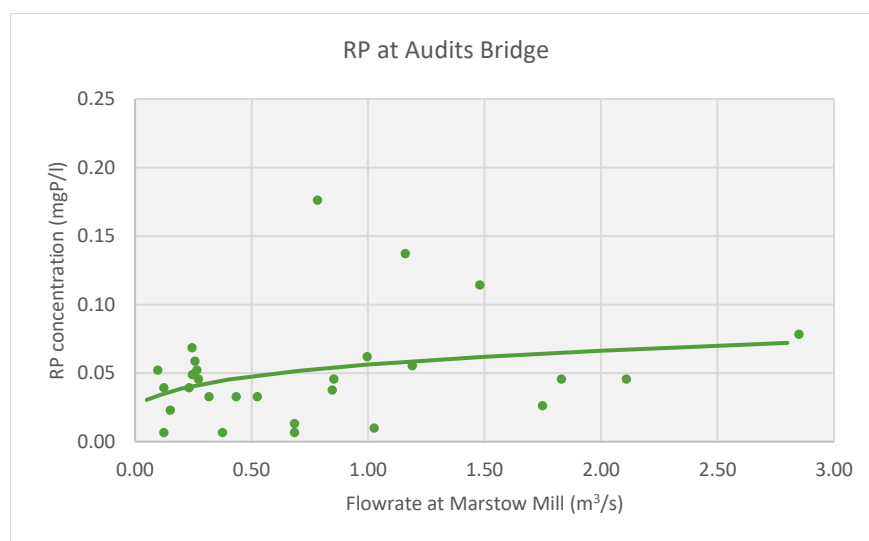


Figure 23: RP v flowrate on the Gamber at Audits Bridge; the solid line is the LAM curve.

*This site shows no evidence of a fixed flow RP load and is entirely diffuse source dominated.*

*LAM has low statistical significance.*

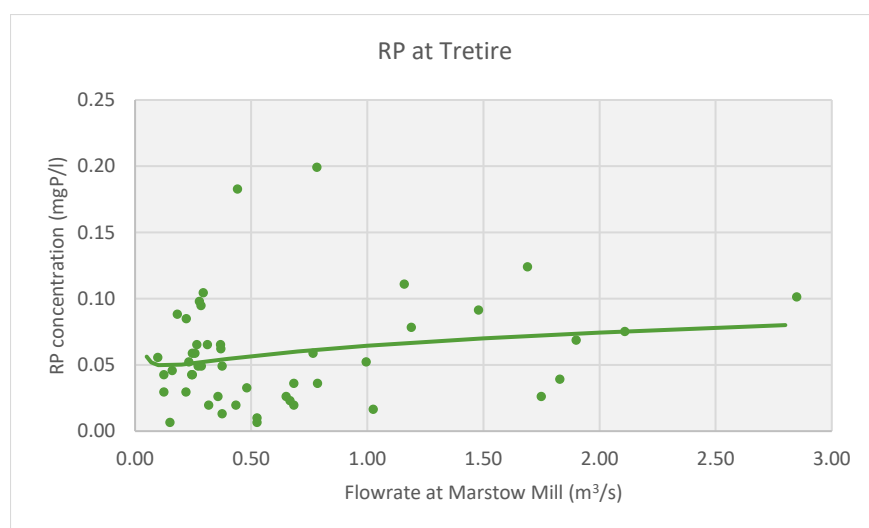


Figure 24: RP v flowrate on the Gamber at Tretire; the solid line is the LAM curve.

*This site shows a small, fixed flow RP load but is largely diffuse source dominated.*

*LAM has low statistical significance.*

Downstream at Tretire, we find a similar low gradient curve (figure 24), but with a small amount of fixed flow signal. In similar fashion to the Upper Garren sites, we calculate the fixed flow RP load to be 0.03kgP/day.

$Q_e$  is calculated to be  $0.01\text{m}^3.\text{s}^{-1}$ , which is very low indeed. At no point does the Gamber become point source dominated.

A further point to note is that the two LAM curves for the Gamber are flatter than the Upper Garren curves. This is similar to our findings on electrical conductivity earlier and suggests that the lower topography of the Gamber sub catchment leads to drainage pathways being more groundwater dominated. This in turn leads to low statistical significance on the LAM model – which is attempting to fit a flow-dependent relationship.

In Table 25 below, we summarise the RP mass flows at the two monitoring sites within the Gamber sub catchment, based on the LAM parameters.

|               | $Q_e$<br>( $\text{m}^3 \cdot \text{s}^{-1}$ ) | Fixed flow RP<br>(kgP/day) | Diffuse RP<br>(kgP/day) |
|---------------|-----------------------------------------------|----------------------------|-------------------------|
| <b>Gamber</b> |                                               |                            |                         |
| Audits Bridge | 0.01                                          | 0.00                       | 0.60                    |
| Tretire       | 0.04                                          | 0.03                       | 1.56                    |

Table 25: annual average reactive phosphorus (RP) flows on the Gamber, calculated from the LAM parameters

### Lower Garren

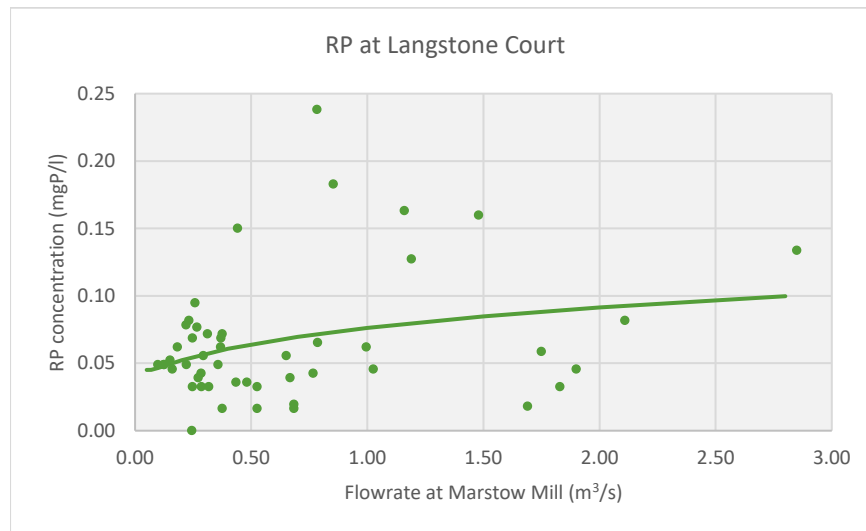


Figure 25: RP v flowrate on the Garren at Langstone Court; the solid line is the LAM curve.

This site shows a very small, fixed flow RP load but is largely diffuse source dominated.

LAM is significant at 70% confidence level.

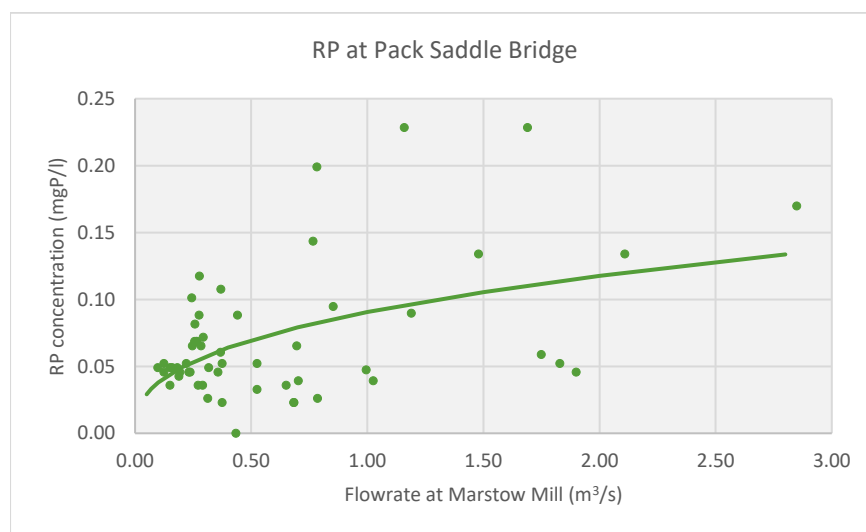


Figure 26: RP v flowrate on the Garren at Pack Saddle Bridge; the solid line is the LAM curve.

This site shows no evidence of a fixed flow RP load and is entirely diffuse source dominated.

LAM is significant at 95% confidence level.

Langstone Court is about 1.5 km below the Gamber confluence and upstream of the Llanerch confluence. Its LAM curve, shown in Figure 25 above, is almost entirely a diffuse source signal. The fixed flow component has all but dissipated at this point. The slope of the curve is intermediate between the Upper Garren and Gamber curves.

Pack Saddle Bridge is immediately downstream of Marstow Mill and thus represents the full Garren Catchment. Its LAM curve is shown in Figure 26 and exhibits a pure diffuse source signal, with a slightly higher gradient than at Langstone Court.

Finally, we present the data for the Luke Brook in Figure 27 below.

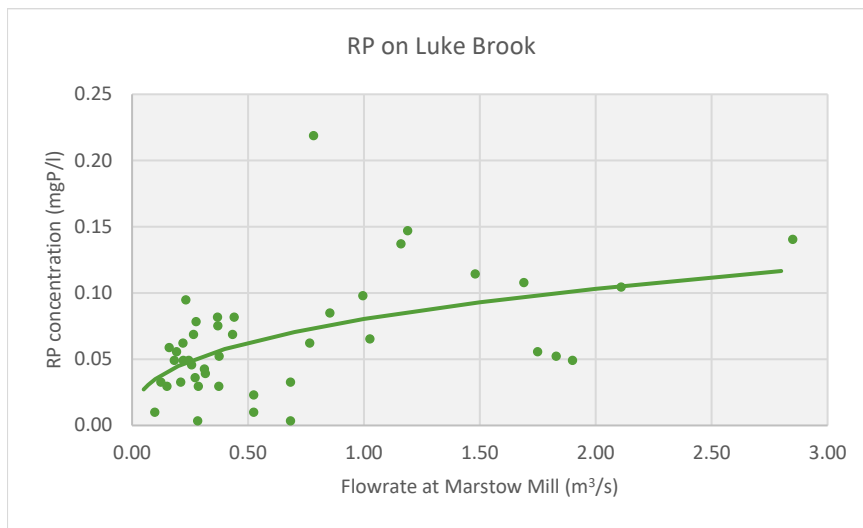


Figure 27: RP v flowrate on the Luke Brook, a Lower Garren tributary; the solid line is the LAM curve.

This site shows no evidence of a fixed flow RP load and is entirely diffuse source dominated.

LAM is significant at 95% confidence level.

The Luke Brook drains 5.9 km<sup>2</sup> of land at the eastern edge of the Garren Catchment. Its LAM curve, shown in Figure 26 above, is again a pure diffuse source signal, with a gradient very similar to Pack Saddle Bridge.

Table 26 below summarises the RP mass flows in the Lower Garren sub catchment.

|                     | $Q_e$<br>(m <sup>3</sup> .s <sup>-1</sup> ) | Fixed flow RP<br>(kgP/day) | Diffuse RP<br>(kgP/day) |
|---------------------|---------------------------------------------|----------------------------|-------------------------|
| <b>Lower Garren</b> |                                             |                            |                         |
| Langstone Court     | 0.02                                        | 0.03                       | 4.07                    |
| Luke Brook (trib.)  | -                                           | 0.00                       | 0.39                    |
| Pack Saddle Bridge  | -                                           | 0.00                       | 6.78                    |

Table 26: annual average reactive phosphorus (RP) flows on the Lower Garren, calculated from the LAM parameters.

### Overview

In order to better understand the RP flows resulting from the LAM analyses above, it is helpful to consider how the RP flow increases as we progress downstream. Essentially, we break down the catchment into sections referenced on the monitoring sites. For each of these, we can then divide the incremental RP flows by the respective catchment area to determine a per-hectare export coefficient. Table 28 below shows the results.

|                     | RP <sub>incremental</sub><br>(kgP/day) | Area<br>(ha) | RP Export Coef.<br>(kgP/ha/year) |
|---------------------|----------------------------------------|--------------|----------------------------------|
| <b>Upper Garren</b> |                                        |              |                                  |
| Un-named tributary  | 0.19                                   | 153          | 0.46                             |
| Orcop               | 0.75                                   | 977          | 0.28                             |
| Llangunnoch Bridge  | 1.45                                   | 2073         | 0.25                             |
| <b>Gamber</b>       |                                        |              |                                  |
| Audits Bridge       | 0.66                                   | 1476         | 0.16                             |
| Tretire             | 0.65                                   | 1083         | 0.22                             |
| <b>Lower Garren</b> |                                        |              |                                  |
| Langstone Court     | 0.60                                   | 989          | 0.22                             |
| Luke Brook (trib.)  | 0.39                                   | 589          | 0.24                             |
| Pack Saddle Bridge  | 2.68                                   | 2360         | 0.42                             |

Table 27: calculation of export coefficients for each area of the Garren Catchment

Notes:

1. Orcop includes the un-named tributary, and Pack Saddle Bridge includes Luke Brook
2. RP<sub>incremental</sub> is the sum of fixed flow and diffuse RP flows

In Figure 28 below, we then plot the RP export coefficients on their respective incremental catchments.



Figure 28: area weighted RP export coefficients for each section of the Garren Catchment, referenced to WSA monitoring sites; numbers with white halos are the respective export coefficients from Table 27.

The three central areas, plus Luke Brook all fall in the range 0.22 – 0.25 kgP/ha/year. The Gamber above Audits Bridge is slightly lower at 0.16 kgP/ha/year. The Garren above Orcop is raised to 0.28 kgP/ha/year by the very high result on the Un-named tributary; excluding this area, the surrounding land has a value of approximately 0.25 kgP/ha/year – the same as the area between Orcop and Llangunnoch Bridge. At 0.46 kgP/ha/year, the Un-named tributary is exporting RP at almost twice the rate of the Upper Garren sub catchment – as is also evident from its mean RP concentration (Table 23). At present we can only speculate about the cause of this. But circumstantially, there are 2 IPU to consider. Within the catchment, we have one IPU with 16,000 layers and a stated intention to spread all manure on site. Additionally, we have an IPU immediately adjacent with 80,000 broilers, with a stated intention of processing manure in an anaerobic digestion (AD) reactor and spreading the resulting digestate on site in order to grow further feedstock for the AD process. Either or both of these IPUs could explain the very high diffuse signal. Sources for the fixed flow signal would most commonly be sewage effluent (meaning direct discharging septic tanks), an industrial leak must also be considered.

The lower catchment area between Langstone Court and Pack Saddle Bridge is also very high at 0.42 kgP/ha/year. This arises through a combination of (a) an 8% increase in mean RP between Langstone Court and Pack Saddle Bridge, (b) the high volumetric flowrate in this section of river and (c) the relatively high gradient on the LAM curve for Pack Saddle Bridge. The superficial implication is that there are some significant diffuse sources in this stretch of the Garren. Alternatively, this imbalance could

simply be indicative of an over-simplification in our modelling. Recall that volumetric flowrates for the Upper Garren and Gamber sub catchments are not directly measured but are inferred from average rainfall, assuming uniform evaporation and transpiration losses (HER fraction) across the Garren Catchment. Speculating that such losses are not in fact uniform, plausibly being lower on the hillier areas of the upper catchments, would have the mathematical effect of increasing export from these regions and correspondingly reducing export from the Lower Garren. In the absence of sub catchment flow monitoring, this cannot easily be resolved.

### Turbidity

Turbidity is not currently a measured parameter. Nevertheless, WSA volunteers do record a subjective *Water Colour* assessment. These entries have been analysed for occurrences of *Clear* versus variations on *Coloured*, *Turbid*, etc. Figure 29 below summarises the results as percentage occurrence of *Clear*.

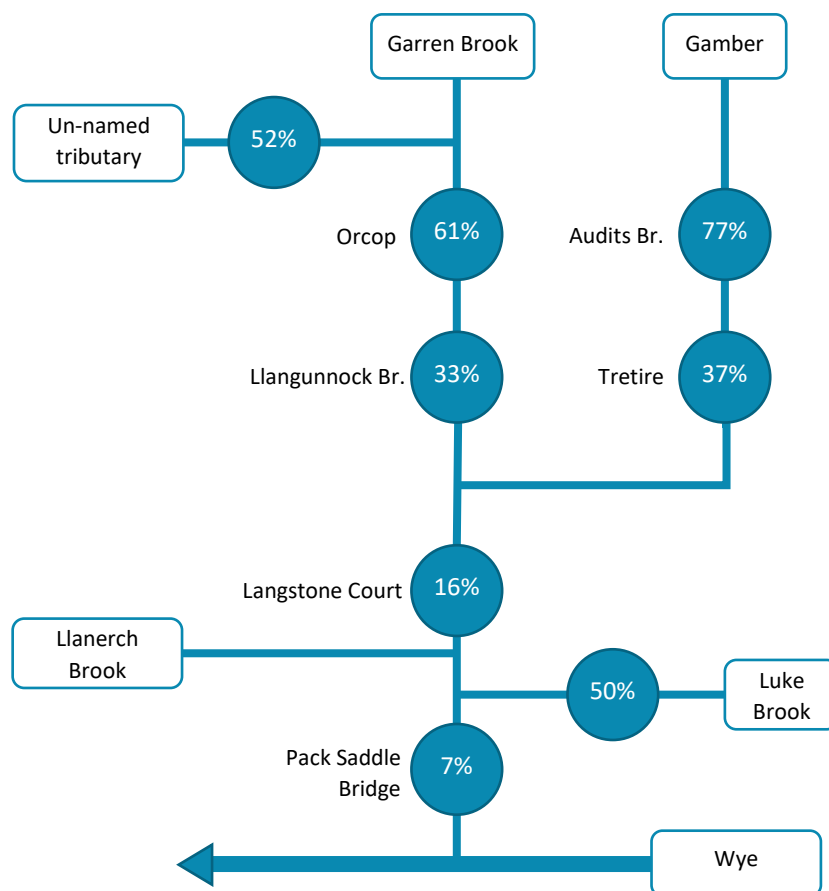


Figure 29: schematic representation of the Garren Catchment showing frequency of 'Clear' water colour observed at 8 monitoring sites, over the period June-21 to June-22

All results have been recorded by a single WSA citizen scientist, so we ought to find a reasonable consistency in the assessments. Optical microscopy performed by our volunteer showed the presence of crystalline material, suggesting that the dominant cause of the turbidity is most probably mineral rather than biological. Looking at Figure 27, there is a clear pattern of turbidity increasing as we go downstream. All tributaries appear to have reasonably clear water in their upper reaches. This would tend to preclude natural turbulence as the mechanism for bringing significant amounts of sediment into the water column; the rivers are clearest where they run the fastest. Thus, we are left with agricultural sources. Surface run-off is precluded by the turbidity being present in dry weather. The remaining candidate would therefore be livestock in and around the rivers. This conclusion concurs with the EA assessment regarding sedimentation.

Given the importance of turbidity on the Garren, WSA should consider options for quantifying this parameter in future.

### Key points

- Soil Olsen-P
  - Average Olsen-P for pasture is currently 2x target range and has increased 50% since 2007
  - Most of the increase is due to 25% of pasture where Olsen-P is 4-8x target range. This can only reasonably be explained by dumping of poultry manure at NVZ maximum rate over many years
  - Average Olsen-P for arable land is also 2x target range but appears stable since 2007
  - Only 11% of arable land (1 sample) found at 8x target level
- Acidity (pH)
  - Generally, in the range pH 8-9 in all areas of the catchment
  - Drops to pH 6-7 observed at Llangunnoch Bridge (Upper Garren), Tretire (Gamber) and all sites downstream; these occur in dry weather during September, October and November; they are most likely associated with slurry spreading for winter arable crops
- Temperature
  - Peak summer temperatures generally in the range 18-22°C; this is 5-6°C higher than the corresponding EA data. This is suspected to be due to allowing insufficient instrument settling time.
  - Diurnal temperature variation is a significant issue when comparing temperature datasets
- Electrical conductivity
  - Corresponds to bedrock variations
  - Shows marked difference in drainage pathways between Gamber and Upper Garren
- Reactive phosphorus
  - Annual average RP concentrations are similar to EA measurements, where these exist
  - RP concentrations on the Upper Garren (livestock dominated) are higher than those on the Gamber (arable dominated)
  - LAM analysis shows the Garren Catchment to be heavily dominated by diffuse pollution sources, although there are some discernible fixed flow inputs: notably in the Upper Garren sub catchment and to a lesser extent in the Gamber sub catchment
  - A small tributary stream of the Upper Garren in the vicinity of 2 IPU's is seen to have 2x the typical RP concentration for the Garren Catchment as a whole; LAM analysis shows its RP signal to be 80% diffuse sourced and 20% fixed flow sourced
- Turbidity
  - Subjective assessment of turbidity shows a pattern of increasing turbidity moving downstream
  - Microscopy suggests that this is caused predominantly by mineral particles
  - The most likely cause is poor management of riparian livestock

## Source apportionment model

### Sewage effluent

Human sewage contributes a variety of pollutants to our rivers. Residual organics (BOD), nutrients (phosphorus, etc.), pharmaceuticals, pesticides, cleaning chemicals, harmful bacteria; the list goes on. The levels at which these substances get into rivers is largely an economic choice, a cost-benefit decision. There are no zeros here. It's about agreeing an acceptable level.

As we saw in the LAM analysis, some sources discharge directly to surface water and manifest as fixed mass flow sources. Others discharge to ground and become part of the diffuse, rainfall dependent mass flow.

The following sections are structured around the different categories of source<sup>13</sup>.

### Sewage treatment works

#### Introduction

STW loads are not all the same. Some STWs serve only residential properties; others might include commercial and industrial effluent. In consequence, the water industry has adopted the concept of

<sup>13</sup> Note that all of the following sections are dealing with total phosphorus (TP) and are not directly comparable with the RP data from water quality data

Population Equivalent (PE) as a standardised unit of load. An input load of 1 PE is commonly defined (within the UK and EU) as follows:

|            |                         |
|------------|-------------------------|
| Liquid     | 180 l/day               |
| Solids     | 250 g/day               |
| BOD        | 60 gO <sub>2</sub> /day |
| Nitrogen   | 12 gN/day               |
| Phosphorus | 2.5 gP/day              |

Table 28: STW input loads per PE.

### Export coefficient

Within the STW, some of the nutrients are taken out with the solids and some are retained in the liquid effluent. In this report I will focus primarily on phosphorus and treated sewage effluent. A typical output stream per PE for an STW with traditional primary and secondary treatments (i.e., no phosphorus stripping) would be:

|            |                           |
|------------|---------------------------|
| Liquid     | 180 l/day                 |
| Phosphorus | 1.44 gP/day <sup>14</sup> |

Table 29: typical effluent flows per PE.

For those STWs discharging to surface water, Table 30 represents what goes into the river every day. For those discharging to ground, this is what goes into the soil. Transport to the river depends on rainfall and a complexity of other factors.

### Current emissions

Data for this section is an asset list from DC-WW, provided via an FoI request.

Due to the absence of large settlements within the Garren Catchment, DC-WW provision of sewage treatment is sparse. There are 6 STWs falling within the catchment, having a total input load of 139 PE between them, representing less than 4% of our population estimate; all have EA permits. Of these STWs, 4 discharge to surface water (*direct discharge*) and 2 discharge to ground. The locations are shown in Figure 30 and details are summarised in Table 31 below, where we also calculate the fixed-flow phosphorus load directly discharging to surface water, and the diffuse phosphorus load discharging to ground. We shall use these data later.

|                                | Input PE   | Fixed flow P load<br>(kgP/day) | Diffuse P load<br>(kgP/day) |
|--------------------------------|------------|--------------------------------|-----------------------------|
| <b>Upper Garren sun totals</b> |            | <b>0.088</b>                   | <b>0.000</b>                |
| Orcop STW                      | 15         | 0.022                          |                             |
| St Weonards STW                | 46         | 0.066                          |                             |
| <b>Gamber sub totals</b>       |            | <b>0.049</b>                   | <b>0.017</b>                |
| Llanwarne STW                  | 34         | 0.049                          |                             |
| Harewood End (No 6) STW        | 12         |                                | 0.017                       |
| <b>Lower Garren sub totals</b> |            | <b>0.030</b>                   | <b>0.016</b>                |
| Llangarron (Herbert Hill) STW  | 21         | 0.030                          |                             |
| Llangarron (3 Garron V) STW    | 11         |                                | 0.016                       |
| <b>Garren Catchment totals</b> | <b>139</b> | <b>0.167</b>                   | <b>0.033</b>                |

Table 30: phosphorus export from DC-WW operated STWs within the Garren Catchment.

P loads are the product of Input PE and the Phosphorus Flow value from Table 30.

<sup>14</sup> This quantity is often referred to as an 'export coefficient'; it can otherwise be stated as 0.54 kgP/PE/year. We will meet several more export coefficients later in this report.

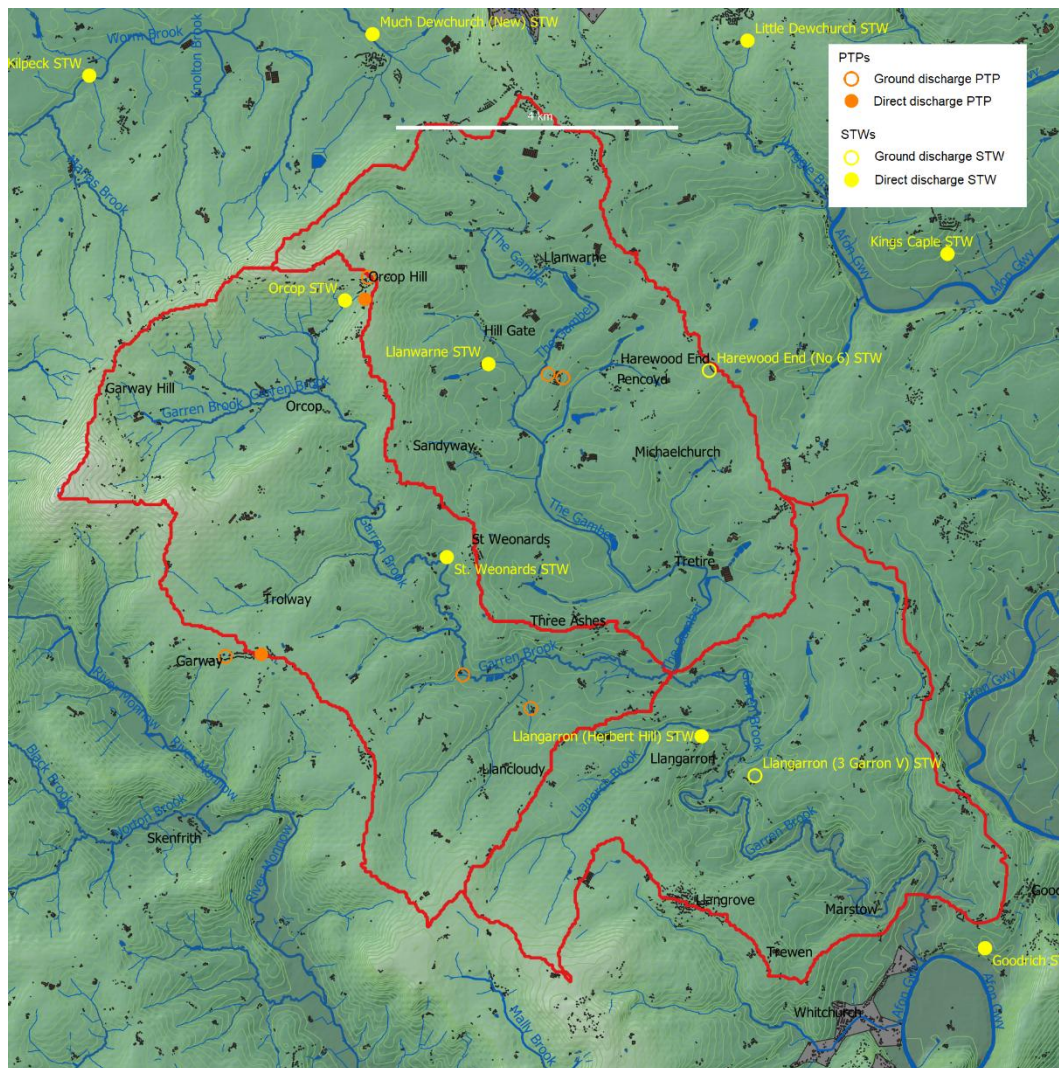


Figure 30: locations of DC-WW operated STWs and licensed, private PTPs within the Garren Catchment.

## Package treatment plants

### Introduction

This section concerns licensed systems only – consequently only a tiny number.

Package treatment plants (PTPs) differ from septic tanks insofar as they incorporate a secondary treatment process and thus have tighter effluent specifications. The technology can be added onto the end of a standard ST or supplied as an integrated unit. This inclusion of secondary treatment allows PTPs potentially to discharge directly into surface water.

Regulation of PTPs is the responsibility of the EA and requirements can be found in the [General binding rules: small sewage discharge to the ground - GOV.UK \(www.gov.uk\)](https://www.gov.uk/government/publications/general-binding-rules-small-sewage-discharge-to-the-ground) and [General binding rules: small sewage discharge to a surface water - GOV.UK \(www.gov.uk\)](https://www.gov.uk/government/publications/general-binding-rules-small-sewage-discharge-to-a-surface-water). For most new, domestic systems, an environmental license is not required – neither for ground discharge nor, surprisingly, for direct surface water discharge. Permitting is only required where the general binding rules cannot be complied with, for example, larger systems, or systems handling something other than domestic sewage. For this reason, the EA have no knowledge of the majority of PTPs and cannot therefore enforce the regulations regarding operation, maintenance and record keeping; or indeed, ensuring that ‘compliant’ systems are actually compliant.

The improved performance of PTPs relative to septic tanks relates primarily to BOD and suspended solids (Broadhead, 2016; May, 2015). Although some PTPs are capable of phosphorus reduction, this technology is not typically deployed. Standard PTPs appear to output only marginally less phosphorus than simple septic tanks. However, Broadhead (2016) suggests that direct discharging PTPs can often have higher phosphorus emissions to surface water than a septic tank when the effect of the drainage field is included.

### Export coefficient

In the light of the above, it seems reasonable once again to use an export coefficient of 1.44 gP/PE/day.

### Current emissions

Here, we consider only PTPs possessing an environmental permit, these being the only ones documented. According to Rivers Trust data, there are 7 such systems located within the Garren Catchment; their locations are shown in Figure 30 above. Again, we allocate these to their respective sub catchment and calculate fixed and diffuse phosphorus export.

|                                 | Input PE   | Fixed flow P load<br>(kgP/day) | Diffuse P load<br>(kgP/day) |
|---------------------------------|------------|--------------------------------|-----------------------------|
| <b>Upper Garren sub totals</b>  |            | <b>0.014</b>                   | <b>0.048</b>                |
| Garway Moon Inn                 | 10         | -                              | 0.014                       |
| Tre Essey Farm                  | 5          | -                              | 0.007                       |
| Wood Cottage                    | 3          | 0.007                          | -                           |
| Velindra Farm STP               | 5          | 0.007                          | -                           |
| Residential Devt. at Orcop Hill | 18         | -                              | 0.026                       |
| <b>Gamber sub totals</b>        |            | <b>0.000</b>                   | <b>0.144</b>                |
| Dhamma Dipa                     | 100        | -                              | 0.144                       |
| <b>Lower Garren sub totals</b>  |            | <b>0.000</b>                   | <b>0.014</b>                |
| Tredunnock Fm. Bus. Centre      | 10         | -                              | 0.014                       |
| <b>Garren Catchment totals</b>  | <b>151</b> | <b>0.014</b>                   | <b>0.206</b>                |

Table 31: phosphorus export from licensed PTPs within the Garren Catchment.

Values for Input PE are our estimates; P loads are the product of Input PE and the export coefficient.

Note that the total input load accounted for by licensed (and therefore documented) discharges from STWs & PTPs combined is 290 PE – less than 8% of our population estimate for the catchment. The remaining 92% of the population are served by undocumented systems which we consider next.

## Septic tanks

### Introduction

This section concerns unlicensed systems – the vast majority of which will be septic tanks, rather than PTPs.

Septic tanks (STs) are a nightmare. They are numerous, often very old and largely undocumented, as noted by (Edwards, n.d.) in his review paper on point source pollution. An excellent review of STs is to be found in (May, 2015) and follow up work reported by (Broadhead, 2016).

A septic tank is a simple two chamber system providing a *primary* treatment (analogous to the primary treatment in an STW). In the first chamber, solids settle to the bottom and scum floats to the top. The remaining liquid flows into the second chamber, from where it is routed to a soakaway in adjacent land – this being considered a type of *secondary* treatment. It is generally assumed that the effluent is ‘cleaned up’ as it percolates through the soil to the river, although there appears to be little evidence to back this up in the context of phosphorus. Indeed, the soil is known to have a finite capacity for phosphorus and one study showed that the phosphorus ‘plume’ extended 30m downstream of a soakaway and advanced by 1m per year. Thus, the performance of soakaways is known to degrade over time.

May (2015) identified two routes for phosphorus to reach surface water. In dry weather, transfer from the soakaway occurs via seepage through the subsoil. In wet weather though, higher levels of phosphorus can be flushed from the soil and from the tanks. At such times, the soil around a soakaway can become a phosphorus emitter, rather than an absorber.

May (2015) also report studies of English catchments where only 3-10% of STs are licensed. In the Garren Catchment, this appears to be approximately 4%. Furthermore, other studies found the majority (80%) of STs appear to be poorly maintained, in the main not being emptied often enough. Old systems often discharged directly to surface water (against current regulations). Others had soakaways too close to surface water (again failing current regulations). Many Victorian systems even route roof runoff via the ST. These three issues all result in 'hydraulic failure' (overload) whereby untreated effluent bypasses the tanks and goes directly to discharge, increasing phosphorus output accordingly.

As discussed in the context of PTPs, current regulation in England do not require registration of an ST / PTP with the EA provided that it complies with the general binding rules.

#### Export coefficient

Given all the practical issues around STs and the lack of evidence to the contrary, the accepted view is to assume that all ST phosphorus eventually gets to the river. May et al. (2008) used an export coefficient of 0.4 kgP/PE/year in their Llangorse study. May (2015) suggests that 0.54 kgP/PE/year is the consensus value in more current literature, a value which is not inconsistent with the findings of Broadhead (2016). In other words, STs are found to have a similar phosphorus output per PE to STWs.

#### Current emissions

It appears that 92% of the Garren Catchment are served by undocumented STs (and the odd PTP). Using the export coefficient of 0.54 kgP/PE/year and the population estimates presented in Table 5, we can calculate the sub catchment phosphorus loads as follow:

|                                | Input PE     | P load<br>(kgP/day) |
|--------------------------------|--------------|---------------------|
| Upper Garren                   | 1,038        | 1.50                |
| Gamber                         | 1,063        | 1.53                |
| Lower Garren                   | 1,299        | 1.87                |
| <b>Garren Catchment totals</b> | <b>3,401</b> | <b>4.90</b>         |

Table 32: phosphorus export from undocumented STs & PTPs within the Garren Catchment.

Table 32 reports the *total* P load but carries no information regarding the split between fixed flow and diffuse components. For this, we need an estimate of the number of systems discharging directly to surface water. For this, we refer to the LAM analyses presented earlier.

Consider the signal at the Orcop monitoring site. This has a fixed flow component of 0.16 kgP/day as reactive phosphorus (Table 25) which, using our RP/TP ratio of 53%, equates to 0.31 kgP/day as total phosphorus. By far the most common source of a fixed flow signal is sewage effluent. So, with this assumption, we can calculate (using the export coefficient of 0.54 kgP/PE/year) that this represents the effluent from approximately 212 PE. Licensed direct discharges upstream of this monitoring site (Orcop STW & Wood Cottage PTP) total 18 PE. We therefore estimate that around 194 PE are undocumented direct discharges to surface water. Now, by a happy coincidence, the catchment above Orcop very closely matches the boundary of Orcop CP, which we know has a population of 455, meaning that 437 people must be served by unlicensed sewage systems; that is 96% of the population. Of these, 44% are discharging directly to surface water upstream of the Orcop monitoring site. Given the picture of English septic tanks painted by May (2015), it seems highly unlikely that many of these are modern, compliant PTPs – as required by regulations.

An analysis of this type is difficult to do further down the catchment. Fixed flow signals eventually dissipate as the river flows downstream, sinking into the statistical noise as more drainage water adds to the river. But we are able to do a similar analysis in another headwater catchment possessing a fixed flow signal: the un-named tributary of Figure 22. The catchment of this stream represents approximately 15% of Orcop CP, so we would expect a population of about 70 people, pro-rata. OS mapping shows seven farms and a number of individual dwellings in the area, so this number seems plausible. The measured fixed flow signal of 0.04 kgP/day (Table 25) as RP equates to 0.07 kgP/day as TP and, hence, represents the processed effluent from 52 PE. There are no licensed discharges in this catchment, so we estimate that approximately 73% of people living here are served by systems discharging directly to surface water.

Obviously, we cannot extrapolate from this to the rest of the catchment. Our LAM analyses indicate that direct discharge is much more common in the hillier areas above Orcop than in the flatter areas elsewhere. Indeed, this would make physical sense in that there is a higher density of small tributary streams to make use of in these areas of the catchment (see Figure 1). Nor can we preclude the possibility of some fixed flow agricultural sources. Moreover, if as May (2015) suggests, many septic tanks are old and poorly maintained, then the per-capita export increases substantially. In which case, our fixed flow signals may in fact result from a smaller number of more dysfunctional septic tanks. The key point here is that nobody knows what the situation is.

Returning to our original question regarding how STs split between fixed flow and diffuse components, we can only base this, rather crudely, on the LAM analyses as follows:

|                                | Input PE     | Fixed flow P load<br>(kgP/day) | Diffuse P load<br>(kgP/day) |
|--------------------------------|--------------|--------------------------------|-----------------------------|
| Upper Garren                   | 1,079        | 0.33                           | 1.16                        |
| Gamber                         | 1,163        | 0.05                           | 1.48                        |
| Lower Garren                   | 1,309        | 0.00                           | 1.87                        |
| <b>Garren Catchment totals</b> | <b>3,552</b> | <b>0.38</b>                    | <b>4.51</b>                 |

*Table 33: phosphorus export from undocumented STs & PTPs within the Garren Catchment.*

In Table 34 above, the fixed flow loads are calculated as the sum of the LAM fixed flow RP signals in each sub catchment converted to TP, using  $RP/TP = 53\%$ .

## Agriculture

### Introduction

On land in a near-natural condition the soil phosphorus content is relatively low, originating mainly from the mineralogy of the soil. Rivers draining such land thus carry little phosphorus – generally showing RP below 0.01 mgP/l. However, modern, intensive agriculture depends upon maintaining elevated levels of nutrients in the soil. It is axiomatic that this will increase the levels of phosphorus in the river. This is reflected in the fact that phosphorus targets for good ecological status in upland rivers, where there are fewer anthropogenic pressures, have lower water quality targets for phosphorus than lowland rivers like the Garren.

In this section, we shall first look at manure loads, before considering the export rate to the river.

### Manure load

We now make use of the livestock population estimates presented in Tables 6 & 7, to calculate manure loads. Rather than tonnage of manure, which is not very informative, we shall list the mass of nutrients, N and P, carried within the manure.

### Cattle & sheep

For cattle and sheep, we start with the age/sex/type population profiles provided in DEFRA's [livestock numbers](#) for Herefordshire. For each category of animal, we calculate manure nitrogen load using standard, per-capita values taken from DEFRA's [NVZ calculator workbook for farmers](#). We then calculate manure phosphorus load using manure nutrient content values (specifically the N/P ratio) from RB209 (ADHB, 2021). The results are shown in Table 35 below.

|                                | Estimated Population | N load (t N/year) | P load (t P/year) |
|--------------------------------|----------------------|-------------------|-------------------|
| <b>Upper Garren sub totals</b> |                      | <b>153.1</b>      | <b>34.2</b>       |
| Dairy                          | 351                  | 22.2              | 5.2               |
| Beef                           | 1,679                | 88.4              | 20.6              |
| Sheep                          | 10,818               | 42.5              | 8.5               |
| <b>Gamber sub totals</b>       |                      | <b>83.5</b>       | <b>18.7</b>       |
| Dairy                          | 192                  | 12.1              | 2.8               |
| Beef                           | 917                  | 48.2              | 11.2              |
| Sheep                          | 5,904                | 23.2              | 4.6               |
| <b>Lower Garren sub totals</b> |                      | <b>86.0</b>       | <b>19.2</b>       |
| Dairy                          | 197                  | 12.5              | 2.9               |
| Beef                           | 944                  | 49.7              | 11.6              |
| Sheep                          | 6,081                | 23.9              | 4.8               |
| <b>Garren Catchment totals</b> |                      | <b>322.6</b>      | <b>72.2</b>       |

Table 34: manure nutrient loads from cattle & sheep within the Garren Catchment.

Populations are from Table 5.

This shows that, subject to our assumption that the grassland of the Garren Catchment is typical of overall Herefordshire statistics, beef cattle are the dominant nitrogen generators making up 58% of the total, with sheep at 28%.

Of particular interest in an NVZ is what this means for nitrogen application rate. Taking the Garren Catchment total N load above and dividing by the area of grassland in the catchment (3,964 ha), yields a catchment average application rate of 81 kgN/ha/year, which is comfortably within the NVZ maximum of 170 kgN/ha/year.

But phosphorus is a slightly different story. The catchment average application rate for phosphorus is 18 kgP/ha/year. Neither the NVZ regulations, nor the Farming Rules for Water<sup>15</sup> provide any numerical limits for phosphorus application rate, so we turn to best practice guidance from RB209. If the land was already at target index 2, then around 9 kgP/ha/year would be an appropriate maintenance level<sup>16</sup> for grazing and 17 kgP/ha/year for one silage cut (slightly more for additional silage cuts). So, given that not all pasture is cropped for silage, we see that the applied manure phosphorus is almost certainly above the recommended maintenance level. But the soil is not at Index 2. Our soil sampling results (Table 19) tell us that the catchment average for soil phosphorus is actually Index 4, in which case RB209 recommends that no additional phosphorus should be applied for grazing or silage growth.

This encapsulates the essential issue surrounding the modern livestock industry. In the days before intensive farming, it was accepted that cattle lost weight over winter, and that milk yields fell. Now, in the interests of 'efficiency', natural food is augmented by high protein feeds in order to maintain growth and milk yields (Rebanks, 2020). These feeds bring with them a significant additional phosphorus load which represents an *import* to the farm which is not offset by milk and beef *exports*. The point is illustrated by RePhOKUs (Withers et al., 2022), who estimate that across the entire Wye Catchment, imported feed for cattle & sheep contains 1017 tonnesP/year, whilst meat and milk exports only 447 tonnesP/year. The balance accumulates in the soil. Thus, pasture phosphorus flow is almost certainly in surplus to some degree, even before we overlay poultry.

<sup>15</sup> [Reduction and Prevention of Agricultural Diffuse Pollution \(England\) Regulations 2018](#)

<sup>16</sup> Maintenance level refers to the application rate that balances typical crop offtake rates

## Poultry

For poultry, we start with the population statistics presented in Table 7. For each bird type, we calculate manure nitrogen load using standard, per-capita values taken from *Water Resources (control of agricultural pollution) (Wales) Regulations 2021: guidance for farmers and land managers*. We then calculate manure phosphorus load using manure nutrient content values from RB209 (ADHB, 2021). The results are shown in Table 36 below.

|                                | Estimated Population | N load (t N/year) | P load (t P/year) |
|--------------------------------|----------------------|-------------------|-------------------|
| <b>Upper Garren sub totals</b> |                      | <b>113.2</b>      | <b>30.0</b>       |
| Egg layers                     | 64,000               | 35.2              | 9.3               |
| Broilers                       | 200,000              | 78.0              | 20.7              |
| Turkeys                        | 0                    | 0.0               | 0.0               |
| <b>Gamber sub totals</b>       |                      | <b>720.9</b>      | <b>191.1</b>      |
| Egg layers                     | 84,000               | 46.2              | 12.2              |
| Broilers                       | 1,653,000            | 644.7             | 170.9             |
| Turkeys                        | 25,000               | 30.0              | 8.0               |
| <b>Lower Garren sub totals</b> |                      | <b>115.2</b>      | <b>30.5</b>       |
| Egg layers                     | 96,000               | 52.8              | 14.0              |
| Broilers                       | 160,000              | 62.4              | 16.5              |
| Turkeys                        | 0                    | 0.0               | 0.0               |
| <b>Garren Catchment totals</b> | <b>2,282,000</b>     | <b>949.3</b>      | <b>251.6</b>      |

Table 35: manure nutrient loads from poultry within the Garren Catchment.

Populations are from Table 7.

As is typical of Herefordshire as a whole, the nitrogen and phosphorus loads are dominated by broilers, which contribute 83% of the total; egg layers are contributing only 14%. As was the case for cattle and sheep, the source of this phosphorus is the imported feed which the birds are reared on.

To put these numbers into perspective, we can again look at spreading rates. Now, the important aspect of intensive poultry is not the location of the IPU so much as the location of spreading. Manure management plans submitted with the more recent planning applications suggest that much of it is either spread on the IPU operator's own land or is spread on neighbours' land. Given that the primary means of transport is tractor and trailer, it is a reasonable assumption that most poultry manure is not moved very far, but some of it could be leaving the Garren Catchment. A relatively small amount is 'exported' to AD plants but some of these are actually located within the catchment. Therefore, unless we understand the disposition of the resulting digestate, we cannot assume that the nutrients are actually leaving the catchment. Indeed, at least one IPU / AD operator spreads digestate on his own land in order to grow feedstock crops for AD<sup>17</sup>. Unfortunately, no one appears to have any systematic data on disposition of poultry manure and AD digestate.

Subject to the above caveat, we can look at the 3,964 ha of grassland. Recall that this is already receiving 81 kgN/ha/year from the resident sheep and cattle and so, within the NVZ limit of 170 kgN/ha/year, there is headroom for a further 89 kgN/ha/year. At this spreading rate, the Garren grassland can only accommodate 351 t N/year – a mere one third of the poultry total in Table 7. If we then look to spread the remaining 598 t N/year of poultry manure on the arable land (4,314 ha) at the NVZ maximum rate (therefore assuming no other organic fertiliser input), this will require 3,518 ha – 82% of all arable land in the Garren Catchment. In summary, when we add poultry manure to the cattle and sheep manure generated within the Garren Catchment, this represents 90% of the NVZ maximum manure application for all agricultural land. By any sensible measure, the Garren Catchment is saturated in manure.

<sup>17</sup> Manure needs to be blended with vegetable matter before it can be anaerobically digested. Thus, land is diverted from useful food production in order to grow feedstock crops for AD plants – in order to 'dispose of' manure. The reality is that liquid digestate presents a larger disposal problem than the original manure.

### Livestock totals

The data from Tables 34 & 35 is summarised as daily totals in Table 36 below and Figure 31 illustrates livestock sector contributions.

|                         | Cattle<br>(kgP/day) | Sheep<br>(kgP/day) | Poultry<br>(kgP/day) | Sub-total<br>(kgP/day) |
|-------------------------|---------------------|--------------------|----------------------|------------------------|
| Upper Garren            | 71                  | 23                 | 82                   | 176                    |
| Gamber                  | 38                  | 13                 | 524                  | 575                    |
| Lower Garren            | 40                  | 13                 | 84                   | 136                    |
| <b>Garren Catchment</b> | <b>149</b>          | <b>49</b>          | <b>689</b>           | <b>887</b>             |

Table 36: manure nutrient loads from all livestock within the Garren Catchment. Poultry is categorised by the location of the IPU and is not necessarily indicative of disposal.

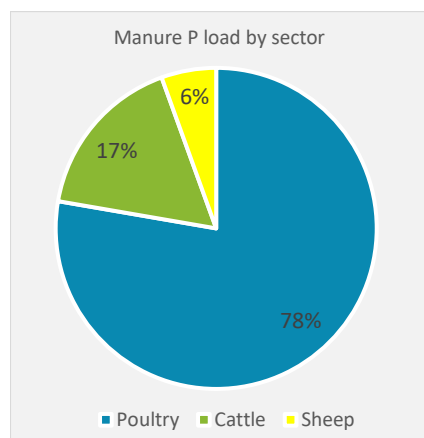


Figure 31: manure phosphorus attribution by livestock sector for the full Garren Catchment

RePhOKUs (Withers et al., 2022) estimated that, across the Wye Catchment, poultry now contributes more manure P than cattle and sheep combined. Given the very high concentration of poultry within the Garren Catchment (3x the average area weighted density of Herefordshire), it is no surprise that this sector generates three times as much manure phosphorus as cattle and sheep combined.

It is useful to see how this poultry dominance has evolved. Figure 32 below shows how the manure phosphorus load has changed over the last two decades.

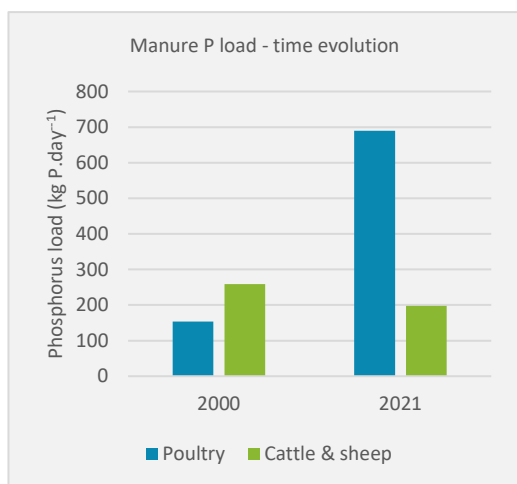


Figure 32: manure phosphorus load comparison between years 2000 and 2021

We find that the contribution from cattle and sheep has fallen by 24%, in line with the changing livestock numbers. The contribution from the poultry sector meantime has increased by 450%.

## Export model

The above analysis concerns nutrient emissions within the catchment. In the case of cattle and sheep, this is predominantly deposited in-situ or spread mechanically on the grassland – thus equating to a soil load. In the case of poultry, the disposition and therefore the soil load is less certain. For this reason, we shall not include poultry as an explicit, per-capita, livestock export at this stage but treat it as a part of the general fertiliser load.

Firstly, we need to convert our measured RP loads to total phosphorus (TP); this is shown in Table 38 below. The first column reproduces RP flow as calculated from the LAM analyses at the downstream monitoring site for each sub catchment. In the second column, we have converted to TP by dividing by 53%, the annual average RP/TP ratio from EA sampling at Trebandy in 2021.

|                         | Site                      | RP flow<br>(kgP/day) | TP flow<br>(kgP/day) |
|-------------------------|---------------------------|----------------------|----------------------|
| Upper Garren            | Llangunnock Bridge        | 2.32                 | 4.40                 |
| Gamber                  | Tretire                   | 1.64                 | 3.10                 |
| Lower Garren            |                           | 2.82                 | 5.34                 |
| <b>Garren Catchment</b> | <b>Pack Saddle Bridge</b> | <b>6.78</b>          | <b>12.84</b>         |

Table 37: measured phosphorus flows for each sub catchment (WSA data); Lower Garren is calculated by subtraction.

In our earlier report on the Llynfi Catchment (Green, 2022a), we applied the export coefficient model developed for the Slapton Catchment in Devon (Burt et al., 1996) and obtained a good match to observed Llynfi RP data. In our favour though were the facts that (a) the hydrologically effective rainfall (HER) is similar to Slapton, (b) the soil texture is not too different (loam and clay to silty loam cf. silty loam at Slapton) and (c) the hilly topography is reasonably similar. However, the Garren Catchment is very different. It has significantly lower HER (66% of the Slapton HER), much coarser soil texture (loam to sandy loam) and has significantly lower topography.

The Burt et al. export coefficients operate in two parts. Different land types have a per hectare coefficient then, on top of this, livestock manure load is added as a per capita value, calculated as a percentage of the deposited manure phosphorus. In Table 39 below, we reproduce the export coefficients from Burt et al. and apply the livestock coefficients to area-weighted manure loads for the Garren. Note that the total grassland export (grazing + cattle + sheep) at Herefordshire stock densities, is 0.79 kgP/ha/year – not dissimilar to the arable figure of 0.68 kgP/ha/year.

| Land use         | Burt et al. export coefficients |               |
|------------------|---------------------------------|---------------|
|                  |                                 | (kgP/ha/year) |
| Arable           |                                 | 0.68          |
| Grazing          |                                 | 0.26          |
| Woodland         |                                 | 0.02          |
| Heath & bog      |                                 | 0.02          |
| <b>Livestock</b> |                                 |               |
| Cattle           | 2.85%                           | 0.39          |
| Sheep            | 3.00%                           | 0.14          |

Table 38: phosphorus export coefficients from Burt et al. (1996). The livestock coefficients apply the % coefficient from Burt et al. to the relevant area-weighted manure loads presented in Table 37.

Now, stating export coefficients as a mass flow is perfectly fine within a single catchment, or when considering similar catchments. But, for the reasons stated above, this dimensionality does not translate well between dissimilar catchments. The most obvious thing to do, therefore is to weight the coefficient by HER, thus expressing it as a concentration rather than a mass flow. This effectively means that we must multiply each export coefficient by the ratio of the HERs (in this case 66%) before applying them to calculate the exported phosphorus flow. This treatment does not deal with all of the variables, but it does deal with perhaps the most obvious and significant one.

### Total exports

We can now examine the total phosphorus export attribution by sector within the Garren Catchment, applying the various export coefficients developed above. This is shown in Table 39 below and illustrated in Figure 33.

|                         | STW & PTP<br>(kgP/day) | ST<br>(kgP/day) | Grassland<br>(kgP/day) | Arable<br>(kgP/day) | Sub-total<br>(kgP/day) |
|-------------------------|------------------------|-----------------|------------------------|---------------------|------------------------|
| Upper Garren            | 0.13                   | 0.98            | 2.66                   | 1.36                | 5.14                   |
| Gamber                  | 0.16                   | 1.09            | 1.45                   | 2.36                | 5.07                   |
| Lower Garren            | 0.05                   | 1.24            | 1.50                   | 1.56                | 4.35                   |
| <b>Garren Catchment</b> | <b>0.34</b>            | <b>3.31</b>     | <b>5.61</b>            | <b>5.29</b>         | <b>14.56</b>           |

Table 39: sector attributions of river phosphorus loads.

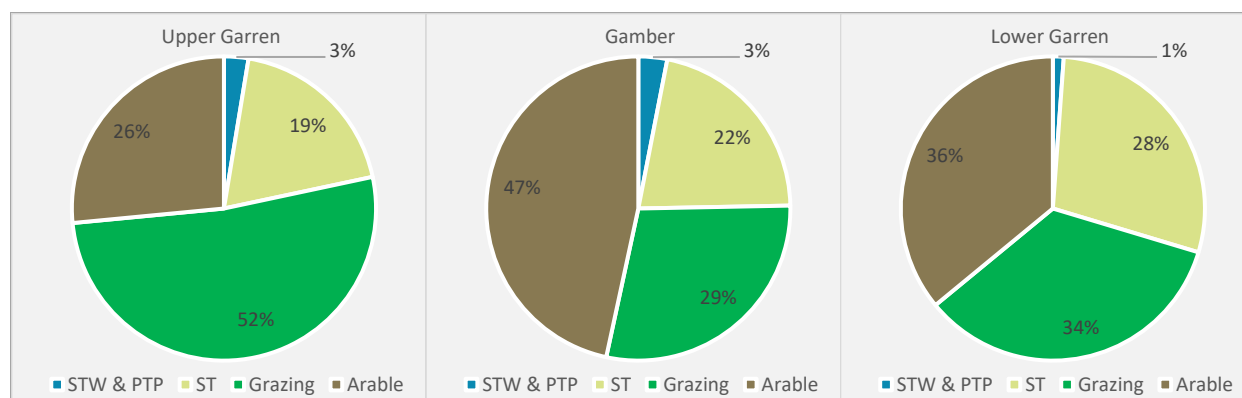


Figure 33: pie charts showing sector attribution of river phosphorus loads within each sub catchment

This shows that DC Welsh Water plant and licensed PTP emissions are responsible for just a tiny fraction of the phosphorus in the river. Unlicensed septic tanks account for the bulk of the sewage contribution and constitute a significant fraction of the total load. Within the agriculture sector, we see the dominance of grassland in the Upper Garren and of arable land in the Gamber.

The model predicts a total phosphorus export averaging 14.56 kgP/day. We can now compare this to EA measurements at the Wye confluence as well as the LAM analysis of WSA data from Pack Saddle Bridge. This is shown in Table 42 below.

|                                               | P load<br>(kgP/day) |
|-----------------------------------------------|---------------------|
| LAM of WSA data from Pack Saddle Bridge       | 12.84               |
| Annual average of EA data from Wye Confluence | 13.11               |

Table 40: average phosphorus flows based on measured water quality data. LAM integrates the model (see Figure 26) over the actual daily volumetric flowrates from Marstow Mill; EA data uses the trend value at 2021 (see Figure 13) multiplied by annual average volumetric flowrate. Both calculations use RP/TP = 53%.

The correspondence with the source apportionment model is very good – within 10%. However, the model so far does not include the poultry contribution. The graphs above are undoubtedly more representative of the situation in 2000, rather than 2021.

### Including poultry

We have been unable to include poultry directly within the above source apportionment model because we have no knowledge of where the resulting manure is disposed of. We can resolve this problem by looking for clues within the data, and fortunately, there is a smoking gun. Recall firstly that both cattle and sheep numbers have reduced substantially since 2000. Our soil sampling suggests that phosphorus concentrations (and by implication fertiliser application rates) have remained reasonably stable on arable land. Furthermore, the [City Population](#) website tells us that the human population has been fairly stable over this period. Thus, we ought to expect a concomitant reduction in river phosphorus

concentrations between 2000 and today. Not so. In figures 11 & 12, we showed that the long-term trend on the EA data, particularly on the Upper Garren and Gamber sub catchments, has been a steady increase. The only conceivable source of this additional river phosphorus is the poultry industry, which as we have seen, has increased rapidly since 2000 (see figure 7) and now outputs 78% of the manure phosphorus load within the Garren Catchment (Figure 31). So, let us examine the two upper sub catchments in more detail.

Table 43 below summarises the changes to ruminant numbers and the expected impact on phosphorus export, based on the above source apportionment model.

|                                    | Change<br>2000 - 2021 |
|------------------------------------|-----------------------|
| Cattle                             | -21%                  |
| Sheep                              | -32%                  |
| Overall impact on grassland export | -14%                  |

Table 41a: change in ruminant numbers and impact on nett grassland export coefficient (land + cattle + sheep)

|              |     | P load change<br>(kgP/day) |
|--------------|-----|----------------------------|
| Upper Garren | -7% | -0.36                      |
| Gamber       | -4% | -0.20                      |

Table 41b: expected change in sub catchment phosphorus load due to reduced grassland export coefficient

Table 44 now summarises the key statistics from EA water quality data. Note that the RP/TP ratio also changed, from 65% in 2000, to 53% in 2021 - which why the changes in TP are larger than the corresponding change in measured RP.

|              | Measured<br>RP change | Calculated<br>TP change | P load change<br>(kgP/day) |
|--------------|-----------------------|-------------------------|----------------------------|
| Upper Garren | +36%                  | +68%                    | 2.93                       |
| Gamber       | +21%                  | +49%                    | 1.24                       |

Table 42: actual measured change in river phosphorus load between 2000 and 2021. RP/TP was 65% in 2000 and 53% in 2021.

From the above two tables, we need to explain where an additional 4.73 kgP/day has come from since 2000. We expected a reduction of 0.56 kgP/day due to falling ruminant numbers, but instead, EA have measured an increase of 4.17 kgP/day. The only conceivable explanation is poultry.

Referring to Burt et al. (1996), we obtain a poultry export coefficient of 2.7%. Applying the 66% HER correction, reduces this to 1.8% (of manure phosphorus). Using this value, we find that the observed rise in river RP between 2000 and 2021, on the combined catchment above the Gamber confluence, corresponds to a manure phosphorus load of 97 tP/year. This represents 39% of the total poultry manure generated within the Garren Catchment – which is an entirely credible value.

Now, our Garren soil analysis project (Green, 2022b) found a significant increase in Grassland Olsen-P since 2007 and no change on arable land. Thus, it is reasonable to infer that spreading as means of disposal, as opposed to displacing other fertiliser inputs, has been mainly restricted to grassland. Taking the area of grassland within the two upper catchments, we can now calculate an average spreading rate of 33 kgP/ha/year.

We present in Table 45 and Figure 34 below a revised source apportionment model for the two upper sub catchments, including the estimated poultry contribution. The two sub catchment totals are now well above our estimates for the respective river phosphorus mass flowrates, but this is unsurprising. We have already discussed the difficulty of transferring export coefficients between dissimilar catchments, and to this we must add uncertainties around livestock numbers and our assumption of uniformity of HER fraction. Importantly though, the absolute error does not necessarily detract from the validity of the proportionality shown in Figure 34.

|              | STW & PTP<br>(kgP/day) | ST<br>(kgP/day) | Grassland<br>(kgP/day) | Arable<br>(kgP/day) | Poultry<br>(kgP/day) | Sub-total<br>(kgP/day) |
|--------------|------------------------|-----------------|------------------------|---------------------|----------------------|------------------------|
| Upper Garren | 0.13                   | 0.98            | 2.66                   | 1.36                | 2.93                 | 8.07                   |
| Gamber       | 0.16                   | 1.09            | 1.45                   | 2.36                | 1.24                 | 6.30                   |

Table 43: sector attributions of river phosphorus loads, including poultry.

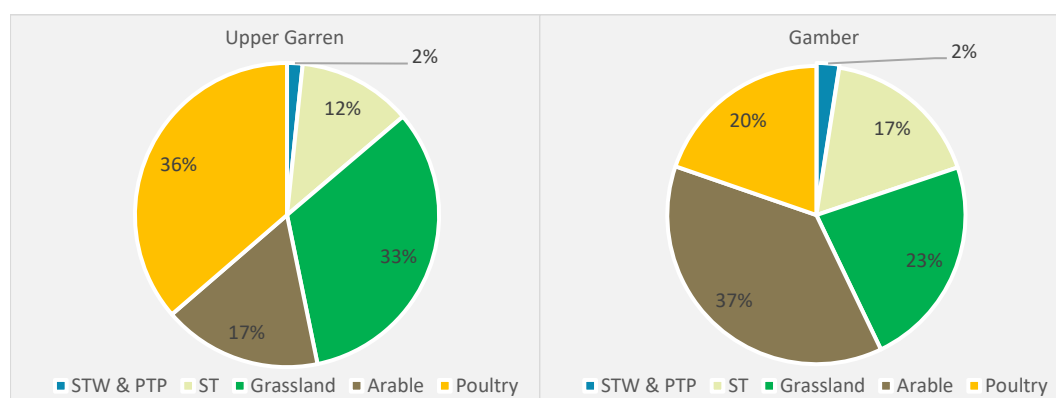


Figure 34: pie charts showing sector attribution of river phosphorus loads within the two upper sub catchments

A final sanity check which we can do is to estimate how Olsen-P might be expected to respond to this phosphorus application rate and compare it to the results we reported above (31mg/l in 2007 and 54 mg/l in 2021: from Green 2022b).

In figure 7, we showed how the poultry population has increased between 2000 and 2021. From this information we can estimate the year-by-year application rate over the period, culminating in the rate calculated above for 2021. RePhOKUs have estimated that Olsen-P for Wye catchment soils increase by around 9mg/l for every 100kgP/ha of applied phosphorus. So, it is then a simple mathematical process (similar to the calculation of Box 2) to convert the cumulative applied phosphorus to a predicted Olsen-P value. The result is shown in Figure 35.

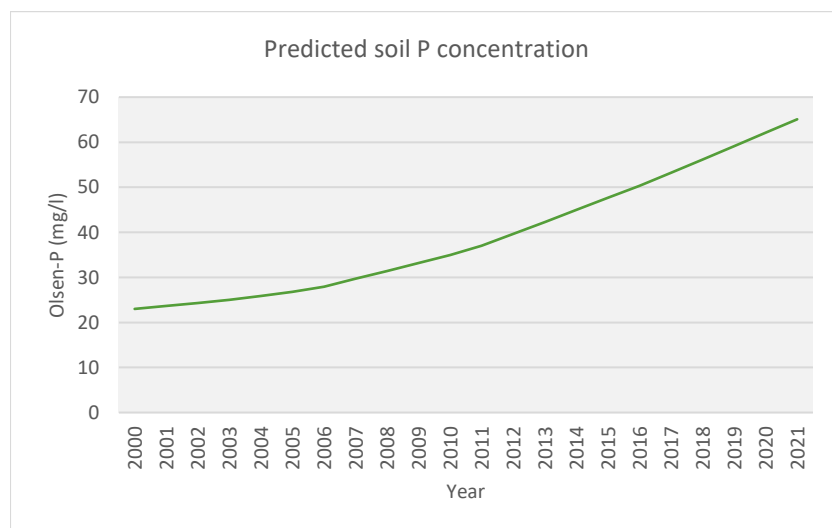


Figure 35: predicted change in Olsen-P resulting from application of 39% of Garren poultry manure to upper catchment grassland

This calculation predicts a value of 65 mg/l in 2021, somewhat higher than the observed value. Reducing the application rate from 39% to 30% gives a close fit to the observed data. But the point here is not to claim precision we do not have, but to establish that the observed increase on soil Olsen-P is remarkably consistent with the increase in upper catchment RP values. The same volume of manure explains both observations. Moreover, the only feasible source of this manure load is poultry manure spreading.

### Key points – source apportionment

- Human sewage
  - Water company operated sewage treatment is provided to only 139 PE
  - Licensed private systems are likewise very few in number (151 PE)
  - 92% of the resident population are served by unlicensed private systems (mainly septic tanks)
  - EA have a statutory responsibility to ensure regulatory compliance of private systems, but do not maintain a list of these
  - Academic studies in other parts of rural England have found very high levels on non-compliance
- Livestock
  - In 2000, cattle & sheep were the dominant source of manure P
  - Since then, ruminant numbers have fallen, and poultry numbers have increased x5
  - In 2021, poultry are producing 3x more manure P than cattle & sheep
- Export model
  - Human sewage contributes less than 25% of P export
  - Agriculture contributes over 75% of P export
  - P export has increased significantly since 2000, particularly in the upper sub catchments, above the Gamber confluence
  - The only plausible explanation for the increase in the upper sub catchments is poultry manure disposal, primarily on grassland
  - The increase in P export in the upper catchment is consistent with the increase in grassland Olsen-P found in our soil survey
  - Poultry manure spreading would appear to be responsible for the rising trend in river RP

## Conclusion

### Discussion

The Garren Catchment is located in an entirely rural area of Herefordshire, within the lower section of the Wye Catchment. Agriculture is highly intensive, with very little remaining woodland and scrub areas. There is no 'un-managed' land of the type we would find in the Welsh uplands. Activities are roughly evenly split between arable cropping and improved grassland supporting cattle and sheep. The two upper sub catchments, the Gamber and Upper Garren, show a divergence, with the slightly hillier terrain of the Upper Garren being biased towards pasture and the flatter terrain of the Gamber being biased towards arable farming. The lower sub catchment is more evenly split.

In the period since the mid-nineties, the intensive poultry sector has expanded rapidly in the upper catchment, particularly in the Gamber sub catchment, which is now a significant hot spot for IPUs. There are now around 2.3M birds within the Garren Catchment, a 5x increase since 2000. Over the same period, Herefordshire cattle numbers have fallen by 21% and sheep by 32%. Meanwhile, the human population has remained stable.

The catchment sits on sandstone lithology, which brings with it a propensity for release of clay particulates when the bedrock is eroded by running water. Although the prevailing soil texture in the area is quite coarse, the riparian zones are modified by fluvial deposition and incorporate a significant clay fraction. The topography is one of gentle rolling countryside, with the Garren falling only 130m from its source to the Wye confluence. Consequently, for the most part, the rivers are predominantly groundwater fed, with relatively little contribution from additional surface pathways during rainfall. A further consequence of the sandstone lithology is high alkalinity (dissolved  $\text{CaCO}_3$ ).

### Abstraction

We have shown how the catchment is heavily abstracted, predominantly along the Gamber (Figure 4). The total licensed volume for direct surface water abstraction exceeds our estimate of Q95 flowrate for the Gamber by over a factor of two. This is a very serious volume of water to be taking from a small

brook. Whilst license conditions may trigger before Q95 flowrate<sup>18</sup>, we note that the only EA flow monitoring site is at Marstow Mill on the Lower Garren. This seems rather inappropriate when all the abstraction is occurring on the Gamber, which only contributes about a third of the Marstow Mill flow volume. There is a clear need for a flow measurement station on the lower Gamber.

### EA water quality data

Despite draining into the Wye SAC, the Garren is not classified as being part of the SAC. Thus, its water quality targets are default WFD targets, rather than the more stringent SAC targets. In the last decade, the Garren has fallen from *good* ecological status to *moderate*, the failing elements being *fish*, *macrophytes* and, in the case of the Upper Garren waterbody, *orthophosphate*. Key causative factors are sedimentation and nutrient management, both of which result from poor riparian land and livestock management. Barriers to fish movement were also cited. In the 2019 WFD assessment, the Upper Garren water body scored only *moderate* on orthophosphate, with the Gamber and Lower Garren scraping into *good* classification. Note however that targets for orthophosphate are heavily relaxed, due to the high alkalinity of the river.

Our review of EA water quality data for the period 2000-2021 revealed the following key points:

- The frequency of EA sampling on the Garren fell off markedly after 2007.
- Although not a WFD element, river nitrate levels are found to be high and correlates with arable agriculture, being highest on the Gamber and lowest on the Upper Garren. Results have been stable over recent years.
- Alkalinity, pH, temperature, electrical conductivity, unionised ammonia and nitrite are generally unremarkable.
- RP (orthophosphate), as expected, falls within WFD *moderate* classification on the Upper Garren and scrapes into the *good* classification on the Gamber and Lower Garren.

But, in addition:

- Since 2000, RP mass flowrate has increased by 36% on the Upper Garren and by 21% on the Gamber
- Since 2000, annual average RP/TP has fallen from 65% to 53%. Thus, the relative stability of the annual mean RP on the Lower Garren is illusory. TP flowrate has increased throughout the catchment: by 68% on the Upper Garren, by 49% on the Gamber, and by 25% on the Lower Garren.

Given the fall in ruminant numbers and dramatic rise in poultry numbers since 2000, it is hard to avoid the conclusion that the huge rise in phosphorus export from the two upper sub catchments is caused by poultry manure spreading. Moreover, the more substantial rise on the Upper Garren suggests that this increase might be driven by grassland practices, rather than arable. It is entirely plausible that this significant shift in the type of manure used within the catchment has also driven the reduction in the RP fraction.

### Soil survey

The Garren Catchment soil survey undertaken by WSA and CPRE showed that most of the land is 2x above target values for Olsen-P. However, within this, we found that arable land had remained substantially constant since the 2007 Countryside Survey, whilst grassland Olsen-P had increased significantly. Furthermore, the rise was found to be driven by approximately 25% of the grassland

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<sup>18</sup> Due to the vagaries of FoI requests, at the point of writing, we have been unable to confirm the reference flow location pertaining to the Gamber abstraction license conditions.

samples which had up to 7x recommended levels of phosphorus. This supports the circumstantial suspicion above that the rise in river RP is driven by grassland manure spreading practices.

More fundamentally, our limited soil survey (1 sample per 4km<sup>2</sup>) has demonstrated the utility of such an approach to help understand the catchment. The results have proved invaluable in this work, in supporting interpretation of the river WQ data (both EA and WSA CS). The cost of a single Olsen-P test is £10 in small volumes and increasing the sampling to 0.5km grid (roughly every other field) would cost £3,000 per year. That's in the region of £25 for each farm in the catchment – surely not a big deal.

### *WSA water quality data & load apportionment*

The higher temporal and spatial frequency of WSA citizen science data (weekly at 8 locations) enables a more detailed analysis than is possible using EA data, especially when combined with the EA daily volumetric flow records from Marstow Mill. Load apportionment modelling is proving to be a very useful analysis tool. Key points emerging from this analysis are as follow:

- Annual means for electrical conductivity are in general agreement with EA data. However, analysis of flowrate dependence clearly illustrates the groundwater-fed nature of the Garren Catchment, especially on the lower topography of the Gamber sub catchment.
- pH is again in general agreement with EA data on an annual mean basis. However, analysis of flowrate dependence highlights clusters of low pH outliers. These correlate with arable agriculture, and all occur during dry weather in autumn. This is highly suggestive of slurry spreading for winter cereal crops directly contaminating surface water.
- Spatial variation in electrical conductivity corresponds to bedrock variations. Flowrate dependence shows a clear difference between the Gamber and Upper Garren in terms of drainage pathways, the Gamber having noticeably less flowrate dependence.
- Annual average RP concentrations are consistent with EA data, where this exists. Importantly, we also find the highest RP concentrations are found in the grassland dominated Upper Garren sub catchment and the lowest in the arable dominated Gamber. LAM analysis shows the Garren Catchment to be dominated by diffuse sources of phosphorus. Flowrate dependence matches the pattern observed for electrical conductivity, with the Upper Garren being more flowrate sensitive than the Gamber. Additionally, there are clear signals from fixed flow sources in some areas, most notably in the Upper Garren above Orcop<sup>19</sup>. Thus, we must conclude that there are a significant number of unlicensed directly discharging septic tanks in these areas – in breach of regulations.
- One small tributary stream above Orcop illustrates the problems in the area. It shows a very significant fixed flow RP signal, consistent with multiple unlicensed direct discharging septic tanks. In addition, the diffuse RP signal on this stream is twice the typical value for the Upper Garren, which suggests some questionable agricultural practices. Immediately adjacent, we find two IPUs, one of whom disposes of manure on site, and the other, who also operates an AD plant, spreads much of the resulting digestate on site. This could undoubtedly be investigated further by means of additional sampling upstream, but it is not the citizen scientist's role to enforce environmental legislation. The problems of this stream must now be investigated by the EA.
- In examining the spatial variation of mean RP, we expressed this as an area-weighted export coefficient for each section of the catchment (Figure 28). This suggested an anomalously high export from the section of the Garren below Langstone Court. The most likely explanation is

<sup>19</sup> The area above Orcop is intersected by a relatively dense network of small streams (see Figure 1), resulting from the hillier terrain. Thus, it is more likely that historic septic tanks would be discharging to surface water in this area than would be the case downstream.

that we have underestimated the flowrates in the two upper sub catchments by calculating these on the basis of uniform rates of evaporation and transpiration losses for rainfall. It is plausible that such losses are in fact lower on the hilly areas in the Upper Garren sub catchment and the headwaters of the Gamber. If so, this would increase estimated export from the upper catchment and reduce the lower catchment export. In the absence of measured flowrates at Llangunnoch and Tretire, this is impossible to resolve with any confidence.

- WSA observational data on turbidity supports the EA conclusions regarding the impact of riparian livestock management on sedimentation of the river. We find a pattern of increasing turbidity moving downstream and basic microscopy suggests that the river carries a significant mineral load.

### *Source apportionment model*

The section on source apportionment proceeded in two stages. Firstly, we estimated phosphorus application rates to land, using the population estimates presented earlier and generally accepted emission rates. Then we estimated phosphorus export flows using export coefficients taken from a published study of a different catchment. We acknowledged that the Garren Catchment differed in many key respects. So, although we might expect a similar proportionality between the various sources, the absolute export prediction may be a little off the mark. We did however adjust the export coefficients for the difference in HER between the catchments, speculating that this might be the most significant difference. With this single modification, the export coefficient model predicted a total catchment export within 10% of the export calculated from measured RP in the lower reaches of Garren. But at this point we had not included export from poultry.

Although we have a good estimate of poultry numbers and locations, and therefore are able to calculate manure phosphorus load, we have not been able to include this explicitly in the export model (analogous to cattle, say) because the location of the birds is not the point. The key factor with poultry is where the manure is spread, and in what quantity – which is entirely unknown. Thus, we have been forced to take a different approach. We observed that both the Upper Garren and Gamber have seen huge increases in total phosphorus load since 2000. There being no other plausible explanation, we attribute this rise to poultry. Furthermore, picking up the point that both soil phosphorus data and relative RP concentrations both point to grassland as being the problem, we calculated what spreading rate would be required on grassland to account for the observed increase in RP, as opposed to the reduction expected due to falling ruminant numbers. The answer is that the grassland of the Upper Garren and Gamber sub catchments is receiving about one third of the total poultry manure generated within the catchment – an entirely plausible result. Clearly, we would expect a considerable amount of poultry manure also to go on arable land. But since arable farmers need to pay for fertiliser, they are much more likely to spread in accordance with crop needs. This assertion is supported by the fact that arable soil Olsen-P does not appear to have increased since 2007. As a cross-check, we then calculated the change in Olsen-P which might result from this cumulative application and found a good match with the results of our soil survey.

In summary then, we find the measured rises in phosphorus load on the Upper Garren and Gamber are in good agreement with the observed change in grassland Olsen-P and that poultry manure spreading is the only plausible explanation for this increase. On this basis, we estimate that poultry manure spreading now contributes 20% of the Gamber phosphorus load and 36% of the Upper Garren load. Moreover, poultry manure spreading is responsible for the large increase in river phosphorus load seen on these sub catchments since 2000.

### Key pressures

We have shown that the Garren Catchment faces significant pressures from turbidity and nutrient load, most particularly phosphorus. Both these findings accord with the EA's WFD assessments. To this we also need to add abstraction, primarily on the Gamber. All three of these pressures originate from agriculture, which is perhaps unsurprising given that there is little else going on within the catchment.

Abstraction is, to a large degree, unavoidable in an arable cropping area. The challenge is to effectively manage it, particularly in summer, during periods of low flow. At present, there is virtually no effective monitoring of actual abstraction volumes by the EA, a point revealed in a recent [Guardian article](#). The whole process relies on honesty and self-reporting. Moreover, we are aware of no local flow monitoring on the Gamber where the bulk of the abstraction occurs. The installation of a flow sensor in the lower Gamber would seem to be an obvious first step in gaining some confidence that agricultural abstraction is under control. The further creation of storage ponds for winter flow is also to be encouraged. Perhaps this is the sort of work which the much-anticipated ELMS payments could address.

On the face of it, turbidity should be the simplest problem to solve. It requires fencing livestock away from surface water, which incidentally would have other benefits in terms of improving riparian biodiversity – a point well understood in the context of Catchment Sensitive Farming. Surely, we can find a pot of money to help this along. Again, it would be a prime candidate for ELMS payments or similar.

Phosphorus is more problematic, coming from multiple sources. The phosphorus load from septic tanks is a small but significant component. The EA's national policy of ignoring their statutory duty to manage regulatory compliance of this area is lamentable and we have shown clear evidence of non-compliant, direct-discharging systems, particularly in the upper catchment, which locally constitute significant P sources in low summer flows. Nevertheless, agriculture is clearly responsible for well over three quarters of the phosphorus burden in the rivers, and crucially, it has increased significantly over the last twenty years in line with the rapid growth in the poultry industry. We have shown how poultry units now produce three times as much manure phosphorus as cattle and sheep combined and that there is barely enough agricultural land in the Garren Catchment to accommodate this combined load within the NVZ spreading limit.

The answer is simple to state. In order to return the Garren Catchment to WFD *good* status for orthophosphate, all poultry manure, from all existing and future IPU's, must be removed from the catchment; the practice of spreading as a means of disposal must cease; conversion by AD and subsequent spreading of digestate must also cease. This conclusion echoes the findings of RePhOKUs (Withers et al., 2022) but is amplified here by the exceptional concentration of IPU's within the Garren Catchment. If these changes do not occur, then river RP concentrations will continue to climb. The situation would also benefit from wider agriculture improving their management of soil phosphorus. To this end, there is a compelling need for comprehensive Olsen-P data across the catchment, which could be used to monitor progress on bringing catchment soils down to agronomic optimum values.

### Proposed actions

Listed below are issues to be addressed, and for which remediation should be monitored as part of the delivery of a formal NMP for the Garren Catchment.

#### WSA

WSA citizen science data has proved its worth within this study but, going forward, there are some opportunities for fine tuning.

- Investigate options for quantifying turbidity

- Consider adjustment of sampling locations
  - Discontinue sampling on the un-named stream – further action here is down to EA
  - Consider adding a location on the Llanerch Brook, currently not sampled
  - Switching the upstream Gamber location from Audits Bridge up to Llanwarne would give better geographical spread within this sub catchment

## EA

Rather than just measuring and documenting a decline, the EA need to take the lead in driving some improvement in the Garren Catchment. Some obvious actions would be:

- Investigate specific issues identified in this report
  - Diffuse phosphorus load on the un-named stream
  - Fixed flow load (presumed non-compliant septic tanks) on the un-named stream and on the Garren above Orcop more generally
  - Investigate the anomalous acidity results, possibly associated with autumn slurry spreading
- Consider installing volumetric flow measurement on the lower Gamber
- Coordinate (and ideally fund) an Olsen-P survey of the catchment, to be published annually
- Provide support to farmers to improve soil phosphorus management
- Re-commence regular water quality sampling on the Gamber and Upper Garren

## Farming community

Given the agricultural bias within the Garren Catchment, it is inevitable that farming needs to make some changes if the river is to recover:

- Poultry sector
  - Stop spreading manure within the Garren Catchment. This is an industrial waste stream which needs responsible disposal - urgently.  
Admittedly, the solutions are not readily achievable at the individual farm level, but this is a highly integrated industry. The meat and egg packers who engineered this situation are fully capable of organising transportation of manure to parts of the UK where it is needed. Of course, there are costs to consider, but all businesses carry the cost of responsible waste stream management – why not this one?
- Arable sector
  - Consider options for expanding storage of winter flow for irrigation
  - Improve soil phosphorus management to bring Olsen-P to target index
- Livestock sector
  - Fence livestock away from rivers and streams
  - Improve soil phosphorus management to bring Olsen-P to target index

## Postscript

As we have already observed, the Garren is not a healthy river system. It carries excessive nutrient loads, it is permanently turbid in the lower reaches, and contains few fish or macrophytes. Referring to the [National Biodiversity Network](#) website and taking a 2km radius around Llangarron, reveals the most recent sighting of kingfisher was in 2011, the last grey heron was seen in 2008 and the last sighting of an otter was in 2000 (on the other hand, EA did record a brown trout in their 2019 survey). On top of this, some parts of the system are heavily abstracted.

It is important to understand that this report does not set out to target farmers. Indeed, we hope that we have shown that most farmers are not doing anything terribly bad. However, in an agriculture-dominated catchment it is unsurprising that the main pressures facing the river originate from farming

practices. For sure, no one set out to harm the rivers. For the most part, the pressures we have discussed are unintended consequences, firstly of the post-war intensification of farming, which brought increased stock densities and additional phosphorus loads from mineral fertilisers and supplementary livestock feeds. But more latterly the addition of intensive poultry units has turbocharged the phosphorus situation. However, we are where we are and, if the situation is to be remedied, stakeholders need to agree some common facts and determine a course of action. This report attempts to collate some facts about the Garren Catchment and propose some improvement actions. Note *improvement* and not *mitigation*. We are attempting to find how best to modify agricultural practices to permanently reduce impacts, as opposed to attempting only to buffer them.

We have obviously focussed on the big picture aspects, especially in the context of phosphorus. We take it as a given that farm scale measures (Catchment Sensitive Farming and the like) will continue to be implemented. But recent studies (Davey et al., 2020) have shown that such measures are not by themselves remotely capable of delivering the scale of improvement required to repair our rivers. This point is picked up by RePhOKUs (Withers et al., 2022), who forcefully demonstrate the impact of diffuse pollution on the catchment scale: the now famous 3000 tonnes annual excess of phosphorus. This is the problem to solve. For the Garren, being excluded from the SAC, targets are more relaxed and WFD compliance can almost certainly be achieved simply by dealing with the poultry sector<sup>20</sup>. But, even here, the rivers are not impacted by poultry *per se*, but simply by the complete absence of responsible waste stream management within the sector. We welcome the admission of some responsibility for the plight of the Wye Catchment, albeit rather belated, from parts of the poultry industry. In particular, we are aware of plans by Avara to process their poultry manure using AD and incineration techniques, augmented by phosphorus stripping. However, we have the following reservations:

- The plan seems to depend on as-yet un-productionised technology for phosphorus stripping
- The plan depends on construction of a huge amount of industrial plant

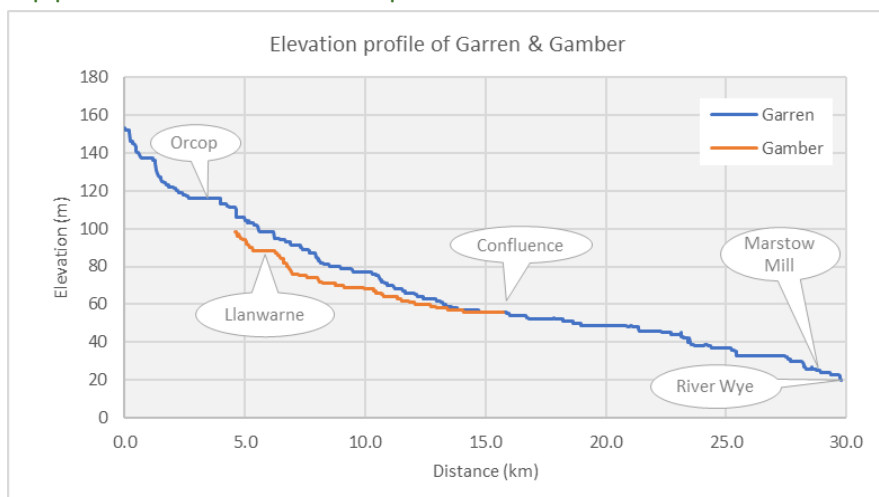
In the meantime, our rivers continue to decline for a few more years while Avara's most cost-effective solution is developed. Furthermore, where such plants are located on farms, there is a clear risk of both diffuse and acute pollution, as has been amply demonstrated on the lower Llynfi where concerns over planning, regulation, enforcement and monitoring have been formally raised by Fish Legal with the local planning authority and NRW. As stated earlier, there is no practical reason why this manure cannot be shipped out of the catchment by Avara today. It is clearly a major hole in our environmental legislation that such a solution cannot be enforced by our regulators.

As in our report on the Llynfi Catchment (Green, 2022a), we also welcome the establishment of the ad hoc Herefordshire Farmers' group, coordinated by Kate Speke-Adams of WUF and would hope that this could be a vehicle for advancing some of the proposals above.

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<sup>20</sup> This contrasts markedly with the situation on the Llynfi Catchment, described in our earlier report (Green, 2022a), where failure to meet the Wye SAC phosphorus target is primarily due emissions from the livestock sector.

## Appendix 1: Elevation profile of the Garren & Gamber



Elevation profile of the Garren Brook and Gamber, with some key locations indicated

## Appendix 2: WFD classification targets

The following tables are derived from the current UK regulations:

(The Water Framework Directive (Standards and Classification) Directions (England and Wales) 2015, 2015)

pH

| High              | Good | Moderate     | Poor         |
|-------------------|------|--------------|--------------|
| 6.0 – 9.0         |      | >4.7         | >4.2         |
| (at 5% & 95%-ile) |      | (at 10%-ile) | (at 10%-ile) |

Temperature

| High         | Good  | Moderate | Poor  |
|--------------|-------|----------|-------|
| <20°C        | <23°C | <28°C    | <30°C |
| (at 98%-ile) |       |          |       |

(Based on salmonid fish type)

Dissolved oxygen

| High         | Good      | Moderate  | Poor      |
|--------------|-----------|-----------|-----------|
| >80% sat.    | >75% sat. | >64% sat. | >50% sat. |
| (at 10%-ile) |           |           |           |

(Based on salmonid fish type)

Unionised ammonia

| High         | Good       | Moderate   | Poor       |
|--------------|------------|------------|------------|
| <0.3 mgN/l   | <0.6 mgN/l | <1.1 mgN/l | <2.5 mgN/l |
| (at 90%-ile) |            |            |            |

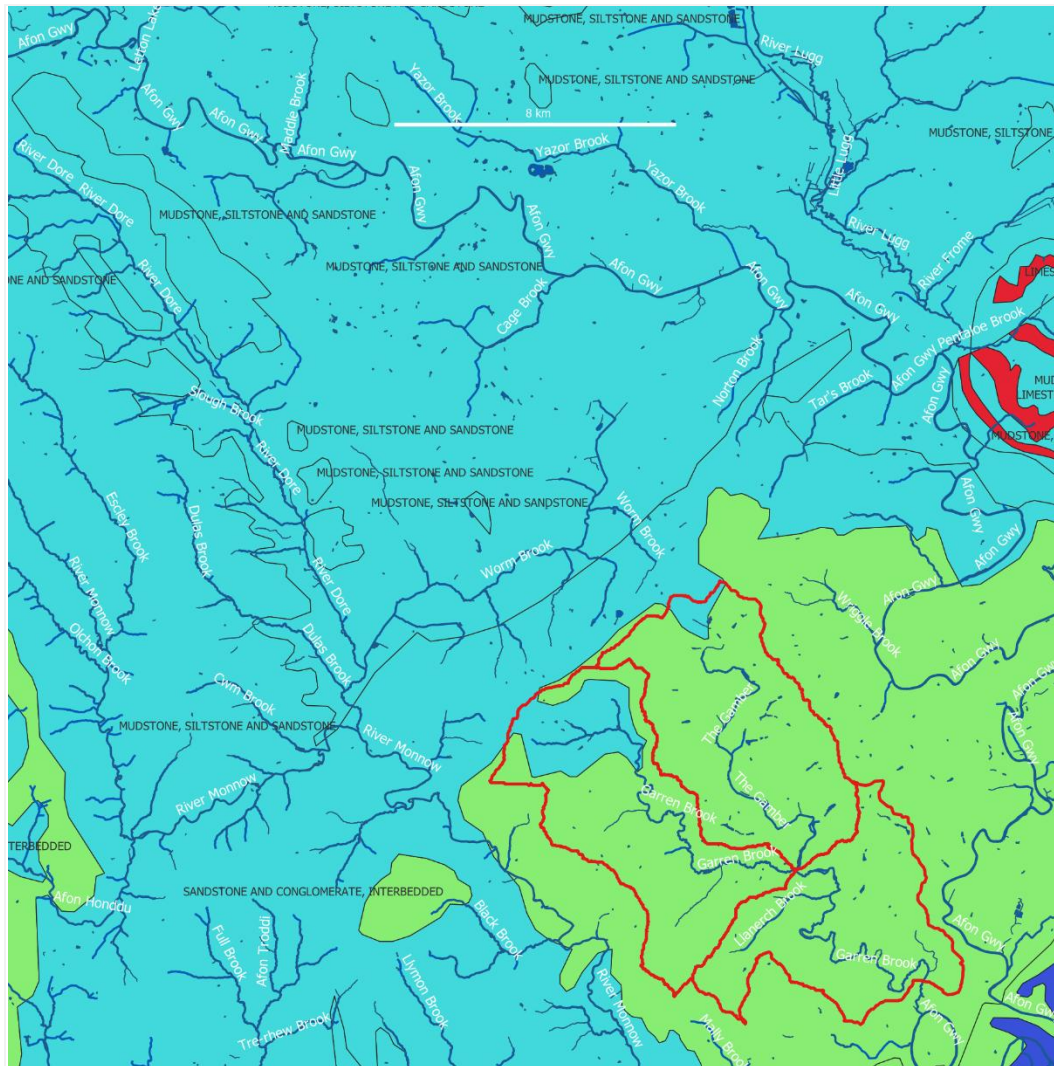
(Based on type 7 river, determined by average alkalinity)

Reactive phosphorus

| WSA sampling location          | High    | Good    | Moderate | Poor    |
|--------------------------------|---------|---------|----------|---------|
| Upper Garren – Orcop           | < 0.033 | < 0.064 | < 0.163  | < 0.979 |
| Upper Garren – Llangunock Br.  | < 0.041 | < 0.076 | < 0.188  | < 1.038 |
| Gamber – Audits Bridge         | < 0.043 | < 0.079 | < 0.193  | < 1.045 |
| Gamber – Tretire               | < 0.045 | < 0.082 | < 0.199  | < 1.063 |
| Lower Garren – Langstone Court | < 0.044 | < 0.082 | < 0.198  | < 1.061 |
| Lower Garren – Pack Saddle Br. | < 0.049 | < 0.089 | < 0.212  | < 1.091 |
| (mgP/l -annual average)        |         |         |          |         |

(Based on local elevation and average alkalinity for sub catchment / water body)

## Appendix 3: geology of the Garren Catchment



Bedrock map of the lower Wye catchment (British Geological Survey 625K mapping); the Garren Catchment is outlined in red

The above figure shows the bedrock beneath a large part of the lower Wye Catchment. The bulk of area sits on mudstone, siltstone and sandstone marl (light blue) (Jarvie et al., 2003). The Garren Catchment is untypical of this wider area, sitting on interbedded sandstone and conglomerate (green). This will result in different erosion rates, affecting both alkalinity and sedimentation rates.

Note the two intrusions at the north of the Garren Catchment. In particular, a significant piece of the Upper Garren catchment above Orcop shares its geology with the Worm Brook to the north and is not typical of the wider Garren Catchment.

## Appendix 4: NMP generic actions

The River Wye SAC Nutrient Management Plan (NMP) proposes that phosphate pollution be addressed on a sub-catchment basis and all sub-catchment plans include the same generic actions:

- Targeted work based on phosphates evidence report but also taking into account other relevant evidence
- Identify potential problem point sources and loading discharging to the river network including DCWW, private works, septic tanks, Combined Sewer Overflows, agricultural units, anaerobic digester plants

- Identify sources discharging to ground including septic tanks, land spreading etc.
- Identify and tackle pathways of runoff from land
- Where large changes to farming practice are occurring, use catchment officers to approach farmers and encourage changes to farm design, including much wider buffer strips, interception ponds to trap runoff and slow flow etc.
- Need for river restoration work along main river and tributaries to reduce pollution risk and ecological resilience
- Ditch blocking and wetland restoration to increase baseflows
- Encourage natural flood management and natural flooding including creation of floodplain meadows

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## List of abbreviations

|        |                                                                                   |
|--------|-----------------------------------------------------------------------------------|
| AD     | Anaerobic Digestion                                                               |
| CEH    | UK Centre for Ecology and Hydrology                                               |
| CP     | Civil Parish                                                                      |
| CPRE   | Campaign to Protect Rural England                                                 |
| CPRW   | Campaign for Protection of Rural Wales                                            |
| CS     | Citizen Science                                                                   |
| DO     | Dissolved Oxygen                                                                  |
| DEFRA  | Department for Environment, Food & Rural Affairs                                  |
| EA     | Environment Agency                                                                |
| EC     | Electrical conductivity                                                           |
| EC-TDS | Total Dissolved Solids (determined from a measurement of Electrical Conductivity) |
| ELMS   | Environmental Land Management Scheme                                              |
| FoI    | Freedom of Information                                                            |
| FOL    | Friends of the Lugg                                                               |
| FOUW   | Friends of the Upper Wye                                                          |
| HER    | Hydrologically Effective Rainfall                                                 |
| IPU    | Intensive Poultry Unit                                                            |
| LAM    | Load Apportionment Modelling                                                      |
| NMP    | Nutrient Management Plan                                                          |
| NRW    | Natural Resources Wales                                                           |
| NVZ    | Nitrate Vulnerable Zone                                                           |
| OS     | Ordnance Survey                                                                   |
| PE     | Population Equivalent                                                             |
| SAM    | Source Apportionment Modelling                                                    |
| SC     | Specific Conductivity (electrical conductivity at 25°C)                           |
| RP     | Reactive Phosphorus                                                               |
| TDS    | Total Dissolved Solids (determined gravimetrically)                               |
| TP     | Total Phosphorus                                                                  |
| WFD    | Water Framework Directive                                                         |
| WSA    | Wye Salmon Association                                                            |

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